project | OUTCOME
MEASURING THE TRUE IMPACT OF LIBRARIES

ASSESSMENT AND ADVOCACY

USING PROJECT OUTCOME FOR ACADEMIC LIBRARIES

edited by
Gena Parsons-Diamond

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Association of College and Research Libraries
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Foreword

Today more than ever, academic libraries serve as partners in fostering student success, achieving institutional missions, and as leaders in assessment and continuous improvement on their campuses. Despite their undeniable impact, libraries often struggle to capture and present compelling data that effectively showcase their contributions. The introduction of the Project Outcome for Academic Libraries toolkit in 2019 has provided a free, easy way for library workers to measure learning outcomes, compare with their peers, drive strategic decisions, and demonstrate their value.

Library assessment metrics have historically focused on easily quantifiable inputs and outputs, such as visitor counts, book circulations, and the number of instructional sessions. However, these metrics fall short of capturing the true impact of libraries on their users and the broader community. Project Outcome for Academic Libraries has emerged as a pivotal tool in this landscape, driven by the dual objectives of providing an accessible assessment model and generating impactful data that benefits the profession at large. By offering straightforward and user-friendly surveys and resources, Project Outcome lowers barriers to entry, enabling libraries of all sizes to embrace outcomes assessment and continuous improvement.

The toolkit hopes to help foster a culture in which outcomes measurement is integral to library practice, ensuring that libraries can effectively communicate their value at institutional and national levels. By simplifying the collection and sharing of outcomes data, Project Outcome allows libraries to benchmark their performance against peers, articulate their impact to stakeholders, and drive ongoing enhancements to their services.

To celebrate the five-year anniversary of the toolkit, ACRL is pleased to publish *Assessment and Advocacy: Using Project Outcome for Academic Libraries*, which highlights the importance of outcomes measurement and demonstrates how libraries can use the toolkit to refine their work and definitively show their impact. The ten case studies included in this book showcase how data-driven insights can lead to improvements in library programs and services. Whether it's a minor adjustment in study room arrangements or substantial national research and benchmarking projects, the Project Outcome toolkit provides the necessary tools for libraries to advocate for their indispensable role.

I hope this book inspires and equips more libraries to adopt and benefit from the Project Outcome toolkit, ultimately enhancing their ability to support students, faculty, and the broader academic community. The journey towards a comprehensive understanding and demonstration of library impact is ongoing, and with tools like Project Outcome, library workers worldwide can contribute to creating and capturing its future.



— Dr. Leo S. Lo
ACRL President 2024-2025
Dean and Professor of the College of University Libraries and Learning Sciences
University of New Mexico

Introduction

Academic librarians know that their work makes a difference: They set students on the road to success and researchers on the road to results. They provide valuable skills and access to knowledge that help our colleges and universities thrive. The challenge for libraries can be capturing meaningful data to support that story. Since 2019, the Project Outcome for Academic Libraries toolkit has helped library workers measure learning outcomes to drive change, make data-informed decisions, and demonstrate the impact and value of academic libraries.

Assessment and Advocacy: Using Project Outcome for Academic Libraries aims to highlight the value of outcomes measurement along with ways the Project Outcome toolkit has enabled individual institutions and the academic library profession to use these measurements to more clearly demonstrate their impact. The ten case studies included in this publication highlight the benefits of the toolkit and how the data can help improve library programs and services. From small changes in a library's study room furniture arrangement to national research and benchmarking, Project Outcome has provided the tools libraries and library workers need to advocate for the valuable work they do.

APPROACHES TO ASSESSMENT: WHY OUTCOMES?

Traditionally, library statistics primarily revolved around easy-to-measure inputs and outputs such as the number of library visits, books circulated, or instruction sessions offered. However, these metrics failed to capture the true impact of libraries on their users and the value they brought to the community. In the 1980s and 90s, the focus shifted towards assessing service quality and user satisfaction, which was a step in the right direction. Still, these measures often lacked the compelling evidence needed to persuade library decision makers, administrators, and external audiences of a library's broader value within the institution and higher education.

More recently, academic libraries have adopted two value categories for capturing and communicating their contributions: financial value and impact value. Financial value involves going beyond basic input/output numbers to assess the library's monetary value through return on investment (ROI), willingness to pay for services, continual evaluation and culling of underutilized or underperforming resources and services, and analyzing the secondary

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economic impact of library activities in local markets. Impact value, including outcomes measurement, examines benefits to library users, typically expressed as changes to behavior, ability, skills, knowledge, or status. These methods have yielded convincing results for public, special, and academic libraries, helping demonstrate their value to stakeholders. That value is often communicated through means such as data visualizations or data storytelling. This shift—from focusing solely on use and satisfaction to outcomes and impact assessment—was influenced by various stakeholders, including government bodies, trustees, administrators, parents, students, and the Institute for Museum and Library Services. This transition has prompted libraries to reevaluate how they assess themselves and emphasize the broader impact they have on their institutions and communities, aligning more with the practices of other social services non-profits dedicated to improving the lives of those they serve.

OUTCOMES MEASUREMENT AS COMMON PRACTICE

The work of Project Outcome for Academic Libraries has been driven not only by the desire to provide the toolkit, but also to provide a valuable assessment model[1] and data that can have a positive impact on the field. The toolkit, model, and data together provide an important, free service to academic libraries, consortia, and the wider profession. The Association of College & Research Libraries (ACRL) would like Project Outcome to serve as the standard for the field, enabling more libraries to undertake outcomes measurement by reducing barriers to entry.

This assessment model, called the theory of change,[2] envisions moving toward outcomes measurement as common practice: creating a culture of continuous improvement and using outcomes data to understand and communicate the value of academic libraries at the institutional and national level. Because Project Outcome provides simple and easy-to-use measures, academic libraries of all sizes can find applicability in the tool. This ease of adoption facilitates a greater understanding of the importance of assessment and outcomes measurement. Ultimately, the goal of Project Outcome is that all libraries will view outcomes measurement as common practice. By making it easy to collect and share outcomes data, Project Outcome allows libraries to understand how they are doing relative to peers by benchmarking institutional scores against national averages; communicate their value to campus partners, administrators, and funders; and allow for continuous improvement in the library.

On a national level, responses to Project Outcome surveys provide a unique data set focused on outcomes that bolster advocacy opportunities for ACRL. As seen in Chapter 1, "Characteristics of Academic Libraries, Library Instruction, and Student Outcomes: A Descriptive Analysis Focusing on Minority-Serving and Rural Institutions," using Project Outcome data in conversation with datasets from other sources, such the Integrated Postsecondary Education Data System (IPEDS) and the ACRL Academic Library Trends & Statistics survey, can allow for greater insights and research opportunities.

WHAT IS PROJECT OUTCOME?

While many libraries collect data about their programs and services, it can be more difficult gathering evidence that captures the benefits libraries provide to students and other users. Measuring outcomes can provide libraries with new ways to demonstrate their effectiveness beyond gate counts and anecdotal success stories.

Project Outcome is designed to help libraries understand and share the impact of essential library programs, instruction, and services by providing simple surveys and an easy-to-use process for measuring and analyzing outcomes. ACRL believes so strongly in the value of this tool for the profession, and places such a high importance on libraries implementing outcomes-based assessment and improvement, that the association invests $60,000 annually in making the tool freely available. This support ensures the tool is available to all post-secondary libraries without a subscription fee.

Project Outcome provides libraries with access to quick and simple patron surveys, an easy-to-use survey management tool, custom reports and interactive data dashboards for analyzing the data, and various resources to help libraries implement the surveys and then use the results. Libraries are encouraged to use their results to support and promote future action—allocating resources more efficiently, advocating for new resources more effectively, and providing support for future library funding, activity reports, and strategic planning. Whether new to outcomes measurement or advanced in data collection, all academic libraries can access the standardized surveys, national and peer benchmarking, and data analysis tools to effect change within their institutions and beyond.

Project Outcome's standardized surveys allow libraries to aggregate their outcomes data and analyze trends by service topic, program type, and over time. Academic libraries can see how the outcomes of their programs and services compare across their institution, Carnegie classification, and nation. Project Outcome for Academic Libraries surveys help libraries measure outcomes and assess their impact in seven key service areas (see Figure 1).

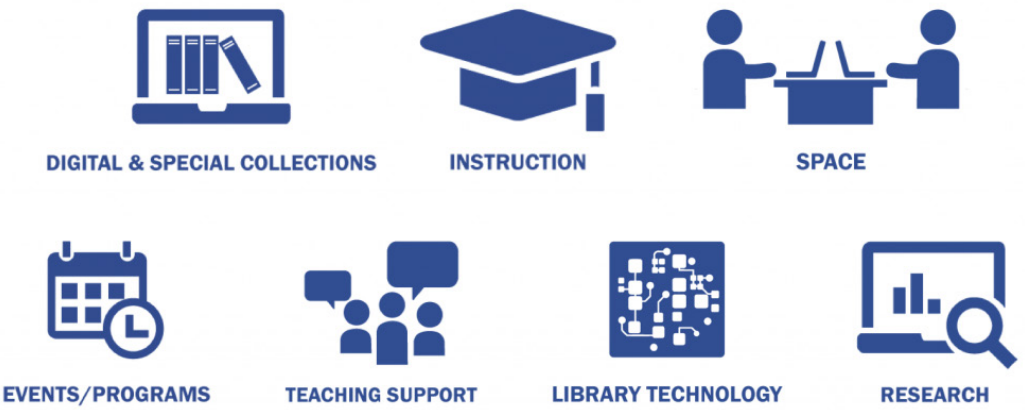


Figure 1. The seven survey topics offered in the Project Outcome for Academic Libraries toolkit.

The outcomes measured in the Project Outcome surveys are one piece of the assessment puzzle. They help libraries understand the specific benefits that result from their services or programs. Outcomes can be quantitative or qualitative and are often expressed as changes that individuals perceive in themselves. They answer the question: "What good did we do?" (See Figure 2.)



Figure 2. Four key outcomes measured in the Project Outcome for Academic Libraries toolkit.

Project Outcome provides three types of tools for libraries to measure their patron outcomes (see Figure 3).



Figure 3. The three types of tools provided in Project Outcome and what they measure.

Each Project Outcome immediate survey is six questions long and includes both Likert-scale and open-ended questions. Libraries can add up to three custom questions to help them gather additional evidence. The immediate surveys are designed to be distributed right after a program or service is completed and aim to help libraries better understand the immediate impact that program or service had on patrons and their intention to change their behavior as a result. Results from immediate surveys are ideal for informing program or service changes and providing a snapshot for advocacy and reporting. The

standard immediate and follow-up surveys for all topics are available in Appendices C and D at the end of the book.

Each Project Outcome follow-up survey has five questions and follows a Yes/No/Not Applicable/Please Explain format. The follow-up surveys are designed to be used four to eight weeks after a program or service is completed and aim to help libraries better understand if patrons have changed their behavior or continued to benefit as a result of the program or service. Results from follow-up surveys are ideal for assessing the lasting impact of a program or service, informing internal planning, measuring progress toward strategic goals, and providing evidence for advocacy.

The Outcome Measurement Guidelines[3] included in the toolkit resources provide additional support for outcomes-focused data collection. The guidelines focus on four areas: developing outcome measures, alternative data collections methods, measuring outcome data over time, and working with partners. They, along with the many other resources, are designed to help libraries conduct advanced methods of outcomes measurement and demonstrate long-term, collaborative impact on their users and institutions.

Development Timeline

Project Outcome was first developed by the Performance Measurement Task Force of the Public Library Association (PLA). The task force collected information about outcomes measurement practices in the 2013 Public Library Data Service (PLDS) survey and conducted pilot tests of new outcomes-based surveys in 27 libraries in late 2014. Using the resulting data, the task force identified the seven essential public library service areas that could be easily and directly linked to improving or changing a patron's knowledge, application, confidence, and awareness.[4] PLA launched the Project Outcome immediate surveys and online toolkit in June 2015. PLA's follow-up surveys were released to the public in June 2016 at the ALA Annual Conference in Orlando. The work of the task force caught the eye of the Bill & Melinda Gates Foundation. Their generous support enabled PLA to accelerate and expand the reach of Project Outcome.

PLA noticed a strong interest in the Project Outcome assessment tools from academic libraries and approached ACRL in 2017. The ACRL Board of Directors voted to appoint a Project Outcome for Academic Libraries Task Force in November 2017 and charged it with adapting Project Outcome to an academic library context. ACRL's then-Executive Director Mary Ellen K. Davis said of the tool, "Project Outcome builds on ACRL's Value of Academic Libraries initiative and provides a tool for comparing outcomes nationwide. We appreciate the opportunity to expand PLA's Project Outcome to serve academic libraries and look forward to our continued collaborations with them."[5] The task force decided on seven academic library service areas and 54 institutions engaged in field-testing the new surveys between June and October of 2018.[6,7] The Project Outcome for Academic Libraries toolkit launched at the ACRL 2019 Conference, held that April in Cleveland. In the summer of 2019, the ACRL Board voted to approve a standing Project Outcome

for Academic Libraries Editorial Board. The editorial board works with ACRL staff to develop new resources, create and present training materials, and advise on new features and improvements to the toolkit.

WHAT WE'VE DONE

Toolkit Improvements

ACRL continuously seeks user feedback and makes improvements to the toolkit with our partners in PLA, along with the assistance of our web development team Community Attributes Inc. Community Attributes provides research and analysis to increase awareness, inform decisions, and improve results for their clients and the communities they serve. They are responsible for the technical upkeep of the toolkit, including resolving user issues, implementing improvements and fixes, and maintaining the data dashboards and survey instruments.

Ongoing work has improved the user experience, site functionality, and account management options. The following improvements have been made to the toolkit since its launch:

- Google translate integrated throughout the site.
- Carnegie Classification filters added to the data dashboards.
- Case studies published:
  - DeSales University: *Closing the Loop: Using Project Outcome to Assess and Improve a First-Year English Composition Information Literacy Program*
  - Southern Utah University: *Using Project Outcome to Assess Instruction Performance in an Academic Library*
  - Nevada State College: *Assessing the Impact of Study Rooms*
  - Iowa State University: *Assessing and Improving Library Spaces and Technology*
  - Central Piedmont Community College: *Measuring Spaces, Technology, and Events: Improving the Student Experience using Project Outcome for Academic Libraries*
- New resources added:
  - Visualizing Data
  - When to Use Project Outcome
  - Measuring Virtual Programs
  - External Publications
- "Survey responses viewer" allows raw data to be downloaded from multiple surveys at once.

- Qualitative dashboard to visualize open-ended responses.
- Survey collection methods added:
  - QR codes
  - Ability to embed survey directly into another website
- All surveys translated into Arabic.[8]
- Ability to save custom dashboard visualizations as interactive links or static images.
- Added National Survey of Student Engagement (NSSE) High Impact Practices as a taggable feature.

User Engagement

Between the April 2019 launch and July 12, 2023, 5,016 users (librarians, library workers, and LIS students) have registered for the Project Outcome for Academic Libraries toolkit. The users come from 1,584 institutions in 65 countries. There have been 12,154 surveys created in the toolkit that have collected 159,252 responses. See Appendix E for a full list of institutions who have collected survey responses. While Project Outcome is open to all academic and research libraries globally, the majority of registered users (over 76 percent) and survey responses (96 percent) come from institutions in the United States.

Surveys are distributed and responses collected by institutions from all Carnegie classifications. Community colleges and 2-year institutions account for 17 percent of response data, four-year institutions for 16 percent, master's-granting colleges and universities for 33 percent, doctoral-granting universities for 31 percent, and special focus institutions for 3 percent. As shown in Figure 4, Associate's Colleges are underrepresented in Project Outcome data, while Master's Colleges & Universities and Doctoral Universities are overrepresented.

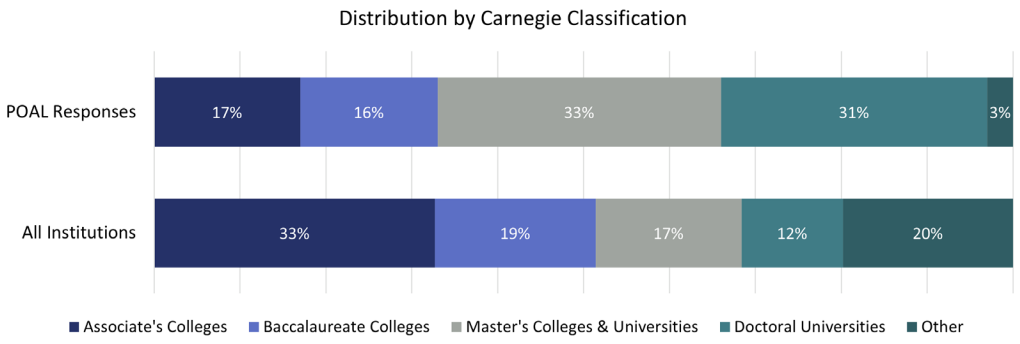


Figure 4. Chart comparing Project Outcome usage by Carnegie Class to overall institutions by Carnegie Class.

WHAT LIBRARIES HAVE LEARNED

Project Outcome provides academic libraries with ready-to-go surveys in seven topic areas. Each survey measures four key learning outcomes—knowledge, confidence, application, and awareness—to help libraries understand the benefits of their programs and services. Looking at aggregate data from across the United States, we can get a big-picture view of the types of programs and services libraries assess, and how they are doing in those areas. The data in Figures 5 and 6 reflect survey activity between the launch of the toolkit on April 16, 2019, and July 12, 2023.

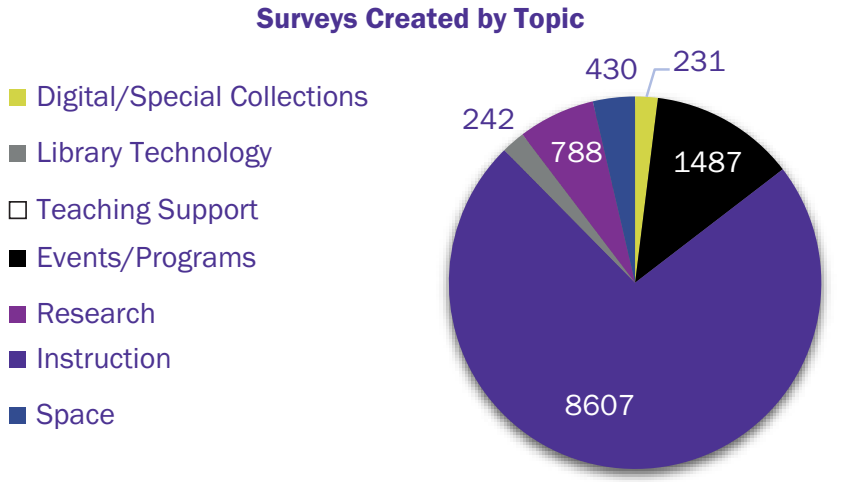


Figure 5. Chart displaying number of surveys created per topic.

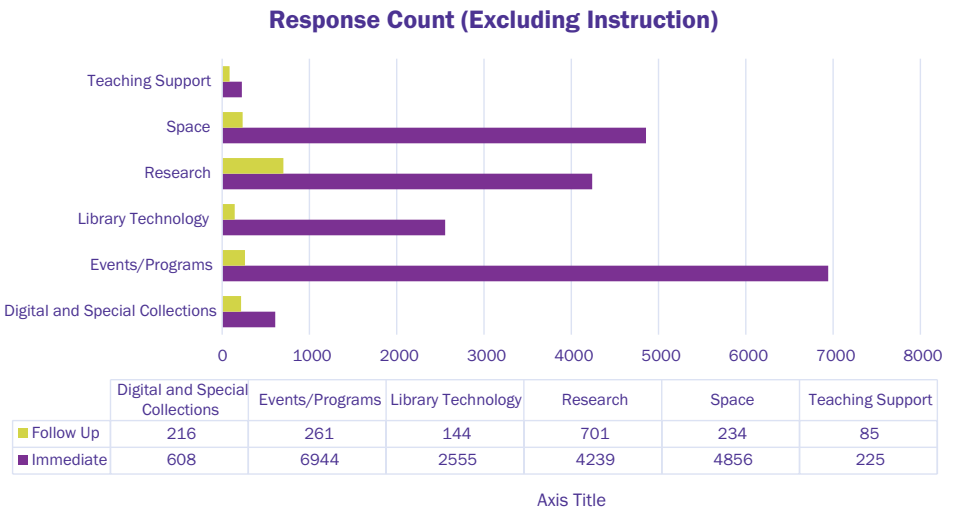


Figure 6. Chart displaying response counts by survey topic, broken up by immediate and follow-up responses. This chart excludes instruction survey response numbers, which are detailed in Chapter 1.

Survey Topics

Below is a breakout of survey usage by type, including total number of surveys and total responses for each topic, as well as a sample of the types of programs or services each survey topic has been used to assess. Surveys have been sent to students, faculty, administration, and other library stakeholders.

- **Digital and Special Collections**: 231 surveys have been created for this topic, with 608 immediate responses and 216 follow-up responses collected. Surveys have been created to assess digital exhibits, instruction on primary sources, special collections materials and methods, and programs related to film, television, and museum studies.
- **Events/Programs**: 1,487 surveys have been created for this topic, with 6,944 immediate and 261 follow-up responses collected. Surveys have been created for book clubs and community reads, library resources and services demonstrations, virtual programs, DEI discussions, and evaluation of therapy dog sessions.
- **Instruction**: 8,607 surveys have been created for this topic, with 133,698 immediate and 6,786 follow-up responses collected. Surveys have been created for library orientation and first-year experience instruction, as well as instruction in information literacy, citations, searching, and specific subject matter.
- **Library Technology**: 242 surveys have been created for this topic, with 2,555 immediate and 144 follow-up responses collected. Surveys have been created to collect data on equipment check-out processes, digital media studio evaluations, and accessing learning management systems.
- **Research**: 788 surveys have been created for this topic, with 4,239 immediate and 701 follow-up responses collected. These have been used to collect appointment or consultation feedback and assess advanced searching technique and, literature review sessions, as well as embedded librarian programs.
- **Space**: 430 surveys have been created for this topic, with 4,856 immediate and 234 follow-up responses collected. Surveys have been created for evaluation of study rooms, including group, conference, and full-day use rooms, as well as new furniture, printing services, and after-hours access.
- **Teaching Support**: 369 surveys have been created for this topic, with 225 immediate and 85 follow-up responses collected. Surveys have been created to assess course mapping, open educational resources (OER) workshops, and information literacy development programs.

Usage over Time

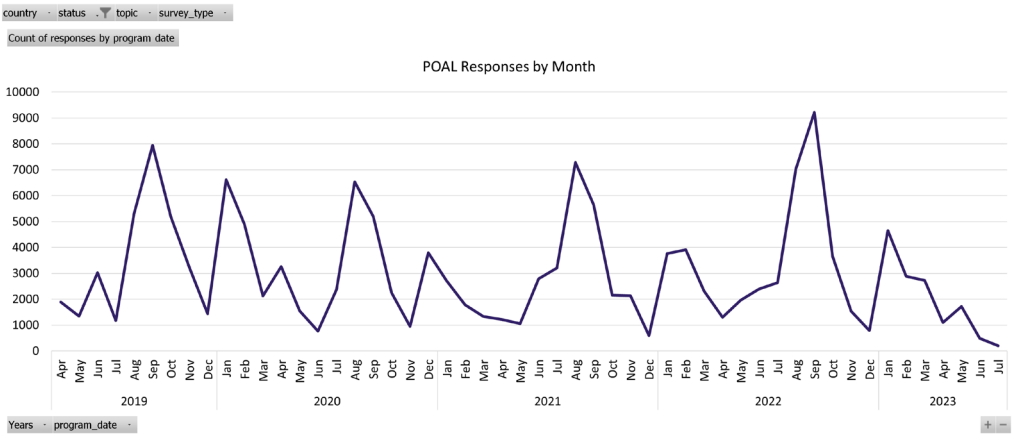


Figure 7. Chart displaying response counts by month from April 2019-July 2023.

In the usage over time data displayed in Figure 7, there are a few interesting points to note. The first is that the toolkit is being widely used at the beginning of the academic year, as the largest spikes in the graph are consistently in September and October. The beginning of the fall semester is typically when many libraries run orientation programs for first-year students, which suggests that Project Outcome surveys are being widely used for this type of program. The smaller spike that appears in January of each year also suggests that library workers often use the survey for one-shot instruction sessions, which typically take place at the beginning of either the fall or spring semester. While there was a dip in survey usage during the COVID-19 pandemic in 2020 and 2021, the 2022 usage numbers show that more libraries than ever before are using the toolkit to measure outcomes.

Value of Academic Libraries

Figure 8 comes from the Project Outcome matrix dashboard, which allows users to explore their data for relationships and correlations and can also be used to drive decision-making and identify gaps in service or impact. The matrix diagram makes it possible to view aggregate average scores for each survey topic in connection with each outcome. This figure presents the aggregate national outcomes measurement for each survey topic. The four quantitative questions for each survey are on a Likert scale from "strongly disagree" (1) to "strongly agree" (5), and this matrix diagram shows the positive difference that libraries are making on user learning across all outcomes and survey topics. The lowest

average score is 4.1 in digital and special collections and library technology. The research, space, and teaching support relationships are the strongest, with an average score of 4.4.

Topic	Outcome: Increased Knowledge	Increased Confidence	Application New Skills	Awareness of Resources	
Digital & Special Collections	4.0	4.1	4.0	4.1	4.1
Events / Programs	4.4	4.2	4.1	4.2	4.2
Instruction	4.4	4.2	4.4	4.4	4.3
Library Technology	4.1	4.1	4.2	4.1	4.1
Research	4.5	4.3	4.5	4.4	4.4
Space	4.3	4.3	4.6	4.1	4.4
Teaching Support	4.5	4.4	4.4	4.5	4.4
	4.3	4.2	4.3	4.3	Averages

Figure 8. Matrix diagram displaying the relationship between survey topic and outcome.

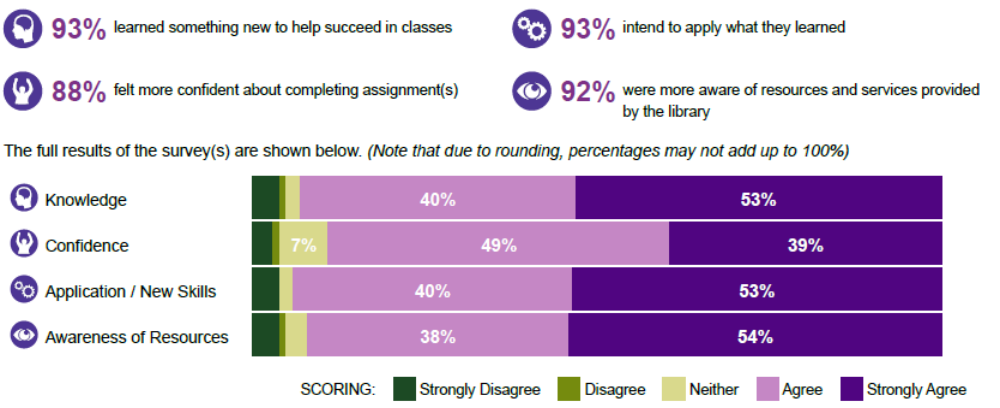


Figure 9. Image displaying results for immediate instruction survey responses.

Figure 9 shows the immediate instruction survey results for all institutions and responses. The data indicates that 93 percent of respondents learned something new that can help them succeed in their classes and that they intend to apply what they learned. These findings indicate that academic libraries play a crucial role in enhancing students' knowledge, confidence, and practical application of new skills. They also demonstrate libraries' contributions to increasing students' awareness of available resources, thereby reinforcing their academic success and development. Registered users can pull a similarly formatted graph collecting responses from their institution in the custom reports section of Project Outcome.

Each survey also has two open-ended questions, and the comments survey respondents share in response to those questions attest to the value of academic libraries—and library workers—in supporting the mission and community at their colleges and universities. Many of the responses align with and reinforce the standard outcomes assessed in quantitative questions. Here are some unedited, representative quotes, with emphasis added and organized by outcome.

KNOWLEDGE

- I liked that the librarian didn't just tell us what resources we have but she also **showed us** and did examples so that we could be engaged.
- I liked **learning how** to search for things in the databases and using that to find more keywords to search for.

CONFIDENCE

- Made me feel **more confident** about my future work and how I could gain good, trustworthy information.
- I learned that there are so many resources in the library to help me with lesson planning and coming up with teaching strategies. It made me **feel much more confident** about planning lessons.

APPLICATION/BEHAVIOR CHANGE

- I liked how we used an example as we learned. This helped me **apply what I learned** with the website given.
- The exchange of ideas shared by the panelists. Topic was relevant and timely. I will begin to review ads and political commercials **in a new way**.

AWARENESS

- Learning more about **all the resources** our library has to offer.
- Helped me get a better understanding of **what the library has to offer** and how to narrow my search. The library also provides many different resources that may

> cost money somewhere else. Having many different resources within one search engine can help me decipher what I'm specifically looking for in a study as well.

Being able to link the open-ended responses to the standard responses allows libraries to emphasize the effectiveness and impact of library services, programs, and resources using respondents' own words. Additionally, open-ended responses can assist library workers in adapting and enhancing library offerings to better meet the needs of their users.[9]

The open-ended responses also reveal more intangible benefits of using the library, such as the benefit of collaboration, the importance of the library to the campus and wider community, and the care it can provide to its users. These affective responses may indicate that librarians and library workers improve students' sense of belonging and strengthen their identity as emerging scholars, areas of strategic focus at many higher education institutions. Here are some unedited, representative quotes highlighting these intangible benefits, with emphasis added.

- The library can help me **succeed by just existing**, because I can rent books with info on topics I need to learn more about.
- That she **cares about us**.
- It was kinda fun and **made me feel better** about going into my next project.
- I feel **less stressed** about this paper.
- I am really enjoying these community reads. It's great to hear different perspectives and I really enjoy the **collaboration** between the campus community and the community at-large.

Taking Action

There are several key areas where libraries have used Project Outcome data to improve their advocacy efforts, resources, programs, assessment, and campus partnerships.

ADVOCACY

Outcomes measurement data provides evidence of library value that can be shared within the library and with campus administrators. It helps libraries reinforce the importance of their work and create messages for library marketing. One library administrator says, "Our [university] administration loves the ability to compare our scores with national averages. That has proved powerful in our ROI conversations." Another library administrator highlighted how easily Project Outcome data can be used in advocacy, saying they use the data "to show administration the effect on student employees; to show academics the effect on student research; both are used in our annual assessment report."

ASSESSMENT

While any form of assessment can help libraries gauge the strengths and weaknesses of their programs and services, measuring outcomes can help focus attention on what users learn

and therefore the effectiveness of those programs. In addition, libraries report sharing their Project Outcome data with campus-level assessment and accreditation offices and including results in promotion and tenure packages as evidence of impactful teaching practice.

The case study *Using Project Outcome to Assess Instruction Performance in an Academic Library,*[10] available in the Resources section of the Project Outcome website, discusses library instructors at Southern Utah University using the instruction surveys to evaluate their performance. Using both the qualitative and quantitative questions, instructors were able to both improve the quality of their teaching and use the quick reports and visualizations provided in the toolkit to add compelling evidence to their tenure and promotion packets.

PARTNERSHIPS

Libraries have used Project Outcome to gather evidence related to campus-wide initiatives, including first-year experience programs and other initiatives geared toward improving student success.

The case study *Two Academic Libraries, One Study: Using Project Outcome for Assessment and Advocacy across Partner Institutions,* Chapter 5 in this publication, discuss how two institutions used the toolkit in tandem. Both Seminole State College of Florida (SSC) and the University of Central Florida (UCF) used the immediate instruction survey to formalize assessment of information literacy instruction across the two institutions, as there is a high transfer rate from SSC to UCF. By partnering together, they were able to better demonstrate the positive impact of information literacy instruction by librarians at both institutions.

PROGRAM IMPROVEMENTS

Having good, consistent evidence can inform changes to library programs and services. Based on Project Outcome data, libraries have made improvements to teaching practices and strategies. As one instruction librarian shares, "We will be redesigning our education curriculum this summer using the [Project Outcome] data from Fall 2022 and Spring 2023."

In the case study *Revitalizing our Instructional Assessment: Embracing a New Approach Using ACRL Project Outcome*, Chapter 7, librarians from the University of Houston-Clear Lake discuss how they used the immediate instruction survey to improve library instruction at both the individual and team level. This led to a standardization of the instruction program and the establishment of new learning criteria to better support first-year and transfer students.

RESOURCES

A consistent feature of the open-ended response data is users asking for more of the types of resources the library offers. Having that evidence can help libraries make specific funding requests and prioritize changes.

In the case study *Assessing the Impact of Study Rooms,*[11] available in the Resources section of the Project Outcome website, Nevada State College used the space survey with students who had reserved study rooms during one semester. They discovered that their study rooms were used more often for individual quiet study than for group work. This led the library to update their policy to allow individual students to reserve study rooms. They also installed sound absorbing panes and converted a nearby classroom into a quiet computer lab. The actionable feedback gathered with the space survey allowed Nevada State to provide more of the specific resources their students needed.

ORGANIZATION OF *ASSESSMENT AND ADVOCACY*

This book begins with the chapter *Characteristics of Academic Libraries, Library Instruction, and Student Outcomes: A Descriptive Analysis Focusing on Minority-Serving and Rural Institutions* by Kara Malenfant and Sara Goek. Malenfant and Goek discuss several national-level academic library survey tools and datasets and how the Project Outcome data can be used in conversation with these sources. Their ultimate goal is to encourage greater participation in national-level surveys and using this data for future research opportunities.

Following this are ten case studies, each of which describes how use of the Project Outcome for Academic Libraries toolkit has benefitted the authors' library and institution.

- *Three Newly Joined Libraries Seeking to Better Serve Students and Grow Advocacy Efforts* by Joe Pirillo
    - The University of Wisconsin Oshkosh (UWO) aimed to assess the impact of library support on student success, particularly in Writing 188 classes, to contribute to a new library strategic plan. Using Project Outcome surveys, the university's instruction librarians gathered feedback from students and individual research appointment participants, leading to workflow adjustments and insights into possible future improvements for the library.
- *Collecting Iowa State University Library Spaces Feedback Using Project Outcome* by Greg Davis
    - Iowa State University (ISU) uses the Project Outcome space survey to assess library spaces and make data-driven decisions. The survey, conducted semi-annually since fall 2018, involves collecting both quantitative and qualitative data from students who have reserved library study rooms.
- *Using the Instruction Survey as Part of a Three-Tiered Approach to Information Literacy Instruction Assessment* by Karoline Manny
    - The Centre College library's information literacy instruction program has used Project Outcome instruction surveys since the 2020-2021 academic year as part of its three-tiered assessment plan: course-level,

satisfaction-based assessment; competency-based assessment; and program-level, competency-based assessment. The goal of this assessment is to provide student feedback to library instructors and use the results to demonstrate the value of library instruction to faculty and administrators.

- *Two Academic Libraries, One Study: Assessment and Advocacy across Partner Institutions* by Katy Miller, Rachel Trnka, and Karen Kaufmann
    - Librarians from two institutions, Seminole State College of Florida (SSC) and University of Central Florida (ICF), collaborated on a project to study library instruction outcomes across both campuses using the Project Outcome instruction survey. The resulting data was used to align information literacy instruction at SSC and UCS, strengthen their partnership, and support student success.
- *Small Changes, Big Impact: Assessing Group Study Rooms Using the Space Survey* by Jennifer Arnold, Christy Case, and Retha Hall
    - In July 2022, Central Piedmont Community College opened their new library, the largest library space of their six campuses. Staff used the Project Outcome space survey to assess the study rooms as a part a larger assessment plan, considering in particular potential changes to the study room configurations that could improve student knowledge, confidence, and awareness of library resources.
- *Revitalizing our Instructional Assessment: Embracing a New Approach Using Project Outcome* by Martha Hood and Juliana Espinosa
    - After two years of virtual instruction due to the pandemic, the librarians at the University of Houston-Clear Lake undertook the task of updating and modifying the first-year experience course. They used the immediate instruction survey to gather a clear vision of areas which needed improvement and used responses to promote the value of the library's instruction program to faculty and other stakeholders.
- *Using Instruction Surveys to Establish Baseline Data for Library Instruction Improvement* by Laura D'Aveta
    - Before January 2022, the library at Hiram College had not collected any assessment or performance data about library instruction sessions. Following an administrative decision that made library instruction a mandatory part of the college's first-year seminar courses , they used the instruction survey to determine if the mandate resulted in an improvement in student performance.
- *Using the Instruction Follow-Up Survey to Advance Library Integration in the First-Year Experience Program* by Courtney Drysdale and Paul Betty

    - Regis University used the follow-up instruction survey to build a stronger argument in favor of mandatory library instruction for all sections of the first-year experience course at their institution.
- *Using the Instruction Survey for an Undergraduate Library Orientation* by Kaitlyn Clohosey
    - The research and instructional services librarian at Felician University used the immediate instruction survey in the library orientation session for all first-year, first-semester students to yield data on how well students remembered information about the session and if their library awareness increased. The resulting data was also included in end-of-year reporting and to advocate for the permanency of a mandatory library session.
- *Assessing Information Literacy Programs for First-Year Writing Courses at a Public University* by Kate Langan, LuMarie Guth, and Dylan Juhl
    - Librarians at Western Michigan University used the immediate instruction survey to assess the effectiveness of library programming for first-year writing courses in different colleges at a mid-sized public university. They used the results to assess modes of instruction, learning outcomes, and student affect.

CONCLUSION

Project Outcome provides valuable data, benchmarking, and feedback from students, faculty, and other library stakeholders, which can help you and your library advocate for continued investment, improve instruction, strengthen partnerships, and much more. Register for free today at https://acrl.projectoutcome.org and let us know how you've implemented the ideas in this book and your new ideas to effect change at your library, institution, and in the profession.

NOTES

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7. Ackerman, Goek, and Plagman, "Outcome Measurement in Academic Libraries."
8. ACRL staff worked with the ALA International Relations Office to offer a session on Project Outcome for Academic Libraries at the 2020 Sharjah International Library Conference, held in the United Arab Emirates. The surveys were translated as part of this session.
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CHAPTER 1

Characteristics of Academic Libraries, Library Instruction, and Student Outcomes:

A Descriptive Analysis Focusing on Minority-Serving and Rural Institutions

Kara Malenfant and Sara Goek

INTRODUCTION

Using Project Outcome data in conversation with datasets from other sources can provide greater insights into the contributions of academic libraries to student learning and success, as well as evidence that can be used in advocating for institutional, state, and federal funding. This chapter aims to encourage greater participation in and use of national academic library survey tools and raise awareness about how these datasets can be used for future research. As described in the Introduction, Project Outcome for Academic Libraries (POAL) has gathered 159,252 survey responses from 1,584 academic institutions in seven topic areas since its launch in April 2019. This chapter examines responses

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for the instruction survey topic at an aggregate level alongside other library characteristics from different datasets. It describes characteristics of U.S. academic libraries (e.g., spending, staffing, and material usage), their instruction programs, and reported student learning outcomes from library instruction. This chapter complements and extends annual academic library survey highlights and insights from the Association of College and Research Libraries (ACRL, a division of the American Library Association, ALA),[1] presenting more in-depth results akin to the Public Library Association (PLA, a division of ALA) annual reports.[2]

We compare characteristics and responses by broad Carnegie classification with those at minority-serving institutions (MSI) and rural institutions to see whether and how they differ,[3] as the students they serve are covered populations under the Digital Equity Act (DEA).[4] Though there have been gains in educational attainment for BIPOC students, race is a prevailing factor in many educational outcomes and these students face persistent barriers to educational access and success. In particular, Black students' educational outcomes are deeply concerning (low persistence, high undergraduate dropout rates, and the highest borrowing rates and largest debt burdens of any group). There are troubling outcomes for American Indians or Alaska Natives and Native Hawaiians or other Pacific Islanders as well.[5] The data in this chapter can help minority-serving institutions as they make the case to apply for state and federal funds for digital equity.

Because library instruction improves information literacy, including digital literacy, examining these two specific institution types may inform digital equity advocacy by ALA and ACRL. This chapter, then, could fulfill a goal of the POAL theory of change (see Introduction): that the data gathered may be used to bolster national advocacy efforts and create opportunities. Beyond ALA and ACRL advocacy, this chapter could inform grant-writers at libraries, consortia, and associations who may be seeking grants through the DEA and Broadband Equity and Deployment (BEAD) programs to further their digital literacy efforts.

Lastly, this chapter serves to demonstrate how different national aggregate datasets can be analyzed together to describe what has happened. We compiled data from multiple sources related to higher education institutions and academic libraries and analyzed it alongside results from Project Outcome instruction surveys. Instruction, the most-used survey topic within Project Outcome, encompasses the core work of libraries in teaching information literacy and research skills. We examined data gathered via the Integrated Postsecondary Education Data System (IPEDS);[6] the Carnegie Classification of Institutions of Higher Education (CCIHE) public data file;[7] the Minority Serving Institutions Directory from the Rutgers Center for Minority Serving Institutions (CMSI);[8] the ACRL Trends and Statistics annual survey;[9] and the instruction survey from POAL.[10] Additionally, beyond descriptive analysis, we use inferential statistics to review the relationships between two variables (e.g., independent variable: awareness and dependent variable: material usage) to determine if there is a correlation.

Our interest in conducting this descriptive analysis comes from our time as staff members at ACRL, where we supported the Value of Academic Libraries initiative.[11] An important component of that work was developing POAL and promoting it through education and outreach, as described in the Introduction. We also have deep experience with the ACRL Trends and Statistics survey and the related subscription tool ACRL Benchmark: Library Metrics and Trends, including extensive work analyzing peer groups for external reviews. This has given us a unique vantage point on these instruments, definitions of variables, data collection practices, and the potential for analyzing results. We also have a desire to see more libraries contribute, participate, and use POAL and Benchmark tools to inform and improve practice. Our hope is to raise awareness in the field about these surveys, encourage participation in the ACRL Trends and Statistics survey, motivate greater use of POAL, and spark interest among LIS researchers in using these datasets for future research.

LITERATURE REVIEW

Assessing Library Instruction

The ACRL Standards for Libraries in Higher Education (SLHE) includes, as one of nine principles, "Educational Role: Libraries partner in the educational mission of the institution to develop and support information-literate learners who can discover, access, and use information effectively for academic success, research, and lifelong learning."[12] Most often, academic libraries accomplish this through library instruction in a variety of formats and modalities. The scholarly and professional literature on assessing instruction in academic libraries is voluminous. A recent overview of literature from the fields of higher education and academic libraries on assessing student learning outcomes (SLOs) notes, "the range of assessment options available, the teaching and learning process, and the time it takes to manage assessment programs, are all factors that contribute to the challenging nature of assessing SLOs."[13] An older literature review on information literacy assessment identified nine types of assessment tools with multiple choice questions as the most popular, used by more than a third of the 91 studies.[14] While many academic librarians develop their own surveys to administer to students after instruction, several standardized instruments have been used in North America, each with benefits and drawbacks.

The Standardized Assessment of Information Literacy Skills (SAILS), conceptualized in 2002 by Rick Wiggins and Carolyn Radcliff at Kent State University, was put into production in 2006 and offered through Carrick Enterprises in 2012.[15] It has 162 test questions across eight skill sets and is based on the ACRL *Information Literacy Competency Standards for Higher Education* (ILCSHE).[16] The Information Literacy Test (ILT), also based on the ILCSHE, was developed in 2002 at James Madison University as a web-based, multiple-choice test that could be used by other institutions.[17] Importantly, the ILCSHE

were replaced in January 2016 by the ACRL *Framework for Information Literacy for Higher Education* and rescinded in June 2016.

The Threshold Achievement Test for Information Literacy (TATIL), also developed by Wiggins and Radcliff, uses the ACRL *Framework for Information Literacy for Higher Education* as a guide, with performance indicators combined into four modules and a total of 101 test questions. Launched in 2018, TATIL has been used by more than 75 educational institutions and was acquired by ACRL in 2023.[18]

The Educational Testing Service developed a standardized test to measure college students' information and communication technology skills in 2004.[19] Later renamed iSkills, the 60- to 75-minute outcomes-based assessment of real-world performance tasks[20] was discontinued on June 30, 2018.[21]

Dissatisfied with the options of SAILS, ILT, and iSkills, a group of academic libraries in Alberta, Canada, developed the Information Literacy Assessment and Advocacy Pilot (ILAAP) instrument. It contains two demographic questions (student's program and year of study), 17 multiple-choice questions (from which participating librarians picked two to three), and two open-ended questions.[22] While subsequently used by some institutions, it has not been widely adopted by the field.

The National Survey of Student Engagement (NSSE) developed the Experiences with Information Literacy Topical Module as a short set of add-on questions institutions may choose to ask undergraduates about information use. This module was updated for 2024 in collaboration with college and university librarians from ACRL.[23]

Project Outcome differs from these other approaches by providing libraries with a simple and reliable method for capturing outcomes of their programs and services. As described in the Introduction, the Project Outcome for Academic Libraries surveys were developed by a task force of experts in the field and tested by over 50 academic libraries. The surveys are based on a successful model developed by the Public Library Association, in use since 2015.[24]

Multi-Institution Studies

These standardized information literacy assessment instruments are typically used for internal assessment and improvement, and results are not shared broadly. When librarians employing these tools do report out, it is often at conferences or in the practice literature about use at a single institution, their own. Relatively few research studies have reported on multi-institutional use of these standardized instruments.[25]

This is consistent with the broader nature of scholarship about academic libraries and student outcomes, where much of the scholarly writing is not empirical research and the vast majority has a single institution setting. In 2017, researchers commissioned to investigate how libraries can increase student learning and success examined a broad swath of evaluation and assessment literature about academic library impact on students. They

coded a total of 535 documents, with about 68 percent (n=369) of all documents coded as research. Of those, only fourteen percent (n=51) of studies spanned multiple institutions.[26]

Because the literature about academic library value is rich and complex, a proposed levels of analysis framework helps to group the research into three broad categories: small-group studies (typically a class), single-institution studies, and multi-institution studies.[27] The latter have examined aggregate national academic library statistics and student achievement data from IPEDS, including the Academic Libraries component.[28] Some have examined IPEDS data along with one or more additional sources, such as survey data from ACRL and the Association of Research Libraries.[29] We found only one study examining national aggregate data specifically on information literacy. Kuh and Gonyea analyzed responses from more than 300,000 students between 1984 and 2002 to the College Student Experiences Questionnaire. The questionnaire included seven questions about library experiences and information literacy, along with student and institutional characteristics.[30]

Library Workforce and Student Workers

An expectation of all academic libraries from the SLHE is the principle of "Personnel: Libraries provide sufficient number and quality of personnel to ensure excellence and to function successfully in an environment of continuous change." Advancing and sustaining the role of libraries as partners in educating students requires library staff who can develop and support information-literate learners. This educational role of academic libraries extends beyond instruction to other types of interactions, such as reference transactions and research consultations. These duties, historically the sole purview of academic librarians, are now undertaken by other types of staff. Librarians invest in training student workers and entrust them with core roles such as peer research coaches (cross-trained with other co-located departments, e.g. writing centers). Savvy library leaders recognize that student employees are often better ambassadors to peers than librarians or library staff can ever hope to be.[31] Additionally, well trained peer research coaches allow librarians and other professionals to refocus their efforts from front-line triage to providing more in-depth support.

Staffing patterns shifted greatly in U.S. academic libraries between 2012-2021 (during which time the number of postsecondary schools decreased by 342). In concert with broad budget cuts in academic libraries in this decade, full-time equivalent (FTE) librarians decreased 9.3 percent, other professional staff (e.g., computer scientists) increased 31.7 percent, other paid library staff (i.e., classified or paraprofessional) decreased 21.7 percent, and student assistants decreased 25 percent.[32] Of note, a 40 percent drop in student workers in academic libraries from 2019-2021 "likely reflects not only reduced need due to pandemic library closures and occupancy limitations but also a reduction in the number of students enrolled and/or on campus, among other potential factors."[33] Academic libraries are often the largest campus employer of student workers, and an opportunity to engage students in meaningful work can be an extension of the library's educational role. When

well designed and implemented, on-campus employment is recognized as a high-quality developmental experience. Such programs provide students with meaningful learning and engagement opportunities, boost retention, and build career-readiness skills.[34] Educationally purposeful peer interactions, as defined by George Kuh, are "one of the most underutilized and cost-effective ways of encouraging more undergraduate students to actively engage in the college experience and complete their studies." These interactions (i.e., mentoring, coaching, and peer tutoring grounded in academically oriented social interactions) equitably increase student use of academic support services, thus greatly enhancing student learning and success; peers have considerable influence on fellow students' attitudes and behavior.[35]

METHODOLOGY

The bulk of this chapter is a descriptive analysis seeking to understand the landscape of U.S. community college, college, and university libraries in terms of library spending, staffing, and material usage, instruction programs, and reported student learning outcomes from library instruction. We also used inferential statistics to review the relationships between several sets of variables—for example, whether more instruction was associated with more use of library resources, and if libraries with higher awareness outcome scores also have higher resource use. We examined data gathered from five sources, focusing on the time from the 2019 launch of Project Outcome for Academic Libraries to the time of writing (2023):

1. Integrated Postsecondary Education Data System (IPEDS), a system of interrelated census surveys conducted annually by the National Center for Education Statistics (NCES, of the Institute for Education Sciences in the U.S. Department of Education);
2. The Carnegie Classification of Institutions of Higher Education (CCIHE) public data file;
3. The Minority Serving Institutions Directory from the Rutgers Center for Minority Serving Institutions (CMSI);
4. The ACRL Trends and Statistics annual survey, containing the 24 IPEDS academic library component questions, 66 questions formerly collected through the biennial NCES Academic Library Survey (discontinued in 2012), and trends questions that vary annually; and
5. The instruction survey topic of POAL, with a focus on the immediate surveys.

See Appendix A in this chapter for the source, date, and description of each variable analyzed in this chapter.

To begin, we compiled the most recent data available from these five sources. Each includes an IPEDS unit ID, the unique identifier for each institution, allowing them to be linked. Because Carnegie classifications are central to our analysis and the CCIHE

dataset includes only degree-granting institutions, the institutions present in the CCIHE 2021 public data file became the backbone of our analysis. Institutions present in other data sources with unit IDs that did not match institutions in CCIHE were excluded from subsequent analysis.

We summarized aggregate data by broad institution type, taking the 33 basic Carnegie classifications and grouping them into five classes (associates, baccalaureate, masters, doctoral, and other).[36] To create the minority-serving institutions group, we relied on data provided by the Rutgers Center for Minority Serving Institutions (based on the U.S. Department of Education, Office of Postsecondary Education's 2021 Eligibility Matrix) and aggregated the seven categories identified therein to one indicator. To create the rural institutions group, we used the CCIHE locale variable, which indicates the degree of urbanization of the institution's location.[37] Given the known challenges of reliably identifying minority-serving institutions[38] and rural serving-institutions,[39] our use of the Rutgers MSI list and CCIHE locale variables are heuristic devices with limitations.

Table 1.1 provides an overview of institutions in the 2021 CCIHE, 2022 ACRL Trends and Statistics survey, and Project Outcome datasets by these characteristics. Notably, associate's/two-year institutions are underrepresented in the ACRL and POAL datasets, while doctoral universities are overrepresented. Overall, 37.6 percent of institutions participated in the ACRL 2022 survey, and 6.2 percent of institutions have used the Project Outcome for Academic Libraries instruction survey.

Institution Type	# Possible (CCIHE 2021)	Proportion of CCIHE TOTAL	# Libraries responding to ACRL 2022 Survey	Proportion of ACRL 2022 TOTAL	# using POAL instruction	Proportion of POAL users	# Individual responses to POAL instruction	Proportion of responses to POAL instruction total
Associate's Colleges	1,288	32.7%	344	23.2%	59	24.1%	22,532	17.3%
Baccalaureate colleges	735	18.7%	303	20.5%	45	18.4%	15,183	11.6%
Master's colleges & universities	669	17.0%	357	24.1%	65	26.5%	45,787	35.1%
Doctoral universities	466	11.8%	354	23.9%	66	26.9%	44,992	34.5%
Other (special focus 4-year + tribal)	781	19.8%	122	8.2%	10	4.1%	2,055	1.6%
ALL INSTITUTIONS	3,939		1,480		245		130,549	
Minority serving	837	21.2%	356	24.1%	60	24.5%	32,266	24.7%
Rural	399	10.1%	136	9.2%	15	6.1%	6,748	5.2%

Table 1.1. Overview of institutions in the 2021 CCIHE, 2022 ACRL Trends and Statistics survey, and Project Outcome datasets.

Associate's and baccalaureate institutions predominate the minority-serving and rural groups, making up 59.7 percent of MSIs and 72.9 percent of rural institutions. Minority-serving and rural institutions are also more likely to be public institutions: 68 percent of each type are public, compared to 41.3 percent of institutions overall.[40]

In looking at specific variables, we had many decisions to make. The 24 IPEDS Academic Libraries component questions are asked by both NCES and ACRL, and we chose to analyze data reported via IPEDS because, as Table 1.1 shows, not all U.S. degree-granting institutions participate in the ACRL survey. Using data provided by institutions to IPEDS gives a more comprehensive and accurate picture, though the most recent data available at the time of this research is from 2021, as opposed to 2022 data from ACRL. Also, because NCES has greater capacity, they invest more time in cleaning IPEDS data than ACRL does their data, so the data are more reliable.

For Project Outcome data, we manually excluded surveys that appeared to be user tests of the platform based on the program or survey name. We applied a program cutoff date of May 31, 2023; the data thus represents four full years of survey activity. The following analysis focuses on responses to the immediate surveys, which are designed to be distributed immediately after completion of a program or service to understand its impact. They are more widely used than the follow-up surveys, which are designed for use four to eight weeks after a program or service to help libraries understand if patrons have changed their behavior or continued to benefit as a result of the program or service. Of the 130,549 total responses in Table 1.1, only 5,051 come from follow-up surveys.

We had a choice to make about which data to use to evaluate library staffing, as IPEDS asks about library staffing in two component surveys. When academic librarians are counted as part of the IPEDS Human Resources component, their counts are mingled with media collection specialists (from 2018 to the present) due to the use of U.S. Bureau of Labor Statistics Standard Occupational Classification code 25-4022.00, which currently combines the two categories. The IPEDS Academic Libraries component uses staffing definitions from the National Information Standards Organization, which academic librarians feel is more accurate. Therefore, in this chapter, we use data about library staffing reported via the AL component in 2020 and 2021.

Throughout this chapter, we often present the findings as proportions or ratios for two variables rather than simply the mean for each individual variable. For example, instead of presenting average library expenditures, we present overall library expenditures per student FTE, which is the total of all library expenditures divided by the number of full-time equivalent students for all applicable institutions. This helps with interpretation and sensemaking given the wide variety among institutions. These ratios are calculated using data only from institutions that reported the relevant variables (e.g., if an institution reported student FTE but not library expenditures, it would be excluded from the overall ratio of library expenditures to student FTE).

While the bulk of this chapter is a descriptive analysis seeking to understand academic library outputs and outcomes by institution type, we also wondered about the relationship

between variables. We were curious to what extent more instruction was associated with more use of library resources, and whether libraries with higher awareness outcome scores also have higher resource use. Therefore, we used inferential statistics to review the relationships between several sets of variables to determine if there are bivariate correlations.

FINDINGS

Funding Levels and Expenditures



$7,929,577,860 total U.S. academic library expenditures



Academic libraries spent $173 per student FTE in 2021

Funding levels affect the extent to which academic libraries are positioned to contribute to institutional effectiveness and achieving their institutions' missions. One way to consider library budgets is to consider library expenditures as percentage of expenditures on all academic support,[41] as shown in Figure 1.1. We see variation by Carnegie classification: Associate's colleges, MSI,[42] rural, and other institutions[43] are lower than the total for all institutions, meaning these institution types allocate less for libraries than other academic support. Baccalaureate, master's granting, and doctoral institutions are higher. This pattern was true in 2019 and mostly in 2020 (master's granting library allocations were slightly lower in 2020).

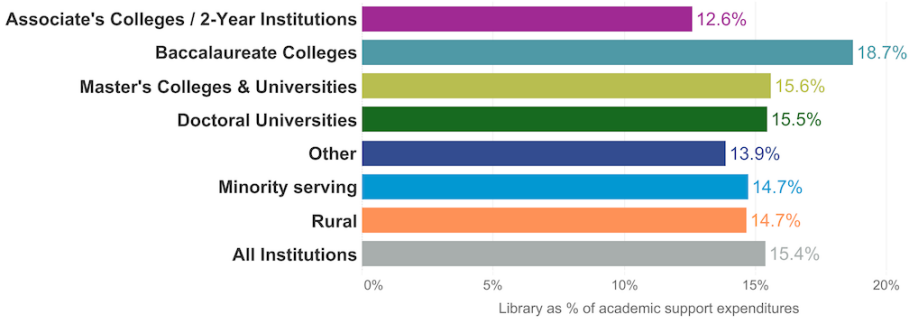


Figure 1.1. Total library expenditures as percent of academic support by institution type (IPEDS 2021).

Another way to view library funding levels is to look at spending on a per student basis. Figure 1.2 shows that the overall U.S. aggregate spending (sum of all library expenditures divided by the number of full-time equivalent students for institutions that reported both) is $173, with a range from $58 at associates to $330 at institutions grouped into the "other" category. It is perhaps unsurprising that doctoral and other institutions have higher per-student library spending; doctoral institutions have a research focus and "other" includes law and medical institutions, all with more costly information resources.[44] This 2021 pattern of expenditures holds for 2020 and 2019.

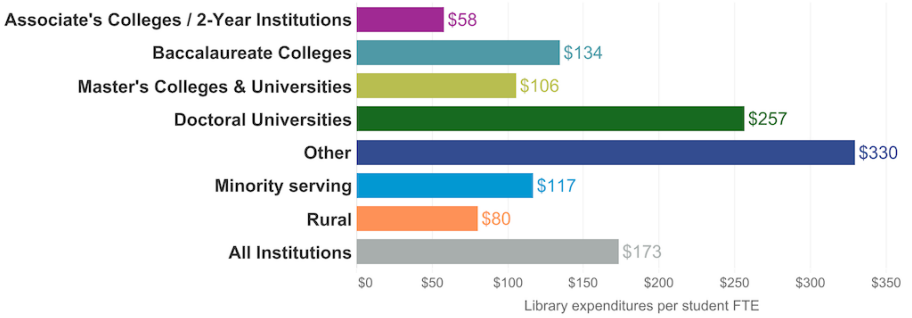


Figure 1.2. Overall U.S. aggregate library expenditures per student FTE by institution type (IPEDS 2021).

Per-student spending is lower than the total U.S. aggregate average at MSI ($117 or 67 percent) and rural institutions ($80 or 46 percent), which can be explained by their large proportion of associate's and baccalaureate institutions. When viewing MSI and rural institutions by Carnegie classification, expenditure trends largely mirrored that of all institutions; that is, associate's have the lowest spending per student and doctoral and other have the highest, as shown in Figures 1.3 and 1.4.

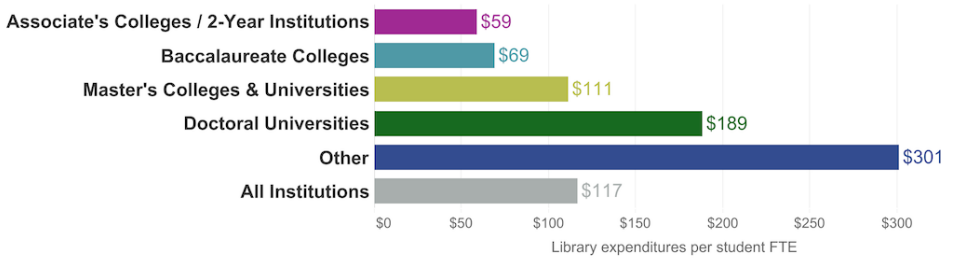


Figure 1.3. Aggregate library expenditures per student FTE for Minority-Serving Institutions by type (IPEDS 2021).

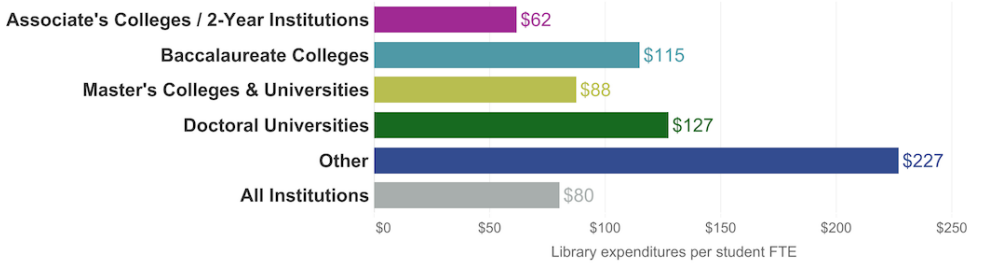


Figure 1.4. Aggregate library expenditures per student FTE for rural institutions by type (IPEDS 2021).

Staffing and Student Workers

To examine how libraries meet the SLHE principle of personnel, Figure 1.5 shows the number of full-time equivalent library staff per 1,000 students. The aggregate average is 1.5 library staff per 1,000 students, with a range from .7 at associate's colleges to 2.7 at other institutions. Again, we see MSI and rural institutions are lower than the total U.S. aggregate average, serving more students with fewer library staff.



Academic libraries employed 70,826 total FTE staff in 2021



Aggregate average of 1.5 library staff per 1,000 students

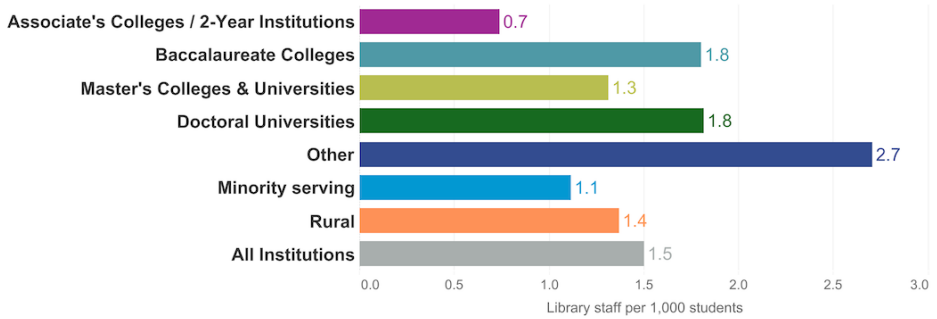


Figure 1.5. Library staff per 1,000 student FTE by institution type (IPEDS 2021).

Institutions report library staffing in four categories, as shown in Figure 1.6: librarians (34.6 percent), other professionals (14.9 percent), other paid staff (28.9 percent), and student workers (21.6 percent). A closer look at student workers shows a ratio of .32 student workers per 1,000 full-time students. As Figure 1.7 shows, associate's granting institutions and MSI rely less on student workers as staff than all other library types. Notably, baccalaureate and other institutions stand out as having a relatively high number of student workers per 1,000 students.

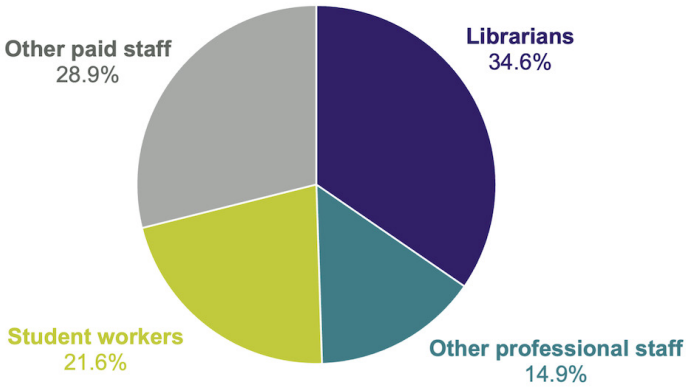


Figure 1.6. Percentage of staff in academic libraries by role (IPEDS 2021).

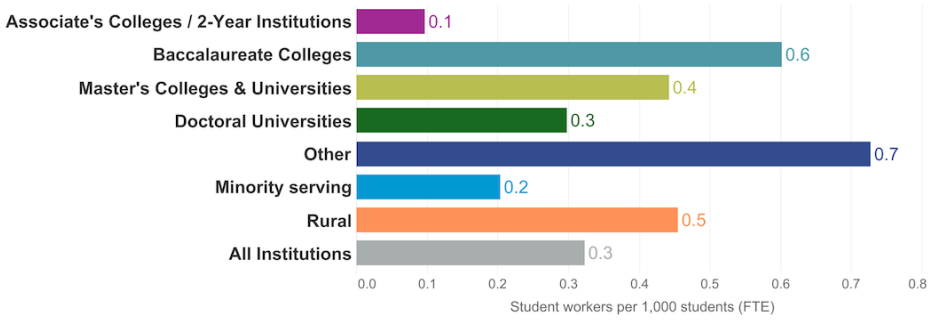


Figure 1.7. Library student workers per 1,000 student FTE by institution type (IPEDS 2021).

Academic libraries spend, on average, $11,859 per full-time equivalent (FTE) student worker, and student workers are 5.7 percent of their staffing costs. In Figure 1.8 we see that baccalaureate colleges stand out again as spending more on student workers as a percentage of staff salaries and wages. This is not surprising given that these institutions often have a teaching-focused mission.

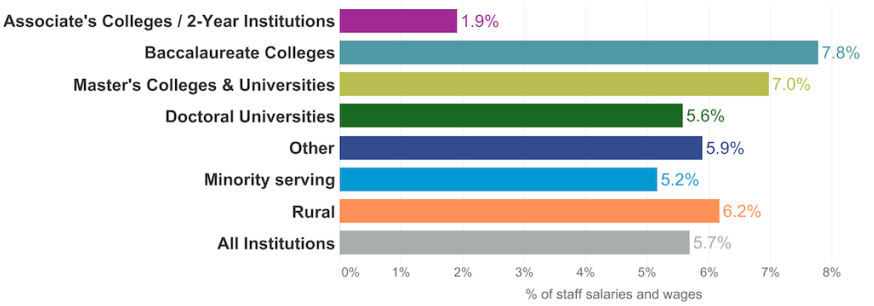
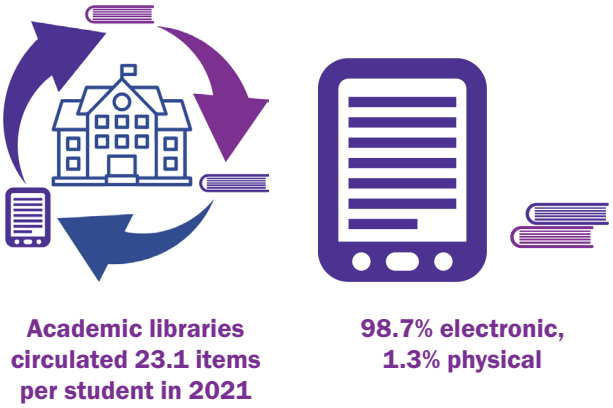


Figure 1.8. Student workers' salaries and wages as a percentage of all library staff salaries and wages by institution type (ACRL 2022).

Collection Usage



The quality of an academic library's collections, both print and digital, is critical to its mission. The SLHE principle for collections states, "Libraries provide access to collections sufficient in quality, depth, diversity, format, and currency to support the research and teaching missions of the institution." For the purposes of this chapter, we were interested in usage as measured by circulation rather than collection size. Again, rather than reporting whole numbers of items circulated, we calculated a per-student basis. This ratio is simply a heuristic that helps standardize the comparison by institution type, and does not reflect that library materials are also used by faculty, staff, and community members. As Figure 1.9 shows, the 2021 overall average is 23.1 items per student, with rural and masters institutions the lowest.[45]

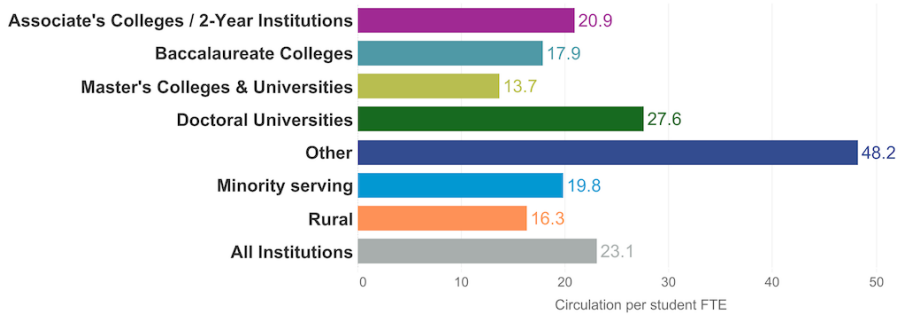


Figure 1.9. Circulation per student FTE by institution type (IPEDS 2021).

Total circulation for all academic libraries rose by 54 percent between 2019 (pre-pandemic) and 2021. The proportion of the total representing usage of digital/electronic materials rose over the same period, from 93.5 percent to 98.7 percent.

Presentations and Attendance



On average, academic libraries each made 248 presentations to groups in 2022, a rate of 14.6 presentations per professional staff FTE

As discussed in the literature review, the SLHE expect academic libraries to contribute to their institutions via their educational role. While this can be carried out in many ways—including reference transactions, research consultations, and provision of online guides—the educational role is fulfilled most directly through instruction and workshops. The 1,368 academic libraries that participated in the ACRL 2022 survey reported 339,491 presentations to groups.[46] In Figure 1.10, we examine presentations per librarian and other professional staff. The average number of presentations per staff is 14.6, with far more at associate's colleges. This is unsurprising given the overall staffing ratio—Figure 1.5 shows associate's colleges have a lower staff to student ratio—but staff at those associate's colleges also do more presentations per year overall. Again, this ratio is a heuristic, which makes it easier to compare institution types but does not reflect that some librarians and other professionals do not make presentations of any type.

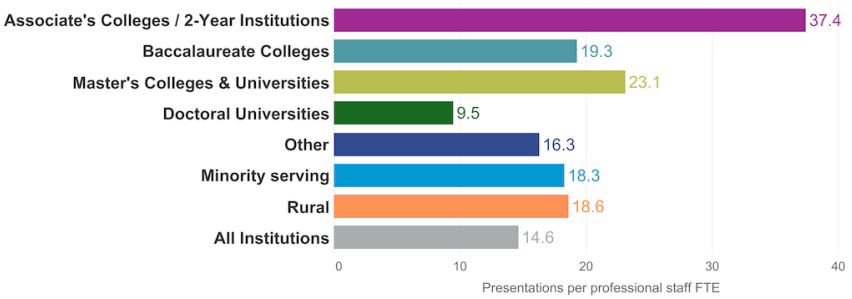


Figure 1.10. Annual presentations per librarian and other professional staff FTE by institution type (ACRL 2022).

Figure 1.11 looks at presentations in light of student enrollment and reported attendance. We see an overall rate of 3.6 presentations attended per 100 students, with institutions in the other category having by far the highest attendance rate.

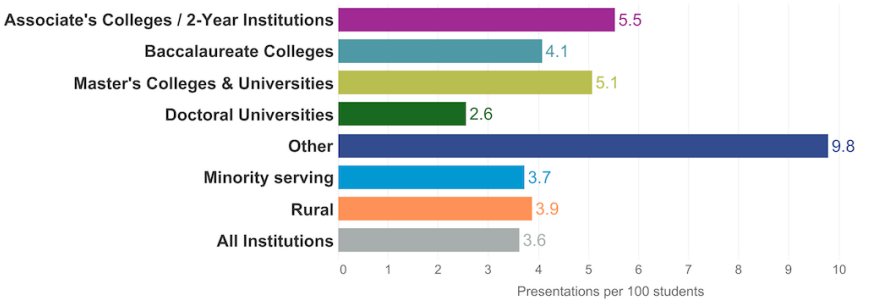


Figure 1.11. Attendance at library presentations by institution type (ACRL 2022).

Learning Outcomes from Library Instruction

Overall, respondents to Project Outcome immediate instruction surveys report overwhelmingly positive outcomes. As shown in Figure 1.12, more than 88 percent of respondents agree or strongly agree with each of the four outcomes: knowledge (I learned something new that will help me succeed in my classes), confidence (I feel more confident about completing my assignment(s)), application (I intend to apply what I just learned), and awareness (I am more aware of the library's resources and services).

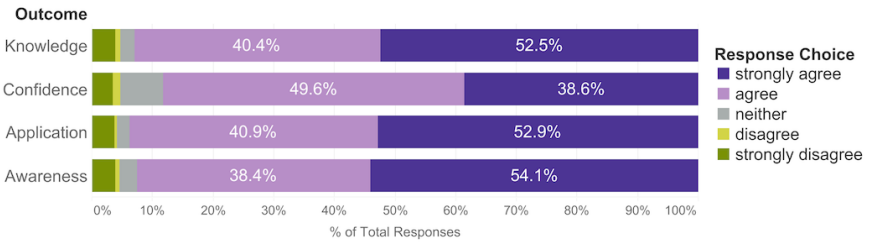


Figure 1.12. POAL immediate instruction outcomes.

The Likert scale responses range from strongly disagree (assigned a score of 1) to strongly agree (assigned a score of 5). Across the board, the confidence outcome returns the lowest overall score. There is little meaningful difference between the scores by institution type; average scores cluster between 4 and 4.5.

Institution type	Knowledge	Confidence	Application	Awareness	Overall	# of institutions
Associate's Colleges	4.35	4.14	4.34	4.37	4.3	55
Baccalaureate Colleges	4.4	4.2	4.4	4.42	4.36	39
Master's Colleges & Universities	4.33	4.16	4.36	4.34	4.3	64
Doctoral Universities	4.34	4.13	4.35	4.36	4.29	60
Other	4.46	4.24	4.54	4.45	4.42	10
Overall	4.35	4.15	4.36	4.36	4.31	228
Minority Serving	4.35	4.15	4.36	4.36	4.31	55
Rural	4.32	4.15	4.36	4.38	4.3	14

Table 1.2. POAL instruction scores by outcome and institution type.

To determine the relationships among variables (e.g., to what extent more instruction was associated with more use of library resources and whether libraries with higher awareness outcome scores also had higher resource use), we tested four bivariate correlations: attendance per student vs. circulation per student; average outcome score vs. average circulation per student; average outcome score vs. average FTE staff; and average outcome score vs. average expenditures per student.[47] We analyzed these variables by individual institution. None of these resulted in a statistically significant correlation. Average outcome score vs. circulation comes closest, with a positive trend and a p value of 0.066 (Figure 1.13). Because the results were so tightly clustered, we chose to display the x-axis scale in Figure 1.13 starting at 2.5 rather than 0 to improve readability. We did not have the capacity to conduct multivariate analysis for this study, and that is an area ripe for future research.

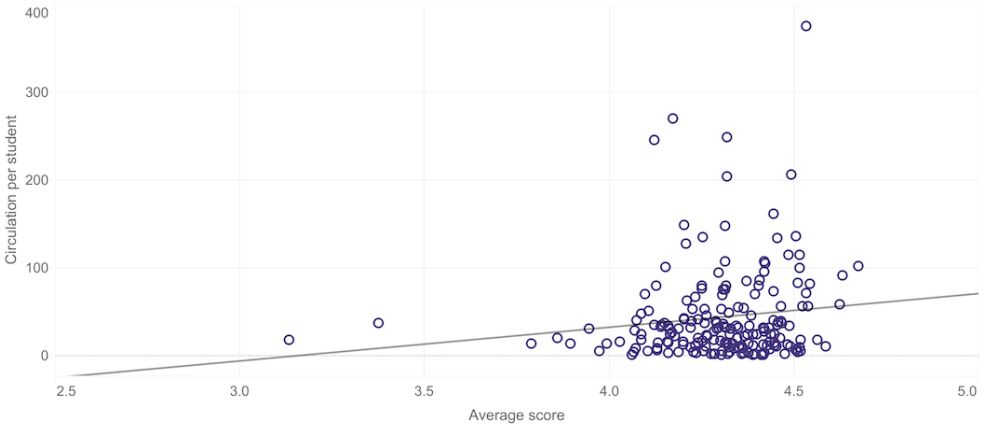


Figure 1.13. POAL immediate instruction outcomes versus circulation per student.

Only institutions with more than 10 instruction survey responses are included. Figure 1.13 shows a positive relationship between reported instruction outcomes (institutional average) and rate of circulation per student. We did not remove any outliers to catch any data that did not belong. Future researchers could set criteria so that any values considerably far away from the rest do not have an undue influence on the regression.

DISCUSSION

The results in this chapter compare characteristics, outputs, and outcomes for academic libraries in institutions of different types. The findings make it clear that academic libraries at MSI and rural institutions are under-resourced compared to the national average for all institutions. Library spending per student at MSIs is only 67 percent of the spending per student for all institutions, and for rural institutions it's only 46 percent. These institution types also spend a lower proportion of their academic support budgets on libraries compared to U.S. institutions as a whole. As noted earlier in the chapter, minority-serving and rural institutions are also more likely to be public than U.S. institutions as a whole, which perhaps accounts for their spending patterns. Public institutions have seen repeated budget cuts in recent years, and academic libraries have often suffered as a result.[48]

In terms of personnel, MSI and rural institutions have a lower than average number of library staff per student, with MSIs relying less on student workers than other institution types. Providing high quality programs and services that improve digital skills requires sufficient number and quality of personnel. Moreover, engaging students in meaningful work can be an extension of the library's educational role, a high-quality developmental experience, and a very effective form of peer support. Additionally, it is a good investment—students represent 21.6 percent of the labor force in academic libraries and just 5.7 percent of the staffing costs. Because of severe cuts to the federal work-study program,[49] academic libraries are no longer able to hire as many student workers. Digital equity grants could help offset this loss. Increased funding and staff capacity would allow MSI and rural institutions to hire, train, and supervise more student workers as peer coaches focusing on digital skills building.

MSI and rural institutions have lower use of library materials than the U.S. average. With the IPEDS Academic Libraries component showing that 98.7 percent of circulation is electronic as of 2021, it is not enough simply to provide access. As acknowledged in the Digital Equity Act, digital literacy skills are necessary to expand the use of the internet so everyone can experience the benefits. Academic libraries are uniquely positioned to ensure students have strong information and digital literacy skills, and library instruction is one effective way to support this skill-building.

In terms of offering presentations, librarians and other professionals at MSI and rural institutions had a heavier workload than U.S. institutions overall. At the same time,

attendance per student was higher at these institutions, which could indicate greater need for such training among these students.

Nationally, those participating in library instruction reported overwhelmingly positive outcomes in terms of increased knowledge, confidence, application, and awareness. Additionally, there is a relationship between positive reported outcomes and higher circulation/use of library materials. Institutions where participants reported stronger outcomes have higher use of library materials. Since 98.7 percent of collection use is electronic, this means greater usage of curated, reputable, online resources—often in subscription databases.

As the following chapters show, much of the value of Project Outcome results can be found at the local level, where libraries can look for gaps in their own performance and specific, actionable feedback for improvements. While students and library users tend to have warm associations with libraries and report positive outcomes, individual libraries can learn much about their own performance as they seek to continuously adapt and improve. Outcomes measurement helps libraries determine the efficacy of their programs and services. The results help them showcase the impact of their work within their institutions and to advocate for the resources they need (including funding) to continue and expand that work.

Future Research

We analyzed a small subset of variables from IPEDS and the ACRL Trends and Statistics survey for a limited number of years as a demonstration exercise. With such large datasets, there is much more to explore than we could cover within the scope of this chapter. Looking at additional variables and trends over time could be quite illuminating. This analysis suggests possible future research questions and directions, such as:

- By more closely examining the variance and nuance within Carnegie classifications, can we determine where investments in institutions and libraries would make the most difference for the educational attainment of BIPOC students?
- How have budgets for institutions of different types fared over a longer period of time, and how has this impacted academic library staffing and performance?
- To what extent will academic library employment of student workers rebound post COVID-19? This will require additional years of post-pandemic data.
- How do outcome measures in other POAL topic areas, like technology, space, or events/programs, relate to library services, staff, and budgets across institutions of different types?
- Could multivariate analysis provide a fuller understanding of the relationships between outcome and output variables and variance in characteristics across institutions?

As with any descriptive analysis, we were able to describe what was reported at a very high, aggregate level and it was out of scope to examine specific instances. Examining a few cases more deeply—such as those presented in this book—would yield a different, complementary understanding of how and why things occurred. These methods and others can be fruitfully combined to create a fuller picture as appropriate to the research question.

CONCLUSION

We endeavored in this chapter to accomplish several aims. First, by better understanding what data is contained in the ACRL Trends and Statistics survey, we hope more academic libraries will participate in this important national survey and make use of the results with Benchmark (librarybenchmark.org). The complimentary version of Benchmark includes a summary of the national trends results as soon as they are available each year, while a subscription offers additional functionality, including interactive data dashboards with all historic data and the ability to conduct custom peer comparisons. Benchmark allows libraries to analyze their own performance compared to other institutions in their Carnegie class or specific peers, using many of the metrics described in this chapter, and the results support data-informed decision-making and advocacy.

Second, we sought to motivate greater use of Project Outcome for Academic Libraries and understanding of the value of outcome measurement data at the national level. The results described in this chapter show the overall positive outcomes of academic library instruction, particularly for students' knowledge, application of skills, and awareness of library resources. For individual libraries, these quantitative results can be combined with respondents' open-ended comments to support a culture of continuous learning and improvement and demonstrate the impact of the libraries on student learning and success to administration and other stakeholders.

Third, by demonstrating how these different national datasets can be analyzed together, we hope to spark interest among LIS researchers in using these datasets for future research. Linking data—though labor-intensive—can lead to deeper analysis. Combining national data with institution-level data also provides a more nuanced and varied picture of the characteristics of academic libraries nationwide.

Finally, an important aim of this chapter was to assist librarians at MSI and rural institutions in seeking digital equity funds to support covered populations in their communities. Because there is unlikely to be another funding opportunity like this in our lifetime, libraries should seek to leverage federal funds to further the good work they are already doing for their communities. The results in this chapter can help MSI and rural institutions make their case in applying for state and federal funds for digital equity. While we specifically focused on skill-building through library instruction and workforce development via student workers, there are other activities academic libraries could propose as they seek digital equity funding. For example, libraries could propose residencies in digital scholarship, supporting workforce development in the broader community, or

vocational training. They could seek federal and state funds to distribute free or low-cost equipment (e.g., Wi-Fi or laptop lending) and to construct, upgrade, or expand public access computing centers.

In 2024, states will begin spending $1.44 billion in DEA capacity grants over a five-year period to implement digital equity plans. Interested librarians should find and read their state plan carefully, proposing activities that meet the goals and objectives as articulated by their state. DEA funds are also being expended at the federal level through a $1.25 billion competitive grant program, offered each year for four years and administered by the United States Department of Commerce National Telecommunications and Information Administration (NTIA).[50] These grants will focus on multi-state, regional, or national efforts that align with federal-level goals and objectives. In addition to DEA, NTIA has another state formula grant, the Broadband Equity and Deployment (BEAD) Program, with $42.5 billion to expand high-speed broadband access via infrastructure, deployment, and adoption—including a goal of ensuring the community has the skills to and understands the benefits of using the internet. The states' BEAD implementation period also starts in 2024.[51]

ACKNOWLEDGEMENTS

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CREDIT[52]

Conceptualization and project administration: Kara Malenfant; Data curation, investigation, and formal analysis: Sara Goek; Methodology, visualization, and writing: Sara Goek and Kara Malenfant

APPENDIX A: VARIABLES

Variable	Source	Variable name (Original Source)	Notes
Institution data			
Unit ID	All		Unique identifier for all institutions across all data sources
Rural institution	CCIHE 2021	LOCALE	For this chapter, institutions with rural locales are aggregated into one variable all institutions assigned to the following value labels 41 Rural: Fringe, 42 Rural: Distant, 43 Rural: Remote.
Carnegie Classification	CCIHE 2021	BASIC2021	For this chapter, aggregated into 5 broad value labels institutions assigned to one of 33 value labels by CCIHE, as follows: Associate's Colleges / 2-year Institutions: values 1-13 Baccalaureate Colleges: values 14, 21-23 Master's College & Universities: values 18-20 Doctoral Universities: values 15-17 Special Focus 4-year Institutions: values 24-33
Minority Serving Institution	CMSI Rutgers	MSI	For this chapter, aggregated seven categories identified into one. (Based on Office of Postsecondary Education 2021 Eligibility Matrix)
Academic Support Expenditures	IPEDS 2019-2021	F2E041, F3E03A1, F1C051	Institution expenditures are reported as separate variables depending on institution control (private non-profit, private for-profit, or public)
Total Institution Expenditures	IPEDS 2019-2021	F2I07, F3G07, F1N07	Institution expenditures are reported as separate variables depending on institution control (private non-profit, private for-profit, or public)
Student FTE	IPEDS 2019-2021	FTE12MN	12-month full-time equivalent enrollment
Student FTE	ACRL 2022	90	Degree-seeking FTE students
Library data			
Total library Expenditures	IPEDS 2019-2021	LEXPTOT	Total expenditures (salaries/wages, benefits, materials/services, and operations/maintenance)
Library staff (FTE): Librarians	IPEDS 2020-2021	LSLIBRN	
Library staff (FTE): Other Professional Staff	IPEDS 2020-2021	LSOPROF	
Library staff (FTE): Other Paid Staff	IPEDS 2020-2021	LSOPAID	
Library staff (FTE): Student Workers	IPEDS 2020-2021	LSSTAST	

Variable	Source	Variable name (Original Source)	Notes
Total library staff (FTE)	IPEDS 2020-2021	LSTOTAL	
Library staff (FTE): Librarians	ACRL 2022	1A	
Library staff (FTE): Other Professional Staff	ACRL 2022	2A	
Library staff (FTE): Other Paid Staff	ACRL 2022	4A	
Library staff (FTE): Student Workers	ACRL 2022	5A	
Total library staff (FTE)	ACRL 2022	6A	
Total staff expenditures (salaries and wages)	ACRL 2022	6B	
Student assistants salaries and wages	ACRL 2022	5B	
Total annual presentations to groups	ACRL 2022	74C	
Total attendance at all presentations	ACRL 2022	75C	
Library Circulation - Physical	IPEDS 2019-2021	LPCRCLT	
Library Circulation - Digital/Electronic	IPEDS 2019-2021	LECRCLT	
Total circulation	IPEDS 2019-2021	LTCRCLT	
Outcomes score	POAL		Likert-scale responses to the four outcome questions on immediate surveys are scored on a scale of 1 to 5, where 1 indicates strongly disagree, 5 strongly agree.

NOTES

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31. Brett B. Bodemer, "They CAN and They SHOULD: Undergraduates Providing Peer Reference and Instruction," *College & Research Libraries* 75, no. 2 (March 2014): 162-178, https://doi.org/10.5860/crl12-411; Maglen Epstein and Bridget Draxler, eds., *How to be a Peer Research Consultant: A Guide for Librarians and Students* (Chicago: Association of College and Research Libraries, 2022); Jennifer Torreano and Mary O'Kelly, eds., *Training Research Consultants: A Guide for Academic Libraries* (Chicago: Association of College and Research Libraries, 2021).
32. Joshua Kim, "How do Academic Libraries Spend Their Money?" *Inside Higher Ed*, March 2023, https://www.insidehighered.com/blogs/learning-innovation/3-questions-academic-library-staffing-acrl%E2%80%99s-mary-jane-petrowski. Additionally, this decline in library student workers is most certainly related to changes to the federal work-study program, which funds part-time campus jobs for those with demonstrated financial need. There were approximately 372,000 recipients in 2020-21 with $703.3 million in federal funding. This is 52 percent of the approximately 704,000 recipients in 2011-2012, when federal appropriations were $974.7 million. National Association of Student Financial Aid Administrators, *2023 National Student Aid Profile*, October 2023, https://www.nasfaa.org/uploads/documents/2023_National_Profile.pdf; National Association of Student Financial Aid Administrators, *2014 National Student Aid Profile*, July 15, 2014, https://www.nasfaa.org/uploads/documents/2014_national_profile.pdf.
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34. Alexa Wesley, Alexis Wesaw, Omari Burnside, and Amelia Parnell, "Employing Student Success: A Comprehensive Examination of On-Campus Student Employment," *NASPA Report*, February 2019, https://www.naspa.org/report/employing-student-success-a-comprehensive-examination-of-on-campus-student-employment.

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36. CCIHE variable BASIC2021 aggregated as follows: associate's colleges / 2-year institutions: 1-13, baccalaureate colleges: 14, 21-23, master's college & universities: 18-20, doctoral universities: 15-17, and other (special focus 4-year institutions and tribal colleges and universities): 24-33.
37. CCIHE variable locale aggregated for value labels of 41 Rural: Fringe, 42 Rural: Distant, and 43 Rural: Remote.
38. In brief, federal recognition as a MSI happens in two ways. First, historically designated MSIs (i.e., Historically Black Colleges and Universities and Tribal Colleges and Universities) were established to provide access to higher education for Black and Native American students. Second, institutions can be recognized as minority serving by enrollment threshold as specified in legislation that defines each MSI type (e.g., Primarily Black Institutions, Hispanic-Serving Institutions, etc.). MSIs determined by enrollment threshold vary greatly in student demographics and institutional characteristics. For example, community colleges, large public institutions, and small private institutions could all be Hispanic-Serving Institutions with very different student bodies and resources. Mike Hoa Nguyen, Joseph J. Ramirez, and Sophia Laderman, "What Counts as a Minority-Serving Institution? Toward the Utilization of a Standardized and Uniform Definition and Typology," *Educational Researcher* 52, no. 3 (April 1, 2023): 174–79, https://doi.org/10.3102/0013189X221105861.
39. We use rural locale for this chapter because rural-serving institutions have no formal definition or measure to comprehensively identify them for research and policy-making purposes. Many are also MSIs and enroll significant proportions of students who receive Pell grants, and they serve communities with high poverty rates, low employment, and population loss. Andrew Koricich, Vanessa A. Sansone, and Alisa Hicklin Fryar, "Rural-Serving Institutions Aren't Who You Think They Are," *Change: The Magazine of Higher Learning* 54, no. 3 (May 2022): 28–33, https://doi.org/10.1080/00091383.2022.2054181.
40. No private for-profit institutions are characterized as minority-serving. Among rural institutions, 4.5 percent are private for-profit and 27.1 percent are private not-for-profit (CCIHE 2021).
41. Per the IPEDS Survey Component Finance Glossary, "Academic Support. A functional expense category that includes expenses of activities and services that support the institution's primary missions of instruction, research, and public service. It includes the retention, preservation, and display of educational materials (for example, libraries, museums, and galleries); organized activities that provide support services to the academic functions of the institution (such as a demonstration school associated with a college of education or veterinary and dental clinics if their primary purpose is to support the instructional program); media such as audiovisual services; academic administration (including academic deans but not department chairpersons); and formally organized and separately budgeted academic personnel development and course and curriculum development expenses. Also included are information technology expenses related to academic support activities; if an institution does not separately budget and expense information technology resources, the costs associated with the three primary programs will be applied to this function and the remainder to institutional support. Institutions include actual or allocated costs for operation and maintenance of plant, interest, and depreciation." https://nces.ed.gov/ipeds/use-the-data/survey-components/2/finance.
42. Note: As previously discussed, there is wide variation among MSIs on all kinds of institutional variables, and while beyond the scope of this chapter there is ample other descriptive educational research on these institutions broadly. For example, see Excelencia in Education, "Hispanic-Serving Institutions (HSIs): 2021-22," March 2023, https://eric.ed.gov/?id=ED627437.
43. Before aggregating into "other" institutions, we noted great variation in spending between the 23 special focus law libraries (39.3 percent of academic support) and 35 special focus medical libraries (12.7 percent of academic support).
44. Again, before aggregating into "other" institutions, we noted great variation in spending between institutions. For example, the 23 special focus law libraries spent $782 per student FTE and the 35 special focus medical libraries spent $426 per student FTE in 2021. The "other" category includes faith-based organizations and Tribal colleges, which have much lower spending, thus affecting the aggregate. These have nonetheless been collapsed into the "other" category for comparative purposes, because on

their own the more detailed categories have only small numbers of institutions (e.g., 35 Tribal colleges, compared to 466 doctoral universities).

45. Prior to aggregating institutions, we noted wide variation among circulation for other institutions: 30 law institution libraries reported 56.6 items per student and 38 medical institution libraries reported 61.4 items per student. Meanwhile, 35 Tribal Colleges and Universities reported much lower circulation per student, at 5.4 items per with digital circulation of 2.8 (52.5 percent of total) and physical circulation of 2.6 (47.5 percent of total).
46. The ACRL survey instructions define Information Services to Groups as "information contacts planned in advance in which a staff member, or a person invited by a staff member, provides information intended for a number of persons. Information services to groups includes information literacy instruction as well as cultural, recreational, or other educational presentations, e.g., library tours, workshops facilitated by library staff, and presentations or lectures given by experts invited by library staff. Presentations both on and off the library premises are included if the library sponsors them." https://acrl.libguides.com/stats/surveyhelp.
47. As the outcome scores are based on multiple years of POAL data—from April 2019 through May 2023—the data drawn from IPEDS in these correlations (circulation, staff, expenditures) represent a 3-year average for the institution.
48. Michael Mitchell, Michael Leachman, Kathleen Masterson, and Samantha Waxman, "Unkept Promises: State Cuts to Higher Education Threaten Access and Equity," Center on Budget and Policy Priorities, October 4, 2018, https://www.cbpp.org/research/state-budget-and-tax/unkept-promises-state-cuts-to-higher-education-threaten-access-and.
49. As previously discussed in note 27, there were approximately 372,000 recipients in 2020-21 with $703.3 million in federal funding. This is 52 percent of the approximately 704,000 recipients in 2011-2012, when federal appropriations were $974.7 million. National Association of Student Financial Aid Administrators, *2023 National Student Aid Profile*, October 2023,
50. National Telecommunications and Information Administration. "Digital Equity Act Programs," accessed March 5, 2024, https://broadbandusa.ntia.doc.gov/funding-programs/digital-equity-act-programs.
51. Learn more in Michelle Frisque and Kara Malenfant, "Keeping Up With… Digital Equity Funding" (Chicago: Association of College and Research Libraries, April 2023), https://www.ala.org/acrl/publications/keeping_up_with/digital_equity_funding; Kara Malenfant, Laurie Blandino, and Anne Craig, "Digital Equity in Higher Education" in *Advocacy and Policy Work for Academic and Research Library Workers: Perspectives and Strengths*, eds. Ray Pun, Sonya Durney, and Tarida Anantachai (Chicago: Association of College and Research Libraries, forthcoming 2025).
52. CRediT, https://credit.niso.org/, is a detailed taxonomy to describe contributor roles in the research process to ascribe more granular context and credit than author order listing.

CASE STUDIES

CHAPTER 2

Three Newly Joined Libraries Seeking to Better Serve Students and Grow Advocacy Efforts

Joe Pirillo

INTRODUCTION

The University of Wisconsin Oshkosh (UWO) is a public university comprised of three campuses: Oshkosh, Fond du Lac, and Fox Cities. It is a member of the University of Wisconsin System, one of the largest systems of public higher education in the country. The Carnegie Classification of Institutions of Higher Education identifies it as belonging to the category Doctoral Universities: Moderate Research Activity.

According to data collected in the fall of 2021 by the Integrated Postsecondary Education Data System (IPEDS) report, the student body is comprised of approximately 13,000 undergraduates and 1,000 graduates. Roughly 44 percent of current students are first generation. Table 2.1 provides a race/ethnicity breakdown of the student body.

The university offers more than 150 undergraduate and graduate programs, with strengths in business, education, and nursing. The university's business school is accredited by the Association to Advance Collegiate Schools of Business, and its nursing school is ranked among the top 100 public nursing schools in the nation.

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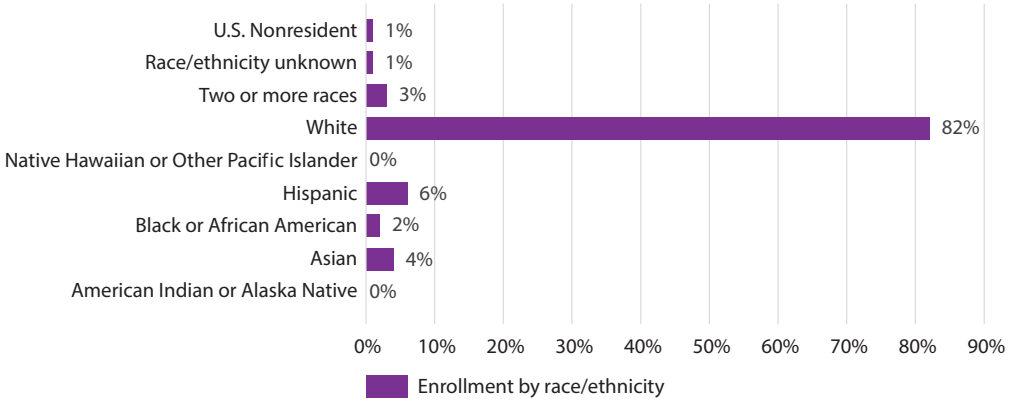


Table 2.1. Percent of all students enrolled in UWO by race/ethnicity in fall 2021. Source: Integrated Postsecondary Education Data System (IPEDS) Fall 2021 Report.

In July 2018, the University of Wisconsin System underwent a major restructuring. The System of UW Colleges dissolved, and those college campuses became affiliated with four-year campuses of the University of Wisconsin System, following that major change, as well as the upheaval resulting from the Covid-19 pandemic, UWO libraries initiated a new strategic plan for 2021-2023. One action item in that plan was assessing the libraries' approach to providing information literacy instruction support in writing classes that were taught by writing professors across all three campuses. UWO instruction librarians chose Project Outcome to help accomplish this action item.

Our project targeted three needs. First, we needed an assessment tool to evaluate student outcomes, especially following our drop-in instruction sessions taking place in Writing (WRT) 188 classes. WRT188 is a required first year course. Professors teaching these courses choose a specific theme for their individual course. Students then explore questions related to the theme while developing their writing, critical thinking, and reading skills.

Second, we wanted to assess general research appointments with students across many disciplines across the three campuses, not just in the writing department.

And finally, we needed to better communicate the value of these two services and their impact on student success. We wanted to use the outcomes to advocate for library inclusion in other institution-wide assessment tools and efforts and acquire evidence of positive student outcomes following our interventions to advocate for additional campus tools and resources that would allow us to support more students.

METHODS

In the fall of 2022, the instruction librarians from each of the three campuses used Project Outcome instruction and research surveys without any alterations or additions to take

full advantage of the standardized tool. The surveys were used exclusively in Writing 188 classes, where UWO libraries' instruction librarians provided drop-in information literacy instruction to support 29 sections of WRT188. This included 25 in-person WRT188 sections and four that were completely online and synchronous. This resulted in meeting with 580 students in person and 38 online.

Students in these first-year writing classes were primarily first-year students from a variety of majors. Drop-in library instruction in first year classes is not entirely uniform across the three campuses, due to the recent merger and flexibility given to individual instruction librarians. In general, however, core outcomes for both in-person and online library instruction for these classes include ensuring students can navigate and use various library tools and services and exploring the instructor's chosen topic while evaluating the authority of sources across a variety of platforms. Our online drop in library instruction supports for WRT188 involved synchronous instruction where the librarian made a visit to the online writing class held within Canvas, UWO's course management system.

These classes were taught on each of the three campuses. Students in participating sections were to take the surveys twice, immediately following the first instruction session and again following the completion of their class project. It was our desire to use the second survey results to identify any change in expressed outcomes after project completion. Links to the instruction surveys were provided to both traditional in-person classes as well as online synchronous classes. Surveys were provided either in a class online course page and posted in the class course in Canvas or shared via a QR code on a handout.

The research assistance surveys were given to students immediately following research appointments and were sent via email. Students could indicate which campus and specific library where they had their appointment.

Instruction librarians informed the students verbally upon the completion of their research assistance appointment that they would be receiving an invitation to voluntarily fill out a survey, where they could enter to win a $50 Amazon gift card. Upon completion of the survey, students were given the option to follow a link to a form on our library website where they could enter their information for the gift card raffle. This step was completely voluntary, and they were assured it was totally separate from the survey they had just submitted.

RESULTS

Survey outcome data was complemented by library's use of Navigate, the institution's success management system and a product of the Education Advisory Board (EAB). The EAB defines a success management system as "a comprehensive technology that links administrators, faculty, staff, and advisors in a coordinated care network to support students from enrollment to graduation and beyond."[1]

Each campus of the University of Wisconsin System currently utilizes Navigate—although not every library. The product enables different departments on campus to coordinate and

measure student interventions. UWO's libraries utilize Navigate to plan, document, evaluate, and schedule both classes and appointments. Polk Library on the Oshkosh campus has also been working with the Office of Institutional Research to identify, partly through Navigate data, various relationships between library interventions and student success outcomes.

The resulting feedback needs to be kept in context. We realize that it is feedback from a small group of students and is not generalizable to all students that receive library assistance. We do not want to overlook the fact that certain inconsistencies in the dissemination of surveys or in some other aspect of the survey process could have inadvertently excluded certain populations. This was a first attempt at a unified assessment process for our newly merged three libraries and we will seek to continuously improve how we deliver and use the surveys to gain even more useful and comprehensive feedback.

By the end of the fall 2022 semester, 120 students had completed the Project Outcome instruction survey immediately after the information literacy drop-in session to WRT188. This was approximately 20 percent of the total number of WRT188 students taught. The highest survey completion rate came when the students were given time prior to the end of their instruction session. We found that emails sent to students after the library instruction session were much less likely to result in completed surveys. That said, the outcome results were very favorable, as seen in Table 2.2.

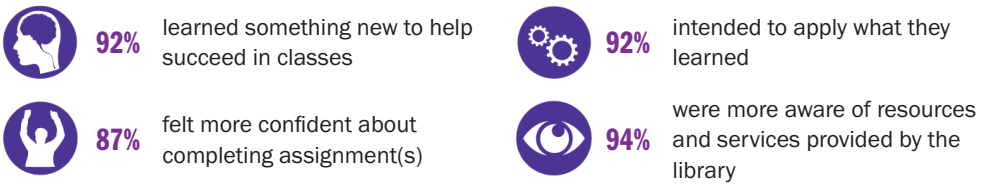


Table 2.2. Project Outcome instruction survey results following WRT188.

Following individual research appointments throughout the semester, students were sent a link to voluntarily fill out a Project Outcome research assistance survey immediately following their appointment. There was a total of 101 individual research appointments for the fall term and 20 students completed the survey, resulting in about a 13 percent response rate.

Table 2.3 shows that the initial research assistance survey results were also favorable.

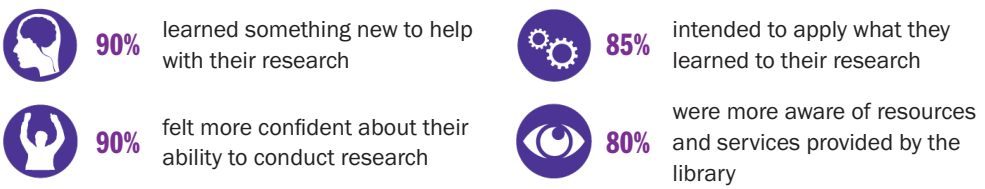


Table 2.3. Project Outcome research assistance survey results.

When compared to other institutions in our Carnegie class, our survey averages were very much in line with national averages. The surveys following the research appointments were higher than the standard average, as shown in Table 2.4.

	Selected Criteria Average	Research Topic Averages			
	Library	Library	Carnegie	National	Total
Knowledge	4.5	4.7	4.5	4.4	4.4
Confidence	4.4	4.6	4.3	4.3	4.3
Application / New Skills	4.3	4.6	4.4	4.4	4.4
Awareness of Resources	4.4	4.6	4.3	4.4	4.4

AVERAGES: Ranges from 1.0 (Strongly Disagree) to 5.0 (Strongly Agree)

Table 2.4. UWO average survey results compared to national averages.

Table 2.5 shows that our instruction surveys were slightly lower in three categories.

	Selected Criteria Average	Instruction Topic Averages			
	Library	Library	Carnegie	National	Total
Knowledge	4.3	4.3	4.3	4.4	4.4
Confidence	4.1	4.1	4.2	4.2	4.2
Application / New Skills	4.3	4.3	4.4	4.4	4.4
Awareness of Resources	4.3	4.3	4.3	4.4	4.4

AVERAGES: Ranges from 1.0 (Strongly Disagree) to 5.0 (Strongly Agree)

Table 2.5. UWO average instruction survey results compared to national averages.

There were some responses that reported both favorable and unfavorable outcomes along with very positive feedback. In these instances, it was suspected that students may have filled out the survey too quickly. In the future we will encourage students to read each question and its choices carefully.

TAKING ACTION

The data from our Project Outcome surveys are useful on several different levels. First, receiving positive feedback and praise can be tremendously affirming for library staff. We want to hear we are supporting students effectively. But the critical and constructive feedback that prompts practical changes is just as important, if not more so. We've been able to make use of both types of feedback following the use of Project Outcome.

The data have helped us adjust our workflows. For example, 14 percent more students noted they were more aware of resources and services after a class versus after a research appointment. We have made more of an effort to note the menu of library services during research appointments. We make a consistent effort to reiterate how they can get additional help and resources in various ways in the future.

We are striving to be more proactive in introducing ourselves to first- and second-year students. We have always taught classes and introduced ourselves and we have been embedded in the course management system for some time; the library has a default tab in every single Canvas course that will direct students to us and our resources. However, we still received responses following research assistance appointments such as, "I can't believe I never knew about this before," or "It would be nice to get reminders, because I forget about the library and its services." For an academic librarian those may be familiar refrains, but it is a reminder that we can never stop thinking about new ways to be present and recommend ourselves to students.

One concrete way we have sought to respond to this feedback is by utilizing a campaign feature within Navigate, our student success management system. This tool allows us to generate campaigns for specific courses. We can create invitation emails at key points of the semester and invite students to follow the link and schedule a research appointment. A few students have commented favorably on this "just in time" type of service reminder. It is most helpful when librarians are aware of a given course and the general timeline of the major project(s).

Project Outcome has also contributed to the addition of data and graphics within our annual library report. They help us communicate key outcomes that are influenced by our library instruction and research appointments. Being able to capture those in succinct fashion and share with shareholders has been essential.

Finally, the data has helped provide evidence to use in advocating for library inclusion in new campus ventures. For example, the Navigate platform is growing in functionality and can enable specific departments to evaluate their intervention effectiveness. Without

the standardized outcome data that Project Outcome provides it would be more difficult to explain why we are a necessary partner in the growing use of this platform to assist students. Since the use of Project Outcome, we have been able to more effectively communicate why we should gain use of this feature. Two staff members from UWO libraries are currently using this tool in a pilot after recent advocacy efforts.

FINAL TAKEAWAYS

As the semester went on, it was obvious that the follow-up instruction survey—the second survey administered after the completion of the WRT188 final assignment—was a little too ambitious for the initial pilot. In the future we may seek to incorporate this into our instruction survey workflow, but it proved to be a tall ask for students.

Another challenge was consistent survey delivery. As discussed, this effort was undertaken by three newly merged libraries. This was our first unified effort at assessment, and, in the future, we will be revisiting how we send the surveys. We want to reach as many students as possible and to reduce bias or inadvertent exclusion of any group, we will be drafting a more standardized delivery protocol. This will include a more succinct and consistent explanation of the tool to students, as well as a standard time or day amid instruction efforts to share the survey. Also, we will emphasize the importance of frank feedback to eliminate the ceiling effect, which is explained on the Project Outcome website as occurring "when a survey receives a very high proportion of positive multiple-choice responses (like "Strongly Agree") or only positive comments."

Lastly, given the ease of use and the suite of Project Outcome survey options, it will be quite easy to incorporate other aspects of library outcome data into our assessment efforts in the future. With the growing need to demonstrate the library's role in student success, making use of a readymade, standardized tool such as Project Outcome has been essential. Depending on the size of the library, creating quality, standardized surveys, identifying how to deliver those surveys, then creating graphics to effectively communicate the information collected can take a lifetime. Even if there is time, expertise may be lacking. Using Project Outcome simplifies that process tremendously.

NOTES

1. "What Is a Student Success Management System?," eab.com, accessed January 3, 2024, https://eab.com/insights/blogs/student-success/what-is-a-student-success-management-system/.

CHAPTER 3

Collecting Iowa State University Library Spaces Feedback Using Project Outcome

Greg Davis

INTRODUCTION

"*In the busy world of academic research library assessment work, can simple and easy-to-use outcome-based assessment tools contribute in meaningful and actionable ways to library spaces-related decision making?*" This is the question at the center of the Iowa State University (ISU) Library's work with the Project Outcome spaces survey.

ISU is a doctorate-granting university in Ames, Iowa that, during the 2022-2023 school year, served 25,241 undergraduate students and 4,094 graduate students. The university offers a wide range of undergraduate and graduate programs across various disciplines. The most popular areas of study include engineering, agriculture, business, computer science, biological sciences, and social sciences. The university attracts students from all 50 U.S. states and more than 100 countries.

Historically, the university has had a predominantly White student population. Both the University's Strategic Plan* and the University Library's Strategic Plan† have aspirational goals to provide a welcoming space to students of all ethnic and racial backgrounds.

* https://strategicplan.iastate.edu/wp-content/themes/strategic-plan/pdfs/StrategicPlan.pdf

† https://www.lib.iastate.edu/about-us/strategic-priorities/strategic-plan

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ISU attracts students with strong academic backgrounds and admitted students typically have competitive high school GPAs and SAT/ACT scores. The university also admits transfer students from community colleges and other institutions.

The ISU campus has several spaces associated with the library. There is one main library, two library reading room locations, and an off-campus library storage facility. The main library provides a variety of study spaces, including 18 group study rooms that can be reserved by students on a first come, first served basis. Group study rooms are equipped with a table, six chairs, a large screen monitor, and a whiteboard.

The library's assessment plan includes a wide range of assessments and metrics, but until recently the library's assessment portfolio had not included outcome measures. The Project Outcome space survey provided an opportunity to add outcomes-based measures to the library's assessment plan.

METHODS

In the fall semester of 2018, the ISU library began conducting survey projects based on the Project Outcome for Academic Libraries spaces survey template. The heart of the survey is constructed from the four core outcome measures (knowledge, application, confidence, awareness) defined for all Project Outcome surveys. Additional outcome measures related to the efficiency and welcomeness (i.e. inclusiveness) of the library spaces were added. Questions to gather open text responses (what students liked about the spaces, what would they improve) were also included.

The ISU project used Qualtrics for the development and administration of the survey because it was a tool already in use by the library's assessment and planning unit. However, the data collection process could also have been accomplished using the built-in Project Outcome survey tool.

A process was developed to identify a convenience sample of students to survey. Each semester, library study rooms are available for students to reserve. The reservation system (the LibCal system from SpringShare*) captures the email address of the student reserving the room. The survey was sent to this collection of student email addresses. While not a randomized sample of students, the sample population are students that have used library group study rooms and are considered a good target audience to provide feedback related to their experience in the library. The survey's introduction indicated to students that the survey was seeking feedback related to their group study room experience.

In addition to the core Project Outcome-related information gathered by the survey, the respondent's email address was used as a key field to link to the following demographic data.

- Respondent's College
- Respondent's Rank (freshman, sophomore, junior, senior, graduate)

* https://www.springshare.com/libcal/

- Respondent's Type (undergraduate, graduate, veterinary medicine)
- Respondent's Gender

The resulting combined dataset was anonymized by removing the key identifier field and then used for the study's data analysis.

The survey process was simple and easy to replicate from one semester to the next.

1. Midway through the semester, identify students that had reserved a library study room since the start of the semester.
2. Create a Qualtrics survey based on the Project Outcome survey template. Use the information collected in step 1 to create a Qualtrics email distribution list.
3. Distribute the Qualtrics survey using the university's email system. Provide a two-week window for responses to be submitted. Midway through the two-week window issue a reminder message to students who have not yet responded.
4. Once the survey window has closed, export the Qualtrics data into an Excel spreadsheet and upload the new data into the ACRL Project Outcome database.
5. Using the exported Excel file, merge in demographic information from the library's management system, matching on the respondents' email address as the key field.
6. Anonymize the dataset by removing the email addresses from the dataset.

The data collection process was designed to maintain the confidentiality of respondents. Once the survey response data was joined with the demographic data, the respondent's identifying key field was removed from the dataset to create an anonymized version to be used in all subsequent data analysis. For this research project, the ISU library principal investigator (PI) did seek institutional review board (IRB) approval. Since the project was focused on improvement of library operations and services, and because no personally identifiable data would be saved or reported from the project, the project was determined to be exempt from IRB oversight.

RESULTS

To date, the ISU Project Outcome survey has been conducted once each semester for six semesters. Over the project lifetime, the survey has been sent to 8,993 students and 1,321 responses have been received, for a 14 percent response rate (see Figure 3.1). The response data obtained is in the form of quantitative information from the survey's Likert questions, and qualitative feedback from the survey's open text response questions.

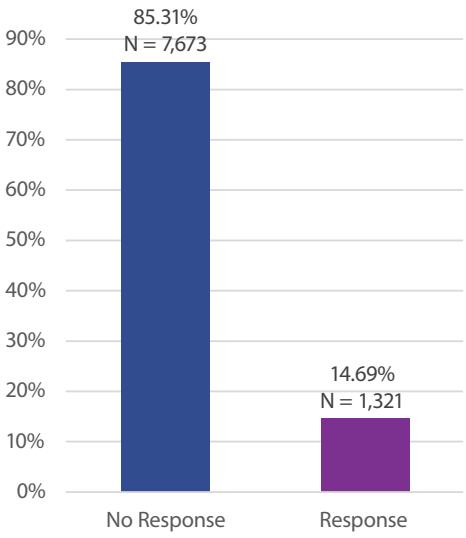


Figure 3.1. Iowa State Library Project Outcome spaces survey overall response rate.

The data collected from the core Project Outcome quantitative questions is uploaded into the ACRL Project Outcome database. This allows the ISU library to compare our results for these questions with other libraries similar to ours in size, demographics, etc. that submit data to Project Outcome using the built-in data dashboard available on the website.

The Project Outcome data dashboard can be used to easily track and compare ISU response frequency metrics. Figure 3.2 shows a screen shot from the dashboard showing the overall response frequencies for ISU data submitted starting in the fall semester of 2018 through the spring semester of 2023.

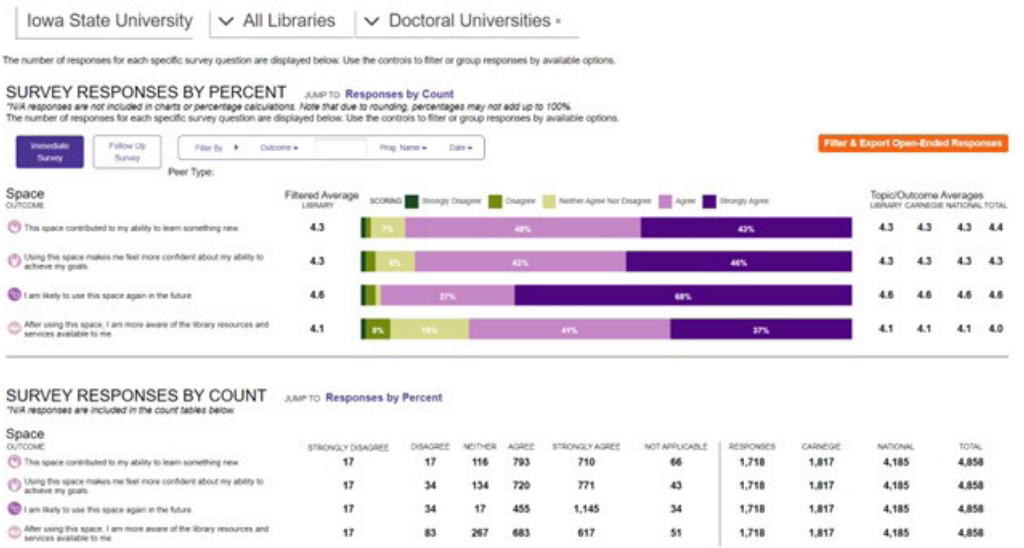


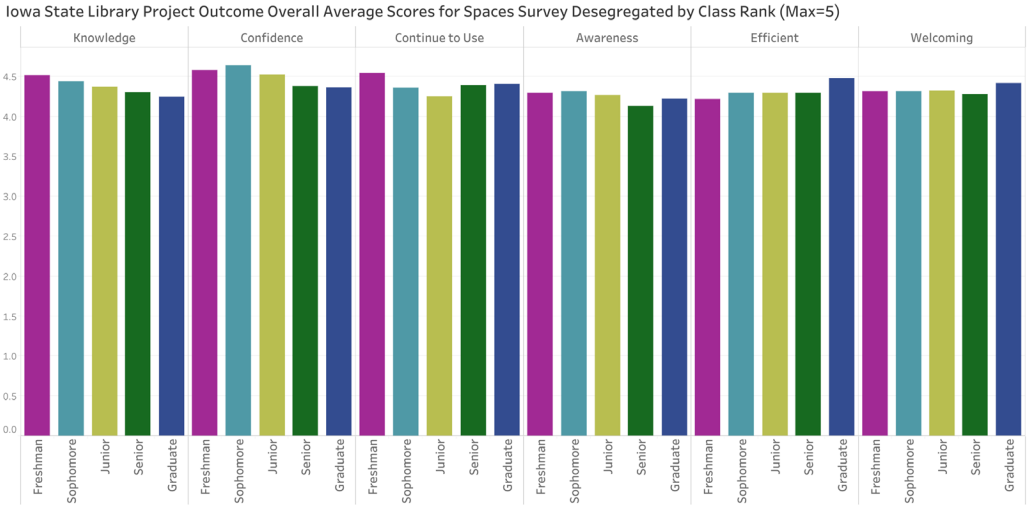
Figure 3.2. Project Outcome data dashboard.

The ISU assessment team also analyzes and reports Project Outcome quantitative data results using the Tableau* data visualization tool. The ISU assessment team shares these data visualizations with the library leadership and management teams for review and discussion.

Following are examples of Tableau data visualizations generated using Project Outcome survey data.

- **Average scores by student rank.** Figure 3.3 shows the average score data segregated by student rank (freshman, sophomore, junior, senior, graduate). The overall average score is shown for comparison. Sophomores rated the library space high for "contributing to their confidence." Graduate students gave high scores for the library's efficiency and welcomeness.

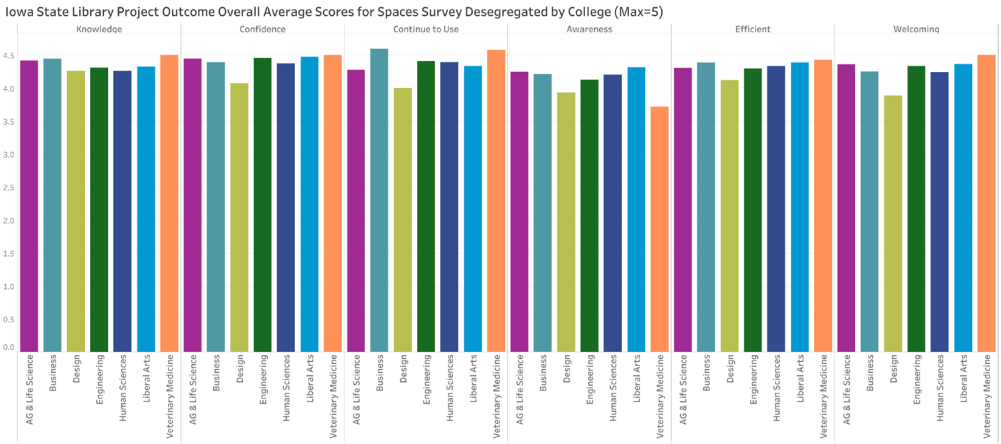
* https://www.tableau.com/



The following prompts were used in the survey:
Knowledge - This space contributed to my ability to learn something new. (Overall Average - 4.3)
Confidence - Having access to this space makes me feel more confident about my ability to achieve my goals. (Overall Average - 4.4)
Continue to Use - I am likely to use this space again. (Overall Average - 4.5)
Awareness - After using this space, I am more aware of the library resources and services available to me. (Overall Average - 4.1)
Efficient - The room reservation process was efficient. (Overall Average - 4.3)
Welcoming - The room reservation staff were welcoming. (Overall Average - 4.3)

Figure 3.3. Average survey scores by student rank.

- **Average scores by college.** Figure 3.4 shows the average score data segregated by ISU colleges. The overall average score is shown for comparison. College of Design students consistently rated the library spaces lower than other college groups.



The following prompts were used in the survey:
Knowledge - This space contributed to my ability to learn something new. (Overall Average - 4.3)
Confidence - Having access to this space makes me feel more confident about my ability to achieve my goals. (Overall Average - 4.4)
Continue to Use - I am likely to use this space again. (Overall Average - 4.5)
Awareness - After using this space, I am more aware of the library resources and services available to me. (Overall Average - 4.3)
Efficient - The room reservation process was efficient. (Overall Average - 4.1)
Welcoming - The room reservation staff were welcoming. (Overall Average - 4.3)

Figure 3.4. Average survey scores by college.

- **Scores over time.** Some students have responded to the survey more than once. In Figure 3.5, responses are grouped by students that responded once, twice, or three or more times. For all questions except the Project Outcome awareness question, the average score for student responses increased (more positive/supportive) for students responding to the survey multiple times.

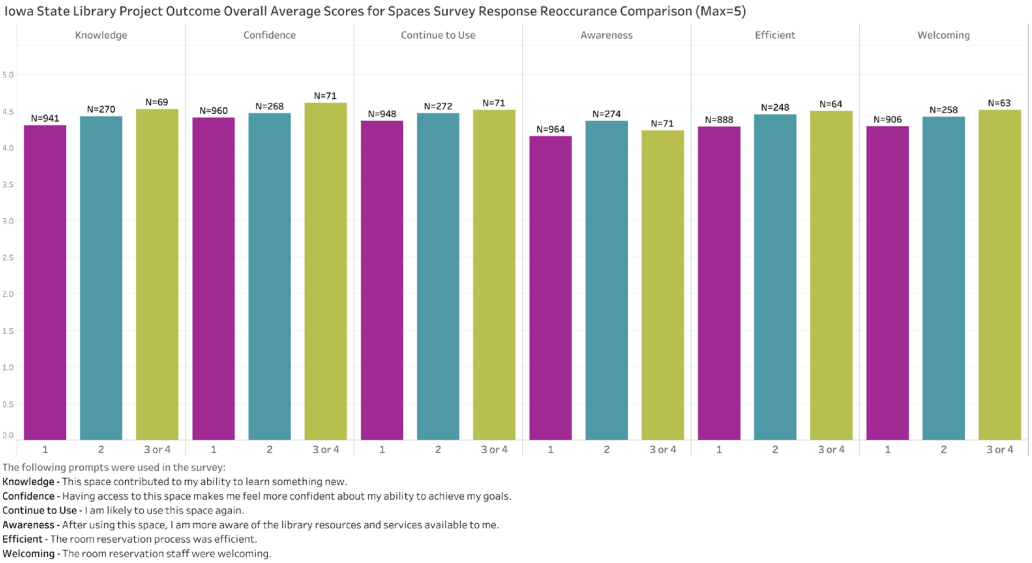


Figure 3.5. Scores over time.

For qualitative response data, the assessment team reviews all responses and removes any identifying information. Then the student feedback is shared in narrative form with the library's leadership team for review and discussion.

TAKING ACTION

The ISU assessment team has found the Project Outcome data useful in library decision-making. For example, at the time we completed our first Project Outcome group study room survey, the library was also developing a request for funding to support a renovation project to create a new, 10,000 square foot collaboration space. The renovation project included adding additional group study rooms to our library. Project Outcome data was included as supporting evidence for including group study rooms in the capital project funding request for the collaboration space renovation. The funding request was approved, and the new library collaboration study area including new open seating options in addition to new group study rooms were in place at the start of the 2019 fall semester.




Figure 3.6. New library collaboration spaces.

Here are some other examples of actions taken in the library supported by results from the Project Outcome spaces survey project.

- More electrical outlets have been added throughout the library.
- Different types of study spaces for both group and individual study have been added.
- Study rooms are cleaned on a more frequent basis.
- Tablets have been added to each group study room to indicate if the room is free or busy, and to also allow walk-in registration for rooms not currently in use.
- Lighting in both group and individual study spaces has been improved.

Due to accomplishments like these, ISU library leaders and decision-makers have come to recognize the value of the data collected from the Project Outcome surveys. This has also contributed to a greater awareness and buy-in to overall assessment activities in the library.

FINAL TAKEAWAYS

Consider the four core Project Outcome questions.

1. This space contributed to my ability to learn something new.
2. Having access to this space makes me feel more confident about my ability to achieve my goals.
3. I am likely to use this space again.
4. After using this space I am more aware of the library resources and services available to me.

These are fundamental questions every academic library should be asking its patrons. Project Outcome makes it easy for libraries interested in collecting spaces-related outcome data to upload their standardized data to the Project Outcome database. By doing this,

the library not only can track their own local trends and results, but also compare their results with other academic libraries by using the free Project Outcome data dashboard.

In 2018, the Association of Research Libraries (ARL) identified research related to library spaces as a high priority area for academic libraries.[1] Research related to library spaces certainly is a priority in the ISU library. For example, the ISU library recently developed a six-phase, multi-year, $90 million renovation plan. In support of this plan, the ISU library's assessment team is conducting ongoing, continuous assessments to measure progress and set priorities related to the renovation of library spaces. As each phase of the renovation plan comes forward, current information about library spaces will be required. It will also be important to show evidence of the impact of existing library spaces and identify the limits as well as the potential of these spaces. The Project Outcome survey has proven to be one simple and efficient tool used by the assessment team to collect this type of information about ISU library spaces.

NEXT STEPS

As expected, the ISU library assessment team found the Project Outcome-based survey easy to administer. By using the Qualtrics survey tool to conduct the survey process, the ISU project gained both agility (for example, the ability to issue reminder messages to survey respondents) and flexibility (the ability to add questions to the survey as needed).

There were recognized limitations with the study's methods.

1. The research population used for the study was a convenience sample based on students who reserved an ISU study room. There were most likely many other students who used a library study room as part of a study team but didn't receive a Project Outcome survey because they didn't reserve the room. Also, students who reserved a room may not have used the room. These students would have still received the Project Outcome survey and potentially provided responses not based on actual library space usage.
2. The surveys were sent out at the midpoint of the semester. Students that used study rooms early in the semester may not have their experience fresh in their mind. And students using the rooms at the end of the semester, which can be an intense time for study, were not surveyed.

To help mitigate these limitations going forward, the ISU library plans to automate the survey process and send the survey requests closer to the time the student reserved the room. The ISU library will also begin to collect information about the number of students using the room for each reservation.

The ISU library has started to administer additional Project Outcome-based surveys each semester to collect feedback related to the library's individual study rooms, technology lending services, and public technology equipment. Looking forward, the library plans to continue to administer these Project Outcome surveys every semester. Data collected

will be uploaded into the appropriate ACRL Project Outcome database. Internally, results of the survey will be shared with the library's leadership team and included in the library's annual assessment report. We expect the Project Outcome work will continue to inform library decision making.

The Iowa State library will also continue to look for opportunities to advocate for the use of Project Outcome surveys with other academic libraries. The hope is to see increased participation in the submission of data from other academic libraries into the Project Outcome database. This will produce more meaningful comparisons of Iowa State response averages with those of other academic libraries.

NOTES

1. "Research Library Impact Framework Initiative and Pilots," Association of Research Libraries, December 31, 2021, https://www.arl.org/research-library-impact-framework-initiative-and-pilots/.

CHAPTER 4

Using Project Outcome as Part of a Three-Tiered Approach to Information Literacy Instruction Assessment

Karoline Manny

INTRODUCTION

Centre College is a private liberal arts college located in Danville, Kentucky. It grants bachelor's degrees in a traditional learning environment. Grace Doherty Library serves a total enrollment of approximately 1,400 full-time undergraduate students and 150 full-time faculty. It is staffed with eight degreed librarians and four support staff. Three librarians act as information literacy instructors, each serving as the liaison for one of the three college divisions—Humanities, Social Studies, and Science and Mathematics. Two additional librarians, the archivist and the digital scholarship librarian, also teach library instruction sessions when a class assignment calls for their specific skills. Upon invitation by a course instructor, librarians teach "one-shot" instruction sessions for any course, at any level, that has a research requirement. One-shot instruction may focus on the use of databases, but it typically also involves an additional learning outcome, such as evaluating, integrating, or citing information. The library's information literacy instruction program has used Project Outcome since the 2020-2021 academic year as part of its three-tiered assessment plan.

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> Tier 1. Course-level, satisfaction-based assessment—assesses student satisfaction with the quality and content of instruction after a specific class session.
>
> Tier 2. Course-level, competency-based assessment—assesses how students' attitudes have evolved (affective domain), what students know (cognitive domain), and what they can do (behavioral/performative domain) after a specific class session.
>
> Tier 3. Program-level, competency-based assessment—assesses student learning (affective, cognitive, and behavioral domains) across a series of connected activities or across a series of courses in order to measure how students are learning as they progress through a particular course of study.

The Grace Doherty Library's Information Literacy Assessment Plan is based on Megan Oakleaf's Information Literacy Instruction Assessment Cycle (ILIAC),[1] which is, in turn, grounded in assessment for learning theory,[2] assessment as learning theory,[3] and assessment as learning to teach theory.[4] As such, the two most important goals of information literacy assessment at Centre are to provide feedback to librarian instructors, who can then improve their teaching, and to provide feedback to students, who can then improve their learning. A third goal of assessment is to demonstrate the value of library instruction to faculty and administration.

Project Outcome helps the library demonstrate value, improve teaching, and demonstrate student competency by providing data for tiers 1 and 2 of our assessment plan.

METHODS

For every instruction session, librarians provide students with both a LibGuide and a digital worksheet. A link to the Project Outcome Immediate Survey for Instruction is on both the LibGuide and the worksheet. The LibGuide also has a QR code to the survey in case students would rather complete it on their phones. Librarians reserve five minutes at the end of each instruction session and ask every student to complete the Project Outcome survey.

Tier 1. Course-Level, Satisfaction-Based Assessment

Satisfaction-based assessment helps us meet two goals. The first is to demonstrate the value of library instruction using a quantitative analysis of the four built-in Likert scale survey questions (that the student learned something new, will apply the lesson, feels more confident about research, and is more aware of resources) and a qualitative analysis of one built-in, open-ended question ("What did you like most about this session?"). The four built-in questions demonstrate to outside constituents like faculty and administration that

students value library instruction and have gained knowledge and confidence to better complete their assignment. Responses to the open-ended question are downloaded in spreadsheet format and then sorted to include examples of specific skills/tools learned, observations on teaching style, and observations on teaching method. These examples add specificity to the Likert scale responses by enumerating a tool or teaching style students appreciated.

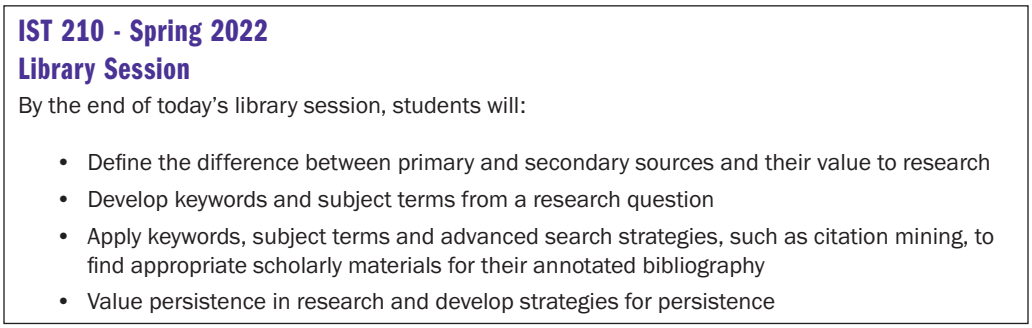
The second goal of satisfaction-based assessment is to improve teaching by allowing us to learn what aspects of the lesson students found helpful or what they would like to learn more about. Project Outcome allows us to gather data to meet this goal through a qualitative analysis of responses to the open-ended question, "What did you like best…," as well as responses to two additional open-ended questions:

- What else could the library do to help you succeed in your classes? (Built-in question.)
- What questions do you still have? (Added question.)

Positive responses from the "like best" question tell us to repeat an activity or continue to emphasize a specific tool or library resource. And, although we cannot tie the question to a specific student for follow-up, responses to "What questions do you still have?" and "What else could the library do…" tell us what we might add to a future lesson to improve it or make it more complete.

Tier 2. Course-Level, Competency-Based Assessment

Competency-based assessment gives librarians insight into how instruction sessions impact the students' understanding of the research process. This, in turn, allows us to meet two goals: improving teaching and improving student learning. Each library session focuses on defined information literacy student learning outcomes (SLOs). Librarians specify the SLOs they intend to address in their lesson planning document and on any PowerPoint presentation or worksheet given to students. For example, Figure 4.1 shows a worksheet header stating the session goals at the top of the page.


IST 210 - Spring 2022
Library Session
By the end of today's library session, students will:

- Define the difference between primary and secondary sources and their value to research
- Develop keywords and subject terms from a research question
- Apply keywords, subject terms and advanced search strategies, such as citation mining, to find appropriate scholarly materials for their annotated bibliography
- Value persistence in research and develop strategies for persistence

Figure 4.1. Sample SLOs on a student-facing worksheet.

The goals illustrated in Figure 4.1 are representative of our typical lesson structure in that they address the cognitive domain (for example, the ability to define terms), behavioral domain (such as the ability to perform tasks like developing keywords and citation mining), and the affective domain (for example, valuing persistence). Librarians primarily use classroom assessment techniques (CATs) to measure student competency in the defined student learning outcomes. CATs may take several formats. To test the cognitive domain, such as understanding the difference between primary and secondary sources in Figure 4.1, we might use polls or one-question quizzes delivered via interactive presentation software. To test the behavioral domain, such as the ability to develop keywords or citation mine, we evaluate responses to specific questions on worksheets, turned in via classroom management software, that require the students to create keywords and synonyms or cite and annotate a useful article they found through citation mining. At the end of the instruction session, the librarian is responsible for recording how many students were able to appropriately complete poll questions or the section of the worksheet identified as the assessment portion.

We acknowledge two weaknesses in this use of CATs as an assessment method. First, student-librarian engagement during class intentionally guides the students through the activity, so unless a student refuses to participate, they are almost guaranteed to adequately complete the assessment activity. Second, though the goal of each session is to change student research behaviors by countering the view that research is a linear process, research shows that single, one-shot library instruction sessions do little to change students' long-term research behaviors,[5] and a single CAT certainly cannot measure long-term behaviors.

Reflection questions help mitigate both these weaknesses. Asking students to reflect on library session content requires them to describe, in their own words, what they have learned, necessitating some degree of understanding. Even more important, reflection has been demonstrated to have a positive effect on long-term improvement of information-seeking behaviors.[6] Therefore, to ensure reflection questions are administered in each library session, we have added two custom, open-ended questions to the Project Outcome instruction survey:

- What is the most important thing you learned in today's library session?
- What is one skill/tool that you have not used for research before that you will use in the future as a result of today's library session?

We perform a qualitative analysis of student responses to these questions. Our goal in our analysis is to find evidence that links the student answer to the lesson SLO and lesson content. Answers to these open-ended survey questions, articulated in the students' own words, serve to validate that the students internalized, at least to some degree, the lesson content.

RESULTS

Each August, the library produces an annual Information Literacy Assessment Summary Report. This report analyzes data collected during the previous fall, Centre (winter), and

spring terms. The report is disseminated on the library information literacy instruction website; a link to that website is sent via email to all faculty when the report is released. The intent of the report is to demonstrate the value of library instruction and demonstrate assessment of student information literacy competency.

Demonstrate the Value of Library Instruction

To demonstrate the value of library instruction to faculty and administration, we include the results from the Project Outcome Likert scale questions—specifically a condensed version of the default report graphic—in our annual Information Literacy Assessment Summary Report (see Figure 4.2).

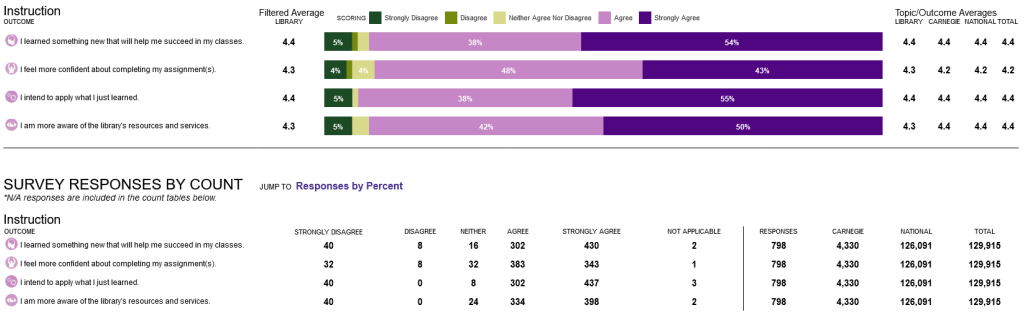


Figure 4.2. Satisfaction-based assessment survey results sample.

Since 2020, when Centre began using Project Outcome, our results from the Likert scale questions have consistently demonstrated that over 90% of students respond "agree" or "strongly agree" with the Likert scale questions. This provides strong evidence that students value library instruction. These results are comparable to the Carnegie and national results also provided in the Project Outcome report.

We also include a selection of responses to the open-ended "What did you like best…" question in our annual Information Literacy Assessment Summary Report, as shown in Figure 4.3. We lightly format the responses for readability.

These responses demonstrate specific aspects of library instruction that students appreciate and note that librarians:

- are effective instructors ("enthusiasm," "energy," "organized").
- use active learning ("applied what we learned," "interactive parts of the lesson").
- provide useful resources ("synthesis matrix," "useful databases").

This student satisfaction data reinforces for faculty that library instruction is a valuable use of their class time.

I liked the in-depth exploration into web of science and its capabilities- I have been getting kind of stuck on Jstor lately, so knowledge of a new resource is refreshing
She was extremely helpful and there were many sites I had no idea about and that I will most likely use for assignments in the future.
I appreciated the variety of credible and easy-to-access sources. I've solely just relied on JSTOR, but now knowing about and having access to more options is a major stress reliever.
I loved that we applied what we learnt in class to practice how to use these databases. It made me much more confident that when I do it on my own later, I will actually know what's going on.
the enthusiasm!
Her energy
I enjoyed the interactive parts of this session.
Hands on practice
It gave me a useful tool not just for this assignment but for all papers
I liked how the worksheets were very organized which will help me stay organized
I liked that we did the lesson together and slowly so that I was able to follow along on my laptop while navigating the sources and not feel like I was behind.

Figure 4.3. "What did you like most…" responses sample.

Demonstrate Student Competency

To gather data to demonstrate student competency in addition to classroom assessment techniques, Centre librarians use two custom, open-ended questions that we added to the Project Outcome survey.

- What is the most important thing you learned in today's library session?
- What is one skill/tool that you have not used for research before that you will use in the future as a result of today's library session?

Student Learning Outcome addressed in class session	Linked responses to additional open-ended survey questions
SLO: Use advanced search strategies	i learned that web of science has options to look at articles that have cited the article that you are looking at or were cited in that article
SLO: Gather and interrogate primary anything.	I learned that I could look at old newspapers without having to pay for anything. This was something new to me and I am excited to research other topics using newspapers.
SLO: Critically evaluate sources based on discipline standards	I learned how to check the validity of a journal by using DOAJ, COPE, NLM, & Retraction Watch.
	That there are multiple ways that you can determine if a journal you are reviewing is legitimate or predatory. Sometimes at first glance some journals will look legitimate but if they are not in major databases, then it is probably not legitimate
SLO: Synthesize multiple sources	Creating a table can help organize your thoughts
	A useful way to organize sources.
	How to make a source x topic table and apply it to a paper
	I learned about making a summary matrix
	The synthesis matrix
	I learned how to create a coherent outline for my debate this semester

Figure 4.4. Custom, open-ended answers samples linking lesson SLOs to student responses.

We are looking for responses that provide evidence of some engagement with the lesson SLO. Figure 4.4 shows examples of responses from four classes during the 2022-2023 academic year.

These custom questions serve two purposes: 1) to encourage students to link immediate learning to future research habits; and 2) to provide data to help validate competency-based assessment data from the lesson CAT. The results from the three years we have used Project Outcome are shown in Figure 4.5.

Assessment Year	Number of student responses sufficiently linked to class SLOs	Number of student responses insufficiently linked to class SLOs	Number of students that did not respond to these questions	Number of students that did not respond to any open-ended questions	Total number of survey respondents
2020–2021	108 (74%)	3	34	30	145
2021–2022	348 (78%)	4	92	87	444
2022–2023	273 (78%)	3	75	70	351

Figure 4.5. Data results from custom, open-ended questions.

While over 90 percent of students successfully completed their lesson's CAT in the 2020-2021, 2021-2022, and 2022-2023 academic y ears, 74-78 percent adequately articulated responses to open-ended questions in the Project Outcome survey that demonstrated they had internalized the lesson's SLOs after the lesson had completed. Only .01-.02 percent provided responses that did not adequately demonstrate they had internalized the lesson SLOs. The remaining approximately 25 percent of students did not respond to the open-ended questions (responding only to the Likert scale questions).

As the Project Outcome survey design suggestions indicate, open-ended questions typically have a lower response rate, especially as the number of these questions increases. Since we began using Project Outcome, Centre has always had a total of four open-ended questions on our survey—the two built-in open-ended questions and two additional open-ended questions used to validate student competency. We believe the approximately 25 percent of students that did not respond to these questions consists of a mix of students who intentionally chose not to respond (due to disinterest or time constraints) and students who were not able to respond (due to lack of understanding). Even with this response rate, in the most recent academic years (2021-2022 and 2022-2023), the Project Outcome survey data does serve to help validate our goal that at least 75 percent of students who attend a library session will demonstrate competency in the lesson's SLO.

TAKING ACTION

We have used the data gathered through Project Outcome to increase faculty engagement with the library, demonstrate the need for improved library instruction opportunities, and improve teaching.

Increase Engagement with Library Instruction

Since we began publicizing data from Project Outcome regarding student satisfaction with library instruction in our annual Information Literacy Assessment Summary Report, the number of instruction sessions that librarians have been invited to teach has increased steadily each year (except during fall 2020, when we were primarily online due to COVID restrictions), as shown in Figure 4.6.

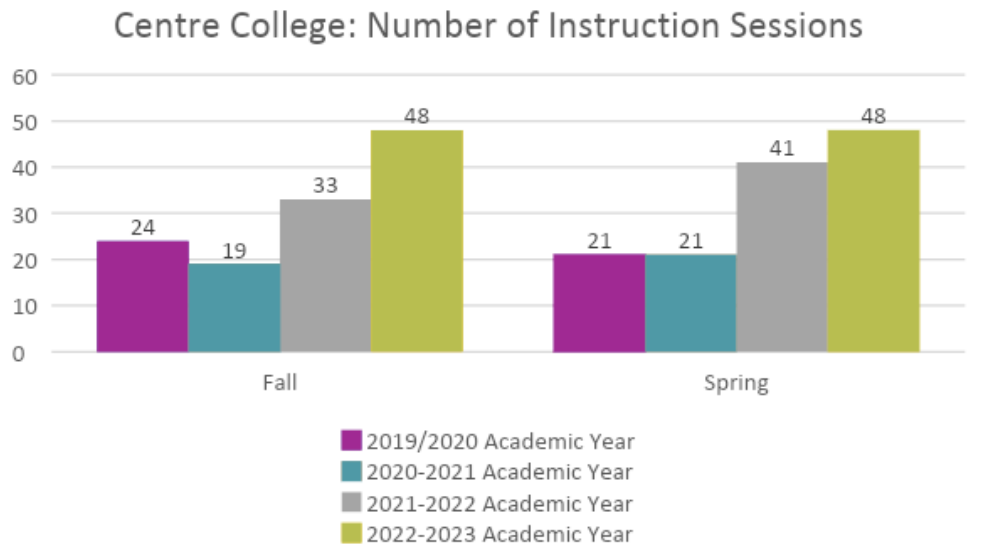


Figure 4.6. Number of library instruction sessions by semester and academic year.

While librarians have improved advertisement methods for instructional services since 2019 which is also helping to drive the increase in library instruction sessions, we do have anecdotal evidence of the impact of our Project Outcome results from conversations with new library instruction users. When a faculty member who has never used library instruction requests a session, we ask what motivated their interest. The response normally involves dissatisfaction with the results of a previous writing assignment or anticipation that a new writing assignment will present challenges. A large portion of faculty also volunteer that they were inspired by the survey results, especially from the Likert scale questions, that demonstrate such a large portion of our students agree they learned something new and feel more confident about their assignment after library instruction.

Going forward, in addition to sharing the overall results from the academic year in an annual report, we will share individual class results with each professor after the library session. That way the instructor will be immediately aware of the value of library instruction to their specific students. They might also be able to follow up on any remaining questions from the open-ended question, "What questions do you still have?"

Demonstrate Need for Improved Library Instruction Opportunities

Course-level student competency data, gathered via CATs and validated via data from the Project Outcome survey open-ended questions, is shared with the library and college administration and faculty in the annual Information Literacy Assessment Summary Report. While student competency data demonstrate that we meet our information literacy goals, it also demonstrates gaps and places where we can improve. For example, our data indicated that, even after one-shot instruction, students still could not reliably distinguish between valid uses for popular sources and peer-reviewed sources in a specific class over the course of several semesters. In the same classes, Project Outcome reflection question responses indicated students didn't want to use peer-reviewed sources because they are "too difficult to read."

Using this data as a starting point, instruction librarians were able to open conversations with the faculty in that program to improve these outcomes. As a result of these conversations, for this class, rather than just teaching one session on using a specific database, we teach four sessions: distinguishing between popular and scholarly sources, finding scholarly sources, reading a scholarly source, and APA citation. The course faculty clearly could not afford to give up four hours of class time, but the assessment data from their class persuaded them to make these four sessions required outside "homework" worth 4 percent of the final course grade. Librarians teach the sessions during Centre's "Common Hour" reserved for outside meetings, grade the session worksheet on a pass/fail basis, and return the grades to the professor, who assigns one point per session to the student's final grade for successful participation. We look forward to seeing evidence of improvement in the student competency data gathered from that library session series this semester. Based on positive student satisfaction data from these sessions, obtained via Project Outcome, other academic departments are currently working with the library to develop similar arrangements to meet their specific class needs.

Improve Teaching

Centre librarians consistently analyze the results of the Project Outcome built-in open-ended questions, "What did you like most…?" and "What questions do you still have?" after each instruction session. We include how we will use what we learn from the student responses to improve that lesson in the "Instructor Reflection" section of our lesson planning document. This allows us to constantly improve our teaching methods and lesson content after each instruction session.

For example, when teaching a new class for an instructor we had not previously worked with, a librarian designed a two-part activity to contrast creative writing without and with research. After teaching the session, the librarian wanted to learn if the example with research that the students analyzed after doing individual writing without research

demonstrated the value of research for creative writing. They looked in the survey results at the "What did you like most…?" responses and found half the class had singled out this part of the session as the part they liked most (see Figure 4.7).

Newly designed activity needing feedback	Linked student comments from "What did you like most about this session" survey question
Student-content engagement: Do a sample writing activity where they write for 2minutes with no research. Ask them to reflect. **Discuss/compare:** Sample paragraph: https://theamericanscholar.org/Joyas-volardores/ – emphasizing what parts of the passage is made from research and how that can tie to other literary elements in the paragraph.	I really liked the activity about the hummingbirds. It helped me realize the importance of research.
	The hummingbird example that could be applied specifically to our class.
	The demonstration of a creative writing piece that had research in it

Figure 4.7. Student feedback on lesson content.

With these results, they have determined to use this activity again.

For the same lesson, when looking at responses to the question "What questions do you still have?" a student responded, "Helping me to cite if I do use research in my creative writing." Acknowledging sources in creative writing was not part of this lesson. Neither the instructor nor the librarian thought to include this issue in the lesson content, but this is a very good question. Going forward, the librarian has decided to add a discussion on this topic to the lesson.

Through this type of analysis of the survey results after each individual instruction session, we improve both the content of our classes and the mode of instruction.

NEXT STEPS

The 2022-2023 academic year marked the end of Centre's 2019-2023 Information Literacy Assessment Plan. The data we have gathered has been greatly valuable to the library and to the college. There are some adaptations we have made to our assessment process for the 2023-2028 assessment plan. These include changes to improve the response rate to open-ended questions and the use of an additional Project Outcome survey.

Improve Student Response Rates to Open-Ended Questions

Student responses to the built-in and custom open-ended questions provide us with important data regarding student satisfaction and student interpretations of the lesson

takeaways. Because we only have an approximately 75 percent response rate to these questions, going forward we intend to emphasize their importance when explaining to the students how to access the Project Outcome survey at the end of each class. We also intend to eliminate one open-ended question from the survey, to reduce the writing burden on the students. We hope, in this way, to increase our response rate.

New Project Outcome Survey Uses

While assessing our assessment plan, we realized an important mode of library instruction is through one-on-one appointments (both scheduled and walk-in). In our 2019-2023 Information Literacy Assessment Plan, we had no means of assessing either student satisfaction or competency after this type of instruction. For the 2023-2028 assessment plan, we have created a separate Project Outcome Immediate Survey for Instruction to administer to students who participate in one-on-one instruction. They receive a link to the survey via Navigate, our student success management system, after signing in to meet with us. We hope data from this survey will help us demonstrate the value of and improve our one-on-one instruction.

FINAL TAKEAWAYS

Centre College Grace Doherty Library's three-tier assessment plan strives to demonstrate the value of library instruction for student success by collecting comprehensive data on students' satisfaction with library instruction and students' information literacy competency as a result of the instruction. The Project Outcome Immediate Survey for Instruction has been an integral part of this assessment plan because it allows us to:

- demonstrate student satisfaction and intent to apply knowledge gained via library instruction.
- validate local student satisfaction by comparing it to peer and national institutions.
- indicate ways to improve teaching by gathering data on what students liked and learned.
- aid in the validation of student competency data gathered via classroom assessment techniques.

For tier one of the assessment plan, course-level satisfaction-based assessment, the survey Likert scale questions have particular value due to their emphasis not only on satisfaction, but also the students' intent to apply what was learned and their improved confidence. Faculty, who we depend on to voluntarily dedicate class time to library instruction, value evidence of students' intent to apply knowledge and their improved confidence greatly. Student responses to the built-in, open-ended question "What did you like most about this session?" strengthen our student satisfaction data by allowing librarians to

enumerate teaching methods, databases, and other writing tools that students appreciate learning about.

For tier two, course-level, competency-based assessment, responses to two custom open-ended questions—"What is the most important thing you learned in today's library session?" and "What is one skill/tool that you have not used for research before that you will use in the future as a result of today's library session?"—allow students to demonstrate they can articulate some aspect of the lesson content in their own words. Connecting these responses to the student learning outcomes for a specific library session helps librarians demonstrate that students have taken knowledge or a performance-based skill from the library session. These responses, recorded after the lesson, combined with responses on classroom assessment tools recorded during the lesson, are additional evidence that students have gained competency during the library session. Because they serve as an opportunity to reflect on the lesson content, they also help reinforce learning and link it to improved future research habits.

Both student satisfaction data and student competency data allow librarians to identify what was good about a lesson design so we can repeat it, and what was lacking in a lesson so we can improve it. The Project Outcome Immediate Survey for Instruction allows us to efficiently gather, analyze and report this data.

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CHAPTER 5

Two Academic Libraries, One Study:

Using Project Outcome for Assessment and Advocacy across Partner Institutions

Rachel E. Trnka, Karen F. Kaufmann, and Katy Miller

INTRODUCTION

The push for academic libraries to demonstrate value to their institutions through assessment data is well documented in professional literature.[1] In particular, the move towards outcomes-based assessment in higher education means that libraries are increasingly required to document their role in student learning and retention.[2] This move has proved challenging for libraries, which have historically focused on quantifiable measurements of inputs and outputs (e.g. number of items in collections or number of instruction sessions conducted) rather than on assessing student learning outcomes.[3] Additionally, while academic librarians agree that formal assessment of library instruction is important, many employ informal assessment measures and would benefit from a more systematic approach.[4]

These factors contributed to the decision of one librarian from Seminole State College of Florida (SSC) and two librarians from the University of Central Florida (UCF) to study outcomes of library instruction (LI) across our institutions using the Association of College & Research Libraries' (ACRL) Project Outcome toolkit. The data from Project Outcome's Immediate Instruction Survey has benefited our libraries individually and collectively, as

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we use this assessment data to demonstrate value to our stakeholders. Another motivation for this project emerged from a robust transfer partnership between Florida institutions, known as DirectConnect® to UCF. Transfer students make up a significant portion of students at UCF; in the year 2022-2023, around 12 percent of all students were transfers.[5] Seminole State College is the second largest feeder school of transfer students to UCF in the Florida College System, with 1,097 SSC students enrolled in UCF in the 2022-2023 year.[6]

Seminole State College serves a diverse student body of more than 20,000 students, and it is classified as a Hispanic-Serving Institution (HSI). The college offers associate's degrees, bachelor's degrees, specialized certificates, and adult education programs; as such, SSC's Carnegie Classification is Baccalaureate/Associate's Colleges. All full-time and most part-time librarians at SSC are required to provide information literacy instruction. This instruction takes the form of sessions to support credit-bearing courses taught by discipline faculty as well as credit-bearing research and information literacy courses taught by faculty librarians in the health sciences and nursing programs.

The University of Central Florida is one of the largest universities in the country with more than 68,000 students. Like SSC, UCF serves a diverse population and is a designated HSI. The university offers bachelor's, master's, doctoral, and specialist degree programs, with its Carnegie Classification being Research Universities/Very High Research Activity. Undergraduates make up about 86% of the total student population.[7] Information literacy instruction at UCF is conducted mainly by librarians in the Academic Engagement division; subject librarians in the Research & Information Services (RIS) department are dedicated to upper-division and graduate courses. A small team of librarians in the Student Learning and Engagement (SLE) department provides library instruction to first-year undergraduate students, particularly in courses like college composition and student success.

For our study, we surveyed students receiving library instruction in face-to-face sections of English Composition II courses during the spring 2022 semester at SSC and UCF. By focusing on the same required general education course, we hoped to explore differences in foundational information literacy skills between transfer and original UCF students. Neither SCC nor UCF had formal, library-wide assessment of instruction programs prior to this project. As a result, we identified a need for formalized assessment to measure LI's institutional and cross-institutional impact. The immediate instruction survey was selected to help us answer our research question: "What are the similarities and differences in the perceived value of the library by students at a baccalaureate state college and a four-year research university?" We hoped that the results of our study could be used to align priorities for information literacy instruction at SSC and UCF, thereby strengthening our transfer partnership and supporting student success.

METHODS

The immediate instruction survey was the best fit for our study, as it offered questions addressing institutional interests at both the college and university. All three authors had

experience piloting Project Outcome surveys to assess library instruction sessions prior to the study. At SSC, Project Outcome immediate instruction and research surveys were piloted across various courses beginning in fall 2020 by one faculty librarian. This pilot phase included English Composition I, English Composition II, College Success, Chemistry, Government (Political Science 2041), a newly launched First-Year Experience (FYE) course in 2023, and an online Research Strategies course in the Bachelor of Science Health Sciences (BSHS) program. When COVID arrived, the on-campus library instruction and survey administration dwindled, but it resumed in the fall of 2021 when classes returned to campus. This pilot at SSC led to informal conversations with UCF librarians about Project Outcome surveys, which in turn led to more formal conversations about developing a cross-institutional study using the Project Outcome immediate instruction survey. In the summer of 2021, three librarians in UCF Libraries' Student Learning & Engagement department piloted the immediate instruction survey through the Project Outcome site in instruction sessions for a first-year college success course. Around that same time, UCF Libraries underwent an external audit that noted the need for increased assessment across the library and recommended gathering feedback from students and faculty. As a result, since fall 2021 the library has required all UCF librarians conducting course-specific library instruction to use a version of the immediate instruction survey, including those librarians in the SLE and RIS departments. The required UCF survey from fall 2021 onward included supplemental questions, identifying course number, section number, course instructor, and librarian; the other change from the initial pilot was that the survey was administered through Qualtrics rather than directly through the Project Outcome site.

Due to the authors' prior experiences with the survey, UCF Libraries' survey requirements, and the benefits of the Project Outcome tools and resources, we selected the immediate instruction survey for our cross-institutional study. We included the supplemental questions from the UCF survey, added an additional quantitative question to identify the institution, and administered the survey through Qualtrics. The additional qualitative question came from the Project Outcome list of optional questions— "please share one thing you learned today"—and was added to gain insight into what students valued most from the session. Librarians distributed the survey at the end of the session using slides approved by the institutional review boards (IRB) at UCF and SSC (Figure 5.1 and Figure 5.2). The slides contained information about the study, including age requirements for participants, along with the Qualtrics survey link.

Since the survey was delivered using Qualtrics with the intention of uploading the results to Project Outcome, our data analysis benefited from employing multiple tools. Qualtrics was used for the descriptive analysis of the combined quantitative data. The data was then exported to Microsoft Excel, separated by institution, and independently uploaded into our respective Project Outcome dashboards. We used the Project Outcome tools and dashboards to create customized reports and visualizations. Finally, Excel was used for the thematic analysis of the combined qualitative data.

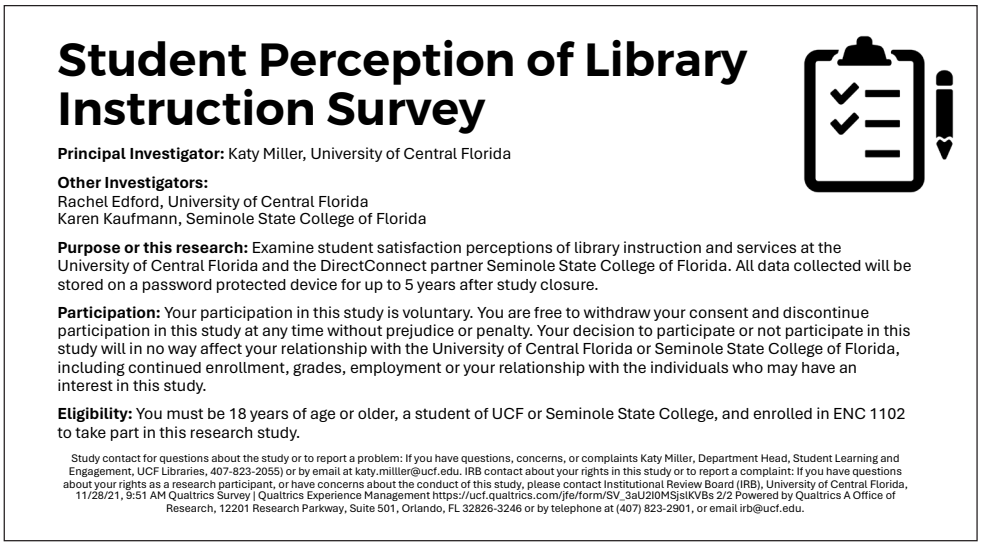


Figure 5.1. Explanation of research slide.

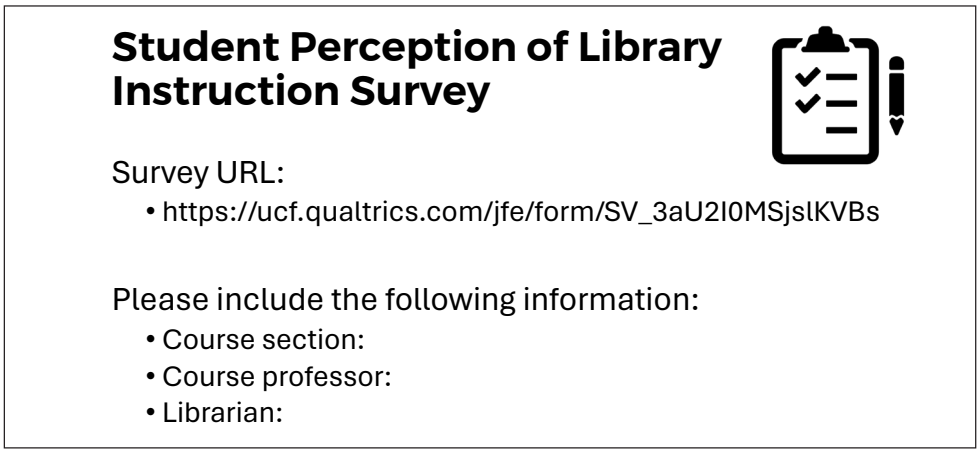


Figure 5.2. Survey link and details. Course section, professor, and librarian were included for each session.

RESULTS

The survey was distributed to 23 English Composition II sessions total (16 at UCF and seven at SSC) by six different librarians, three at each institution. In total, there were 203 responses to our survey; 85.5 percent of respondents were UCF students and 14.5 percent were SSC students. The responses to the four Likert-scale questions, measuring the impact of the outcome areas of knowledge, confidence, application of new skills, and awareness of resources are presented first, followed by responses to the three open-ended questions.

Quantitative

One benefit of our data collection method was that Qualtrics easily generates descriptive statistics for the combined quantitative data (see Table 5.1).*

Outcome	Question	Scale Min-Max	Mean	Standard Deviation	Variance
Knowledge	I learned something new that will help me succeed in my classes.	1–5	4.48	0.78	0.60
Confidence	I feel more confident about completing my assignment(s).	1–5	4.30	0.74	0.55
Application of New Skills	I intend to apply what I learned.	1–5	4.48	0.70	0.49
Awareness of Resources	I am more aware of the library's resources and services.	1–5	4.62	0.66	0.44

Table 5.1. Quantitative descriptive statistics generated by Qualtrics.

The mean for all questions was relatively high, as 5 equals "strongly agree" on the Likert scale; this indicates that most students responded positively to the questions. The low standard deviation in the results shows that the data are clustered closely around the mean, demonstrating that the results can be considered reliable. The small variance in our results shows that the responses had few 1 or 2 ratings.

While Qualtrics was helpful for collecting and analyzing the combined survey data, Project Outcome provided multiple options for reporting the separate quantitative institutional data. The default report includes a summary of the quantitative results, divided up by the four outcome areas, as well as comparisons to outcome averages from other libraries nationwide and by Carnegie Classification (see Figure 5.3).

The Carnegie Classification benchmarking compares survey data to other institutions in the same Carnegie Classification. Benchmarking data gives institutions the ability to observe trends, identify potential gaps and areas for improvement as compared with similar institutions, and identify which outcomes are being met exceptionally well or on par with other institutions in the same classification.

As indicated in the report, the averages for the SSC data responses are high in all four outcome areas, ranging from 4.2 in confidence to 4.5 in knowledge and awareness of resources. The knowledge and awareness of resources averages match those of other libraries in the Baccalaureate/Associate's College Carnegie Classification and surpass the national averages.

The default report also provides the option of including additional visualizations available through the dashboard, including matrix diagrams (see Figure 5.4). The matrix diagram illustrates the aggregate scores corresponding to the 5-point Likert scale responses for each outcome. The visualization is interactive, with additional details on each outcome available by clicking on the corresponding area.

* It is important to note that the "Analyzing Quantitative Data" page on the Project Outcome resources contains instructions on how to calculate descriptive statistics using the raw data (ACRL, 2019).

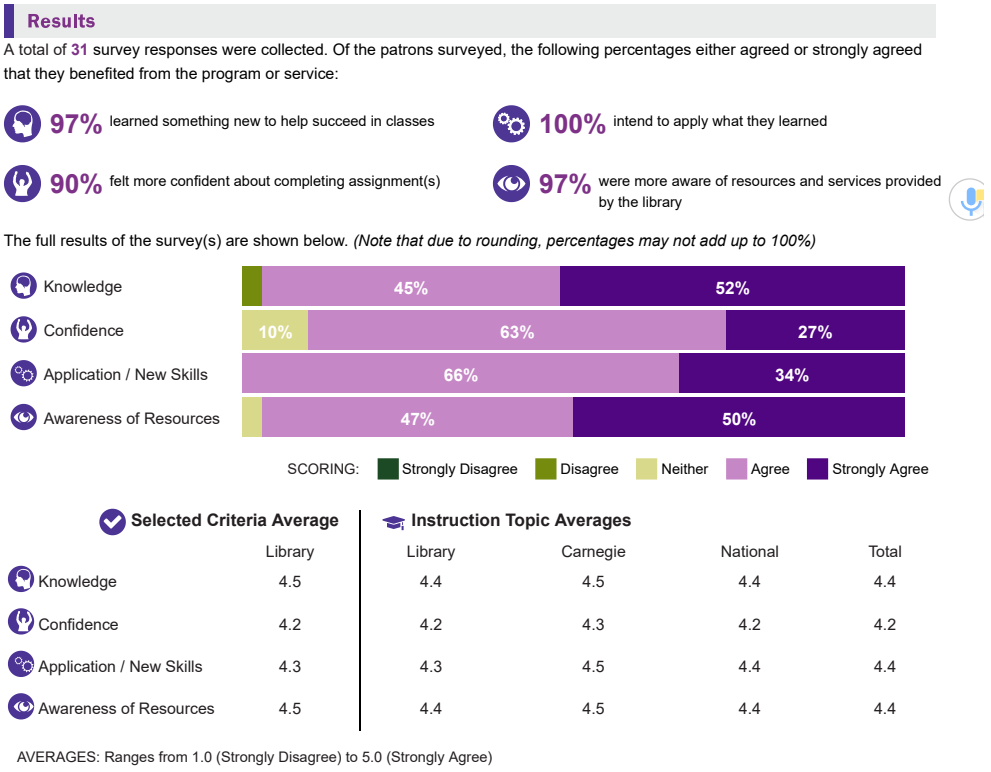


Figure 5.3. Selection from Project Outcome default report for Seminole State College of Florida data.

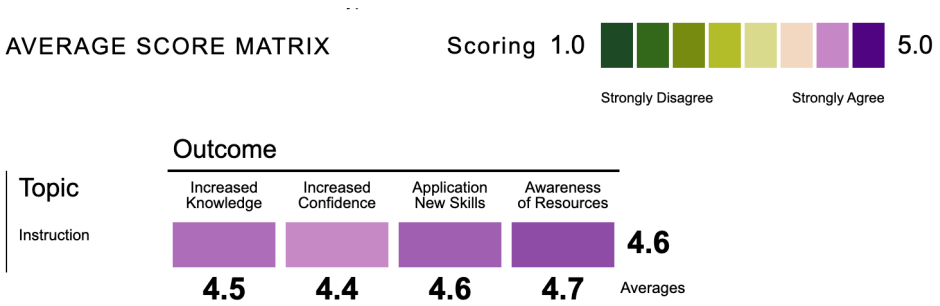


Figure 5.4. Detail of interactive outcome matrix for UCF data.

The color of the diagram indicates the strength of the relationship, with the darker purple representing a stronger relationship. Like the SSC data, the outcome averages for UCF students are high, ranging from 4.4 to 4.7. Similar to SSC, the average for the increased confidence outcome ranked the lowest of the four.

Qualitative

The Project Outcome dashboard includes a feature to create word clouds visualizing responses to the open-ended questions from each institution. However, for our work, a more detailed analysis of the combined data was necessary to understand student responses across institutions. In total, 90 percent (n=182) of the students who completed the survey answered at least one of the three qualitative questions. After the responses from the open-ended questions were downloaded from Qualtrics and uploaded into Excel, each team member independently coded them.

We opted for an open coding approach to allow the themes to emerge from the responses rather than coding from a set of predetermined themes. After the first round of coding, we combined variations of the terms generated and calculated the frequency of those combined terms. This resulted in ten themes: additional services, confidence, database, information literacy, interactive, positive interaction, relevant, resources, satisfied, and search strategies. We also created a separate category for blank or no comment responses. Each researcher then independently recoded the data set using these ten themes; researchers could assign more than one theme to each response.

The question "what did you like most about the session?" generated the most detailed responses of the three questions (see Figure 5.5).

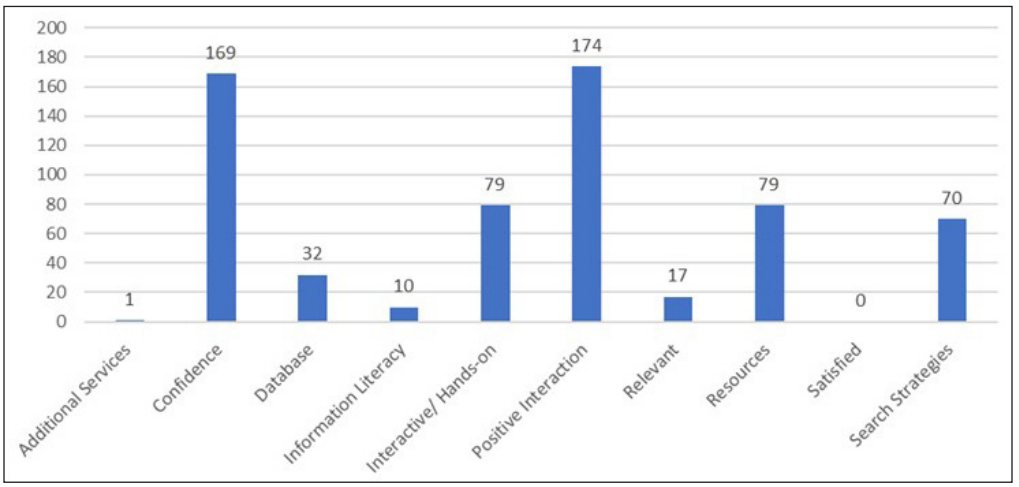


Figure 5.5. Responses to the question "What did you like most about the session?"

In particular, the "positive interaction" and "confidence" themes were the most prominent. The positive interaction comments often referenced the librarian, including words like "helpful" and "friendly," while also describing the sessions as "clear" and "informative."

In the "confidence" comments, students expressed feeling better able to complete their assignments after attending the session.

For the second question, "what else could the library do to help you succeed in your classes," the "no comment/blank" category received the highest number of responses, followed by "additional services" and "satisfied" (see Figure 5.6).

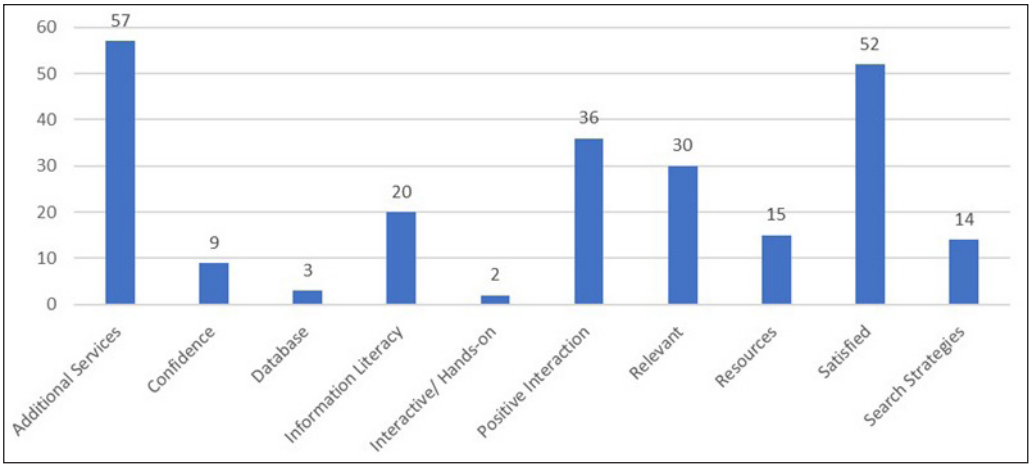


Figure 5.6. Responses to the question "What else could the library do to help you succeed in your classes?"

Twenty-six of the "additional services" comments related to library instruction, while others referenced library collections, spaces, and outreach efforts.

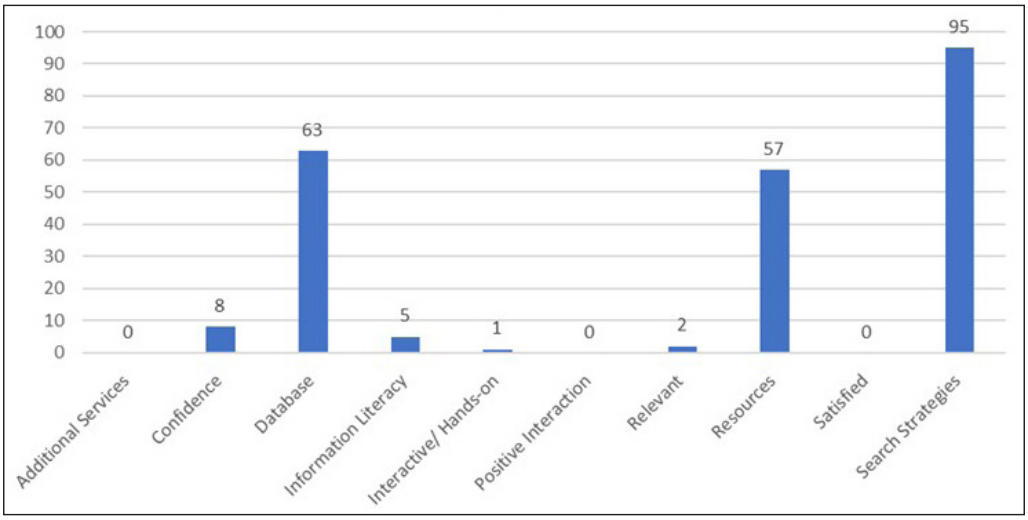


Figure 5.7. Responses to the prompt "Please share one thing you learned today."

The last question, "please share one thing you learned today," revealed what students took away as valuable from the instruction, with search strategies and databases identified as the top themes (see Figure 5.7).

This information acts as an informal assessment of what was retained from the session. In our findings, students mentioned learning skills such as how to navigate the various resources available on the library website, locate peer-reviewed articles, use filters effectively, and narrow down search results. In most cases, databases were not mentioned by name; instead, students appreciated learning about how many were available and how to use the databases to move beyond a basic search on the library homepage.

TAKING ACTION: RESEARCH INFORMING PRACTICE

Demonstrating the value of academic libraries has become increasingly important in the wake of the COVID pandemic and rapid technological changes, as administrators outside the library question the need for library spaces on campus.[8] Meerovitch traces the historical development of library spaces in terms of focus, goals, roles, and measurement.[9]

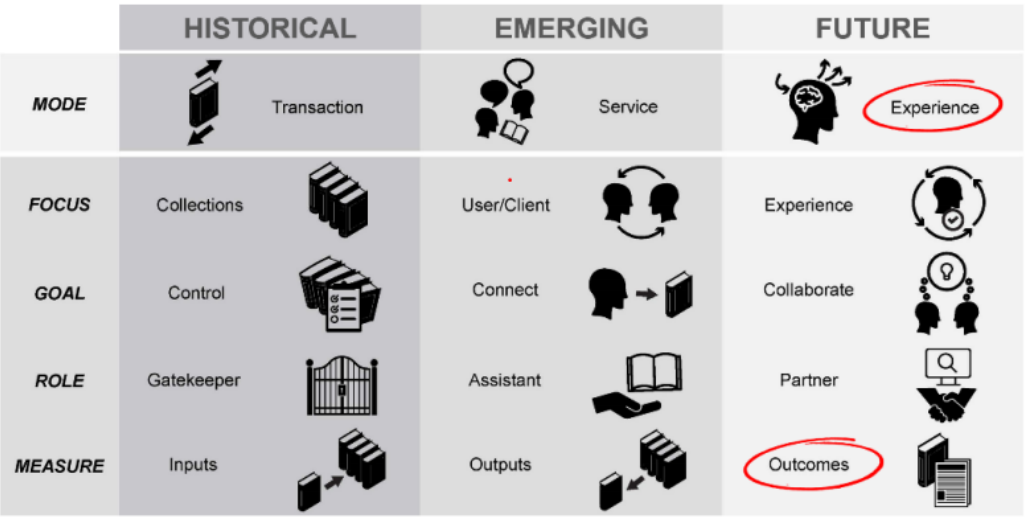


Figure 5.8. "Shift toward meeting the needs of students while creating a positive, equitable learning experience." Image reprinted from G. Meerovitch, "Embracing the Moment - Reconceptualizing the Campus Library," Library Leadership & Management 37 no. 2 (2023): 3. Licensed under CC BY 4.0.

Meerovitch's visualization of the shift from the academic library as a collection-centered gatekeeper focused on measuring inputs to a collaborative, campus partner measuring outcomes further echoes the benefits of Project Outcome surveys to advocacy work (see Figure 5.8). By focusing on outcomes, Project Outcome assists libraries in communicating the impact of information literacy instruction to stakeholders. Since many academic institutions focus on student learning outcomes in their strategic plans, annual reports, and other documentation for accreditation, having outcomes-based data—rather than output or input-based data—included in those documents from the library helps demonstrate the institutional value of libraries. Furthermore, Project Outcome's benchmarking tools enable college, university, and library administrators to make comparisons beyond local or state-wide partnerships with libraries nationwide. Comparing rankings by Carnegie Classification is beneficial for institutions that value national and international rankings. In addition to the institutional and cross-institutional benefits, the results of Project Outcome surveys can also benefit individual librarians by providing evidence of teaching effectiveness for evaluation or promotion purposes.

At SSC, the survey results were shared with stakeholders including faculty librarians and the library director in a few different ways. Over the course of the last two years, the research team broadly disseminated their research findings through two webinars hosted by ACRL Project Outcome for Academic Libraries. The first webinar in spring 2022 focused on the planning and implementation of the study,[10] and the second in spring 2023 on the initial findings and methods for analyzing the quantitative and qualitative data.[11] The full study was recently published in *The Journal of Academic Librarianship*.[12] In summer 2023, an email was sent to SSC research and instruction librarians providing updates on the study and highlighting the two webinars. An additional email was sent to SSC librarians after the publication of the article.

Another way results were shared with SSC librarians and the library director was through the Project Outcome dashboard. The Project Outcome dashboard easily facilitates the sharing of survey data for each institution. All SSC librarians and the library director who have Project Outcome accounts have access to survey results from both the pilot surveys (2020-2021) and this cross-institutional study (2022). Having access to this information is beneficial to both administrators and individual librarians. The SSC faculty librarian conducting the pilot surveys included the Project Outcome survey data from the pilot surveys (2021-2022) as evidence for her annual evaluation. The data is also valuable for individual librarians in terms of self-assessment and reflection on information literacy instruction outcomes and student perceptions.[13] For administrators, Project Outcome is a useful tool for providing longitudinal assessment data, looking at trends, and benchmarking to other Carnegie Class institutions. This tool could provide administrators with a systematic approach for annual evaluations of librarians, program reviews, and accreditation reviews.

Although Project Outcome surveys have not been formally adopted at SSC libraries to programmatically assess information literacy instruction outcomes or as part of a systematic and reflective approach,[14] in 2022, as a result of piloting Project Outcome surveys, there were discussions with the library director about this possibility. The director expressed a strong interest that Project Outcome be part of a multi-pronged approach to assessing information literacy outcomes. Focusing on outcome measurements for information literacy instruction was of prominent interest in these discussions. The SSC library had recently begun the development of a formal information literacy program and was investigating using other outcomes tools to assess information literacy across the curriculum. Using Project Outcome instruction survey data is a valid, replicable, systematic, and programmatic option for assessing information literacy at the college. While the final results of the cross-institutional study were not completed in time to be included in SSC Libraries annual report for 2021-2022, the data from the initial pilot surveys of English Composition I was shared with the library director. The data and selected visualizations available through the Project Outcome dashboard were included in SSC Libraries 2021-2022 annual review, hosted by the Institutional Research department and publicly available on the library website.[15] This is an illustration of one way Project Outcome data can be used by academic libraries when demonstrating institutional value through annual reviews and reports.

Librarians at the University of Central Florida shared survey data with library administration in the hopes that it could be used in the libraries' future reports and as talking points in meetings with other administrators across campus. The inclusion of outcomes-based survey data in the libraries' annual report would represent a significant addition, as the report has typically presented input and output measurements of the various library departments and divisions. Additionally, since the 2021 audit of UCF Libraries and resulting assessment recommendations, the library has focused more on library-wide assessment and now includes university assessment duties as a portion of one librarian's annual assignment. The librarian who handles university-wide assessment, including instruction and other services, is now using the Project Outcome dashboard to upload data for all course-specific library instruction across multiple departments. This represents a shift, as previously the Project Outcome survey dashboard only included selected survey data from a single department at UCF Libraries.

In addition to benefiting UCF Libraries as a whole, the data from the Project Outcome surveys has been beneficial to individual librarians in terms of annual evaluations and promotion. For example, the two UCF librarians on the research team included the data and selected graphics from the Project Outcome dashboards in their annual accomplishments reports and their promotion dossiers to demonstrate their impact on student learning. One author of this study is a supervisor and can attest to the value of having and using the qualitative survey responses in the evaluations of librarians in their department.

Having the Project Outcome survey results also helps convey the value of library instruction to those outside the library who review promotion dossiers. Student perception responses are a common component of promotion dossiers for faculty who teach credit-bearing courses to demonstrate teaching effectiveness. At UCF, librarians are non-tenured faculty, and they do not teach credit-bearing courses. Therefore, the only way librarians have data related to student perceptions is if we decide to survey students in our sessions. As a replicable measurement of student perceptions of learning, the Project Outcome survey is similar to student perception surveys; the survey demonstrates the instructional work of librarians and acknowledges their contributions to student learning. As Alwan et al. (2018) claim,[16] some faculty misunderstand the role of librarians or view librarians as subordinate faculty. As such, sharing the results of student perceptions of learning in LI sessions with other faculty helps librarians communicate their value in a language familiar to faculty.

NEXT STEPS

Our research inquiry of student perceptions of information literacy instruction at two academic libraries, closely connected via a transfer student partnership, revealed that most students either agreed or strongly agreed that they benefited from library instruction. Our survey responses revealed that even though the course content of English Composition II classes differs across institutions—SSC courses are more focused on literary resources than UCF courses—students identified similar benefits of IL in terms of information literacy skills in SSC and UCF sessions in their responses to our open-ended questions. More specifically, we learned what aspects of instruction are important to students, informing our practice for future improvement. The immediate instruction survey is unique in that it provides outcomes measurements that are validated and benchmarked, producing longitudinal data that can be collected and analyzed for a consistent and persistent view of instruction. The survey is engaging; students can reply to the questions that are relevant to them using open text responses. Project Outcome surveys provide a systematic way for students to reflect on their learning.[17] By taking a survey immediately following the information literacy instruction, students reflect and share their perceptions of the four impact outcomes.

The open text responses are insightful in multiple ways. We learn the type of words that make sense to the students, how they describe their research experience, and their ways of thinking about their new knowledge. In addition, the open text responses can be analyzed to identify themes that may inform information literacy instruction strategies and assist in identifying pedagogical constructs that are meaningful to students. After the coding was completed, we noticed two of the themes that we generated organically, confidence and resources, were exact matches to two of the outcome areas identified in the Likert scale questions. For future work, we would like to explore coding the open

responses to all four of the outcomes. However, it should be noted that the open coding and authentic thematic analysis were unencumbered by the four situated Project Outcome outcomes, thereby indicating the value of the outcomes and the survey structure.

Reflecting on our experience using Project Outcome surveys, we wish to share some anomalies that could be useful for future work. When we administered the survey at UCF in the first session for this study, students had difficulty reading the long Qualtrics URL originally included on the presentation slide. In response, we created a shortened Bitly URL, which was unfortunately blocked at SSC; our final solution was to create a QR code. Due to these issues with the survey URL, we advise using the QR code that is now included on the Project Outcome surveys instead of an alternative method. It is also important to consider possible barriers to participation at your institution prior to distributing the survey. SSC has dual enrolled high school students under the age of 18 who were part of the English Composition II classes we surveyed. Due to the IRB requirement that study participants must be 18 or older, those students were prohibited from participating in the survey. An additional recommendation is to be sure to record the number of students who attended each session, so that an accurate response rate can be calculated. While we did have records of the number of students enrolled in each course section that we taught, we did not record how many of those students actually attended the session.

FINAL TAKEAWAYS

Here are a few more tips to assist you on your research journey with Project Outcome.

- Consult the Project Outcome resources and review the case studies for best practices before administering the survey.
- Consider how Project Outcome tools and resources can supplement existing practices at your institution.
- Consider how to expand upon available tools for more data details and analysis.
- Take careful notes and document your process and rationale behind decisions.
- Be open to unexpected results and note them for future work or publications.

ADDITIONAL RESOURCES

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CHAPTER 6

Small Changes, Big Impact:

Assessing Group Study Rooms Using the Space Survey

Jennifer Arnold, Christy Case, and Retha Hall

INTRODUCTION

Founded in 1963, Central Piedmont Community College (Central Piedmont) is a public two-year community college. Beginning with one campus, it has expanded to six full-service campuses and two centers located throughout the city of Charlotte, North Carolina, and the surrounding areas within Mecklenburg County. The college offers two-year associates in applied science degrees, diplomas and certificates in nearly 300 areas of study, and college and career readiness and continuing education programs focusing on university transfer, the workforce, and personal enrichment. In addition, the college supports Charlotte Mecklenburg Schools' middle colleges on four of its area campuses, with an early college opening on the main Central campus in the fall of 2023 (middle/early colleges are high schools that operate on community college campuses for 11th-13th grade and 9th-13th grade, respectively).

With approximately 25,000 full- and part-time curriculum enrolled students (spring and summer 2023), Central Piedmont is the second largest of the 58 community colleges within the North Carolina Community College System. The student population is racially and ethnically diverse and is home to more than 3,000 international students representing

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152 countries. Nearly 50 percent identify as minority students. Fifty-six percent of the students identify as female and 46 percent as male.

The Central Piedmont Library Services Division is comprised of six libraries, archives, and the academic learning center, and resides within the academic affairs unit of the college. Since 2016, five of the six area campuses have new facilities with each providing student access to computers, study rooms, leisure space, equipment, library instruction, collections that support curricula, and more. The Hagemeyer Library, located in the Parr Center on the college's Central campus, is the newest and largest library at approximately 66,321 square feet.

The new Hagemeyer Library opened in July 2022 as a part of the college's larger Parr Center building. The Parr Center represents the college's largest construction project since 2016, bringing student affairs departments (admissions, registration, advising, student engagement), a student union space with dining, a 450-seat theater, an art gallery, and the library together into one building. As a part of the planning process for the new building, library staff successfully advocated for a significant increase in the number of study rooms based on Project Outcome student surveys conducted by the college's planning and research department, as well as usage statistics collected by library staff. Like most community colleges, Central Piedmont has a commuter student population and limited gathering, social, or study spaces in other campus facilities.

As part of the Parr Center plan, the study rooms were intentionally designed with a variety of furniture types and layouts so that, whatever way students liked to study, there would likely be a room that suited them. For example, some rooms had tables and chairs and a computer monitor, while other rooms had couches or smaller, softer chairs and lower tables; the only commonality in all rooms was a whiteboard on one wall of the study room. Within the library itself there are a total of 23 study rooms, 21 of which are designed with one of the above configurations and two of which are family study rooms that include both adult and child-sized chairs and tables along with additional kid-friendly items (DVD player, books, toys).

The library has been an active user of Project Outcome for Academic Libraries, regularly using the instruction, event, and space surveys. The library also participates in the college's assessment program known as Evidence to Action. After the study rooms were in use for one semester, the library used the Project Outcome space survey for the study rooms as a part of the Evidence to Action assessment plan, considering in particular what changes to the study room configurations could be made to improve student knowledge, confidence, and awareness of library resources.

METHODS

The Project Outcome for Academic Libraries space survey was sent via email in the fall 2022 and spring 2023 semesters to all students who had booked a study room using data pulled from Springshare's LibCal, which we use for group study room bookings. Students

are able to book a room at a service desk, online, or via a mobile app; to facilitate ease of booking, the library has placed QR codes that link to the booking page on the door of each study room. LibCal Spaces also provides pictures of each room so the students can see the furniture configuration prior to booking. The fall 2022 survey evaluated the study rooms in their original furniture configurations, while the spring 2023 survey assessed the study rooms after changes were made to the arrangements of tables and chairs. In the case of both surveys, the emails of all students who had booked a group study room were accessible via LibCal's administrative side; student emails were pulled by the library's senior librarian for assessment and instruction, who was then able to email those students the survey. This survey process was approved by the college's planning and research director, who oversees Evidence to Action.

Additionally, Springshare's LibCal Spaces module provides library staff with data on study room popularity, unique student users, and repeat usage, also via administrative access to the statistics feature of the software. For the fall 2022 Project Outcome survey, a total of 998 students were identified as having used a study room in the new Hagemeyer Library during that semester. We received 109 total responses via a Project Outcome survey for a response rate of 11 percent. For the post-change spring semester survey, we identified 1,499 unique study room users. We received 90 total responses for a response rate of 6 percent. These response rates are typical for surveys sent by email to Central Piedmont students. Often when doing Project Outcome surveys, we put print copies of the survey in each study room; we did not do so for these surveys and that may have increased participation.

We added three additional questions to the standard Project Outcome Space Survey.

1. What did you like most about this space? (open-ended)
2. What could the library do to improve this space? (open-ended)
3. Technology in the study rooms is adequate for my needs. (Likert scale)

The 2022 fall survey (preliminary survey) was sent on December 9, 2022, and the 2023 spring survey (post room changes) was sent on April 4, 2023. As an additional assessment measure for Evidence to Action, the library, along with the college's planning and research department, conducted a focus group in September 2022.

RESULTS

Preliminary survey results (see Figure 6.1) show student satisfaction in the study rooms that is in line with national averages. Open-ended questions revealed a strong appreciation for rooms with tables and chairs and a dissatisfaction with rooms with only low chairs and no table. LibCal Spaces data reaffirmed these results: Study rooms with tables and chairs were most frequently booked, study rooms with couches less frequently booked, and study rooms with low chairs the least-booked of all.

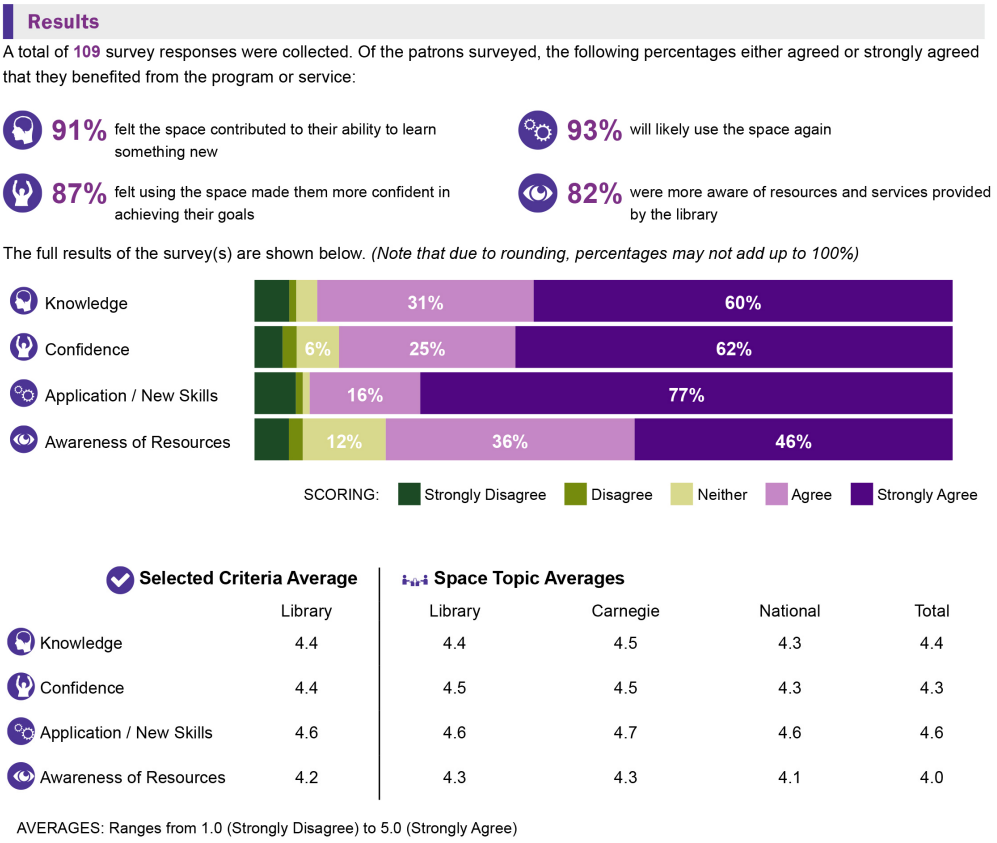


Figure 6.1. Results from the preliminary survey distributed in fall 2022.

As is true a lot of the time, though, it was the qualitative feedback in the open-ended comments that was especially valuable. It provided us with actionable insights into how we could improve students' experiences using the study rooms. For the two open-ended questions (" What did you like most about this space?" and "What could the library do to improve this space?"), we received a total of 161 comments. There were a lot of very positive themes about the group study rooms. Students commonly expressed really loving the new spaces. Our building has glass walls to the exterior, so many of these rooms have great views out to our Central campus quad or to the Charlotte city skyline. They appreciated the privacy and having a space to focus on their work that felt like their own little space within the library.

Notably, however, a significant number of the comments alerted us to an issue with specific furniture configurations. The open-ended responses included the following statements: "More big tables in study rooms instead of the small ones with sofas," "I don't find the blue chair study rooms comfortable. I'm glad all the rooms were not set up like that," "I could have used more table space and more comfortable seating." Table 6.1 illustrates some of the informative comments we received on the surveys.

What did you like most about this space?	What could the library do to improve this space?
It allows a private space where I can dedicate time in between classes to complete coursework without distractions.	Some of the private study rooms are not as comfortable as I would want in that the chair-to-table heights are awkward.
They are spacious and clean and have different designs depending on whether you are studying alone or in groups. The study rooms are great!	In my personal opinion I think some of the rooms could be furnished better. The library spent a lot of money on bougie furniture that is not functional for studying.
The ambiance is peaceful. I have my own area where I don't have to worry about outside influences and distractions.	The chairs could be a lot comfier on average.

Table 6.1. Open-ended question responses.

Using the LibCal Spaces data, we found that the largest study room is the most popular and that demand for rooms suited to larger groups is high. Study rooms that only held one or two people were also unpopular according to booking rankings (see Figure 6.2).

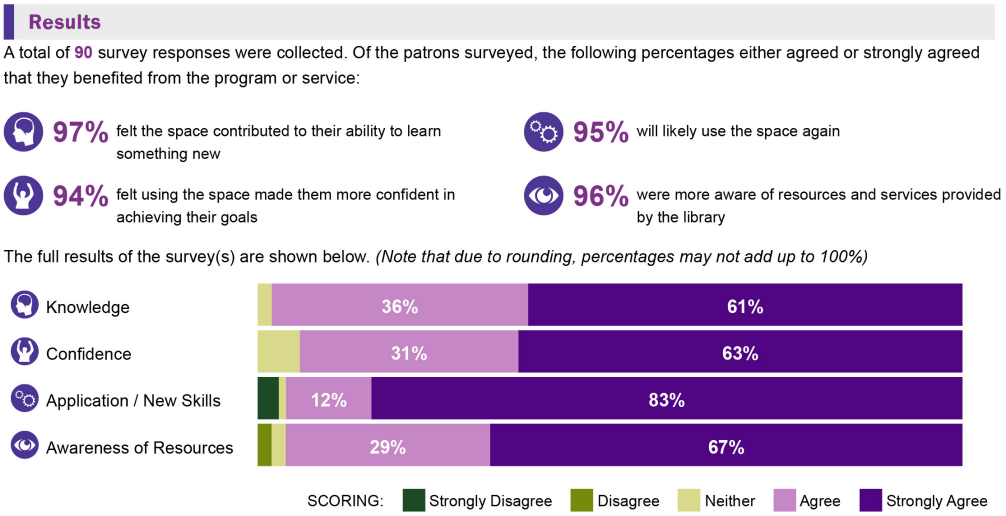


Figure 6.2. Lowly study room 3P, the least popular study room.

TAKING ACTION

The survey results were downloaded into an Excel spreadsheet and the comments regarding the furniture were highlighted. The results were reviewed by various stakeholder groups within the library including the library's assessment committee, campus librarians, and the library's administrative team, which includes the dean of libraries. Based on feedback from these groups, the library's senior librarian for assessment and instruction developed a plan for making changes to the furniture configuration; the plan was cleared with the college's facilities staff to ensure we were not violating any college guidelines.

In March of 2023, over the college's spring break, library staff replaced lower chair configurations—study rooms 2C, 3K, and 3L—with regular tables and chairs from open seating areas of the library. As a result, the popularity of those three rooms increased. For example, study room 2C had been the twelfth most booked room but moved up to fifth after the evidence-based changes. Similarly, room 3K moved up to the thirteenth most booked from twentieth, and study room 3L rose to the eighteenth most booked room from its previous position as twenty-second. Room 3P was repurposed into a specialized virtual reality experience room due to its high availability and open space for moving around.



Selected Criteria Average	Space Topic Averages			
	Library	Library	Carnegie	National	Total
Knowledge	4.6	4.4	4.5	4.3	4.4
Confidence	4.6	4.5	4.5	4.3	4.3
Application / New Skills	4.7	4.6	4.7	4.6	4.6
Awareness of Resources	4.6	4.3	4.3	4.1	4.0

Figure 6.3. Results from the post room changes survey distributed in spring 2023.

As indicated, a follow-up survey after the furniture configurations were adjusted was conducted in April 2023. In that survey, open-ended responses about seating dissatisfaction entirely disappeared. One student pointedly wrote, "There was space for me to spread out my materials. I like sitting at a table." Project Outcome's survey results showed students much more confident about achieving their goals via the study rooms and far more aware of resources and services provided. In this second survey, students ranked the Central Piedmont study room experience above the national average for Spaces (see Figure 6.3).

NEXT STEPS

Perhaps the most important lesson we have learned is that change is possible. Though we had to take the initiative to source tables and chairs from other places and used our own labor, the results were successful enough to be worth the time and effort. Next time, putting paper Project Outcome surveys in the study rooms in addition to emailing known study room users would likely lead to higher student participation. As discussed, we recruited a focus group as a part of the Evidence to Action assessment program via print advertising in study rooms, but that only yielded one student to interview.

For future space survey assessment efforts, the Central Piedmont library would make several changes to the administration of the survey and complimentary assessment efforts. In addition to placing paper copies of the survey in the study rooms, placing signage with a QR code to the survey within the room would likely encourage students to complete the survey while they were in the room. Along the same lines, focus groups would be more effective if the time commitment was shortened.

FINAL TAKEAWAYS

Our use of the space survey highlights the usefulness of Project Outcome surveys as a whole. First, we got some very straightforward, actionable feedback on the furniture in the study rooms. Additionally, the survey really fit in well as part of the college's Evidence to Action plan—it was a natural fit to use this survey to paint part of the picture of how students were reacting to and using our study rooms. Evidence to Action requires the library to come up with an action plan for improvement, which is exactly what the survey gave us. We've found that the Project Outcome pre-programmed questions apply to most of our library operations.

The most useful lessons we learned from this assessment project are:

- A combination of delivery methods for Project Outcome surveys (email, print, QR Codes) will yield higher response rates.
- Mining open ended comments for similar themes is well worth the effort.
- Simple changes like rearranging furniture can have a real impact on student perception of library spaces.

- Project Outcome is a valuable piece of the library's assessment puzzle and fits well with other library and college-wide assessment efforts.

More broadly, we're emboldened to investigate more change possibilities. Students have pressed for larger study spaces; that may be the next task we take on. We can't move walls, but we can find solutions. For other libraries taking on similar assessment projects, we recommend getting into the habit of using Project Outcome surveys for small and large projects alike. At Central Piedmont, we are using the event surveys with all of our library events and are looking into how this data can help shape future library programming. Remember that assessment can be done at the student-facing level as well as the administrative level. Lastly, and perhaps most important, know that even small changes make a real difference in how students use and experience the library.

CHAPTER 7

Revitalizing our Instructional Assessment:

Embracing a New Approach using Project Outcome

Martha Hood and Juliana Espinosa

INTRODUCTION

The University of Houston-Clear Lake (UHCL) is located along the Texas Gulf Coast. Steered by the graduate studies needed by NASA and other nearby aerospace industry employers, the upper division university (no enrolled freshmen or sophomores) opened its doors to the local community in 1974. In 2010 it began serving students in the nearby city of Pearland, which is known as UHCL at Pearland. During that same year, UHCL became a Hispanic-Serving Institution (HSI), with the current population of Hispanic students at approximately 42 percent.[1] In the fall of 2014, the University expanded and began offering freshman and sophomore classes. Today, the University of Houston-Clear Lake is one of the four public institutions in the University of Houston System that grants bachelor's, master's, and doctoral degrees in traditional, online, and hybrid learning environments.

The university has a total enrollment of approximately 11,100 students,[2] with first generation students composing 41 percent of the student body.[3] For fiscal year 2022, 73 percent of the student population were undergraduates and 25 percent were at the

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graduate level.[4] Texas residents make up almost the entire population at 92 percent.[5] Students who attend UHCL are generally part-time (55 percent) and only 2 percent of the student body lives on campus.[6]

UHCL is designated as a Doctoral Universities: Doctoral/Professional Universities by the Carnegie Classification and includes four colleges: business (BUS), education (COE), human sciences and humanities (HSH), and science and engineering (CSE). Most graduate students are enrolled in the CSE programs. Since its founding in 1974, the university has provided classes at various regional instructional sites including Texas Department of Criminal Justice facilities and the Texas Medical Center.

The Alfred R. Neumann Library was named after the university's founding president. It occupies two floors with approximately 80,000 square feet and is integrated into the primary academic, administrative, and student life building on campus. The newly renovated library facilities include a flexible state-of-the-art classroom for library instruction sessions. To maximize active learning, 40 laptops are available and six monitors are strategically placed for better visibility. With over 300 databases, students have access to over one million electronic titles including books, serials, and media. The UHCL at Pearland location also has a smaller library of approximately 900 square feet and is staffed to serve faculty and students at that location.

With a staffing of five librarians under the assistant director for research and instruction (ADRI), Alfred R. Neumann Library historically collected post-instructional data but did not comprehensively analyze it. As part of their new role created in January 2020, the associate director for evaluation and assessment (ADEA) studied previous library instructional statistical analysis reports and noticed that the library needed a deeper analysis of instruction beyond the number of sessions taught each year and the questions answered correctly by students. After creating the Library Assessment Advisory Committee (LAAC) in July 2020, the ADEA shared her recent discovery of the newly launched ACRL Project Outcome for Academic Libraries Toolkit and communicated the need to develop a consistent feedback model used by all instruction librarians, as well as the need to have instructional objectives. The information was shared but implementation was postponed due to the pandemic and the active learning classroom renovation.

As research and instruction librarians prepared to return to a face-to-face instructional format after two years of virtual classes during the pandemic, there was an explicit recognition that the first-year experience course would need to be updated and modified. The newly created student success librarian (SSL) position was filled in February 2022. As part of their responsibilities, this role would join the research and instruction department and oversee Psychology 1100 Learning Frameworks, a first-year experience course. Together with the ADEA, a collaboration began to develop a new assessment framework. In a past role, the student success librarian had used ACRL Project Outcome instruction surveys. When the ADEA learned this, she requested that the SSL give a presentation to the LAAC members which included heads of all departments, including the ADRI. This

unique cross-departmental opportunity was a perfect segue to implementing a case study using ACRL Project Outcome for deeper analysis of the first-year instructional program at Alfred R. Neumann Library.

ACRL Project Outcome's immediate instruction survey was adopted synchronously with the newly revised first-year experience course. In addition to collecting benchmarking data, the survey allowed the library to measure four specific learning criteria which provided a clear vision of areas that needed improvement. These criteria were knowledge, confidence, application/new skills, and awareness of resources. It also provided Alfred R. Neumann Library an avenue to advocate for the value of the library's instructional program with faculty and other stakeholders.

METHODS

Psychology (PSYC) 1100 Learning Frameworks (First-Year Seminar) is a one credit, two-hour course primarily for first-year students, with a designated section for new transfer students. A strong relationship had formed between PSYC 1100 instructors and research and instruction librarians over eight years of continuous instruction collaboration. Instructors are primarily from the Department of Psychology as well as adjacent departments in the College of Human Sciences and Humanities. Every semester, approximately four instructors teach the course from a shared syllabus and lesson plan. An average of eight sections are offered each fall and spring semester. Halfway into a semester, the PSYC 1100 instructors dedicate an entire class period to critical thinking and evaluating information. Instructors bring students to the library classroom where librarians prepare students for a conceptual toolkit assignment by discussing how to evaluate sources, define authority types, and review examples of credentials. The assignment requires students to identify and clarify ten different concepts, such as point of view, civic engagement, and citizenship. The library instructional sessions are intentionally scheduled prior to the assignment due date.

In preparation for fall 2022 semester, the student success librarian worked with a research and instruction librarian teammate who had previously taught the course to gain a better understanding of past assessment goals and strategies. PSYC 1100 had been taught virtually and synchronously due to COVID, but with the return to campus as well as the newly created active learning classroom, the SSL wanted to have a better understanding of this instruction session in order to make any necessary changes. Two requested changes from the instructors were to better incorporate the required common reader book and to make the session more interactive.

The four ACRL Project Outcome learning criteria were the main reason the instruction survey was selected. No specific learning objectives or outcomes had ever been created for the library instruction session nor were any of the PSYC 1100 course objectives incorporated, which made it difficult to analyze past data collected. In selecting

the instruction survey from ACRL Project Outcome, the ADEA and the SSL intended to base the instruction on these four learning criteria. No complementary assessment methods were used and we did not add any custom questions to the survey, since we wanted comparative data from similar Carnegie classified and national universities. Our priority was to establish benchmarks to set a new assessment foundation and to guide the development of future instructional goals.

Time for students to take the survey was built into the lesson plan at the beginning of the redesigned PSYC 1100 library instruction session. This decision was based on the SSL's experience at a prior university library, where electronic instruction surveys had a low response rate for a similarly taught course. Due to this, paper surveys were the preferred option for our instruction program. Since we elected to use paper surveys, we allocated two to three minutes at the end of the library instruction for students to complete them. The instructors were highly supportive of the survey and encouraged their students to fill out the anonymous survey at the end of each session.

After each library instruction session, the SSL collected and reviewed the survey responses to identify any immediate corrections that needed to be made for the following class. The majority of the paper surveys were inputted into ACRL Project Outcome the day of the session by the SSL or the research and instruction librarian in fall 2022 and spring 2023. Surveys were entered using the "enter paper survey responses manually" feature. Open-ended responses were entered using the exact, unedited language as written by the students on the paper surveys.

Throughout the week of library instruction, the SSL reviewed the open-ended responses in order to check whether the concepts were understood by the students. When comments were written about lack of understanding of library resources and services, the SSL made iterative changes for the subsequent classes. For example, the SSL added library resource information on classroom whiteboards to encourage students to use library databases and OneSearch, an online library search engine, during their small group activities.

RESULTS

One of the biggest advantages of using ACRL Project Outcome was that the survey results provided much needed benchmark data for our academic library. In fall 2022, we had an overall 91 percent response rate based on a total of 120 responses from 131 students. In spring 2023 we had an overall 93 percent response rate, with 94 responses from 101 students. On all ACRL Project Outcome surveys, the scoring range is from 1.0 (strongly disagree) to 5.0 (strongly agree) (see Figure 7.1). The scores for the knowledge learning criteria remained the same in both semesters at 4.2. However, the score for application/new skills slightly dropped from 4.3 in the fall to 4.2 in the spring (see Table 7.1). In both semesters, the two lowest scoring learning criteria were confidence, with a score of 4.0 in fall/spring, and awareness of resources, with a score of 4.2 in the fall and 4.1 in the spring.

Although it was discouraging to see scores drop or remain stationary, the intention of implementing ACRL Project Outcome was to set benchmarks in order to establish future instructional improvements.

1. I learned something new that will help me succeed in my classes.

Strongly Disagree | Disagree | Neither | Agree | Strongly Agree | N/A

2. I feel more confident about completing my assignment(s).

Strongly Disagree | Disagree | Neither | Agree | Strongly Agree | N/A

3. I intend to apply what I just learned.

Strongly Disagree | Disagree | Neither | Agree | Strongly Agree | N/A

4. I am more aware of the library's resources and services.

Strongly Disagree | Disagree | Neither | Agree | Strongly Agree | N/A

Figure 7.1. Likert scale questions and results from the immediate instruction survey.

Question	Learning Criteria	Fall 2022	Spring 2023
#1	Knowledge	4.2	4.2
#2	Confidence	4.0	4.0
#3	Application/New Skills	4.3	4.2
#4	Awareness of Resources	4.2	4.1

Table 7.1. Average response scores to immediate instruction survey Likert scale questions.

Additionally, some of the most valuable pieces of data we received were the responses to the two open-ended questions, "What did you like most about this session?," question number five, and "What else could the library do to help you succeed in your classes?," question number 6. We were eager to review this feedback since we spent considerable time redoing the lesson plan. The students' comments to question five were overwhelmingly positive and they repeatedly shared how much they liked how interactive and engaging the session was. One of the benefits of the toolkit is a word cloud option for qualitative analysis. When we generated a word cloud on the ACRL Project Outcome data dashboard for both fall 2022 (see Figure 7.2) and spring 2023, the top words were: activities/activity, liked, engaging, authority, and learning. For the question asking what else the library can do to help, we generated a second word cloud for the top responses. The top words were more, information, resources, library, and help.

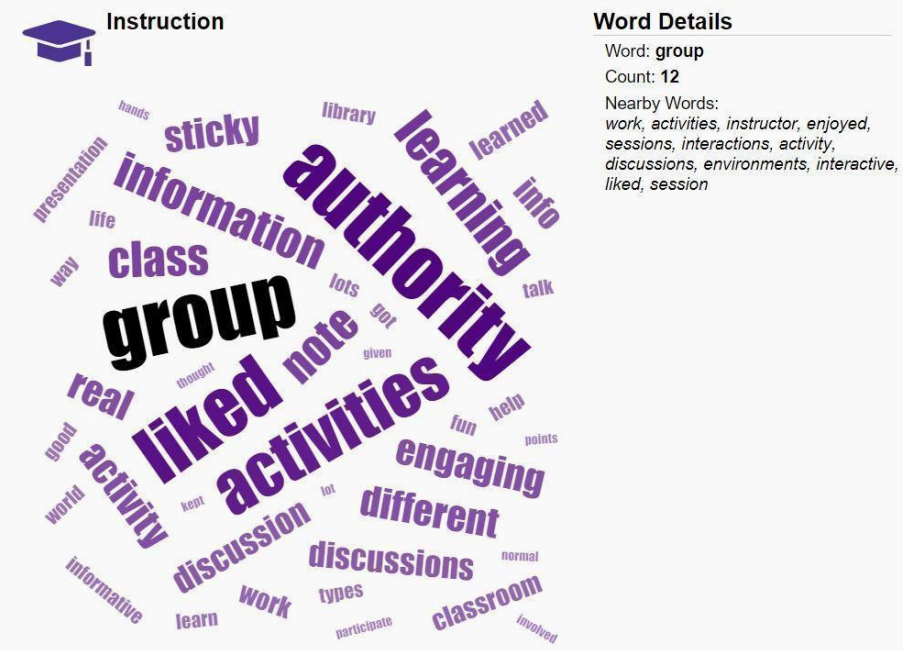


Figure 7.2. Word cloud based on fall 2022 survey answers to "What did you like most about this session?"

In fall 2022 we had a 93 percent response rate to question five, which rose to 95.7 percent the following spring. For question six, the response rate was 77.5 percent for the fall semester. The response rate to this question increased the next semester, as well, to 81.9 percent (see Table 7.2). For question six there were a number of responses of "nothing" or "N/A" (not applicable). Students seem to either be satisfied with the library's current services or unaware of additional resources that could be provided (see Figures 7.3 and 7.4). While the open-ended questions are beneficial in capturing the imagination of students, there is a chance that the response rate could be higher if students choose from a list of options instead.

Semester	Question	# of survey responses	Total responses	Attendance	Question Response Rate
Fall 2022	Q5	112	120	131	93%
Fall 2022	Q6	93	120	131	77.5%
Spring 2023	Q5	90	94	101	95.7%
Spring 2023	Q6	77	94	101	81.9%

Table 7.2. Total number of responses and response rate for questions five and six in 2022 and 2023.

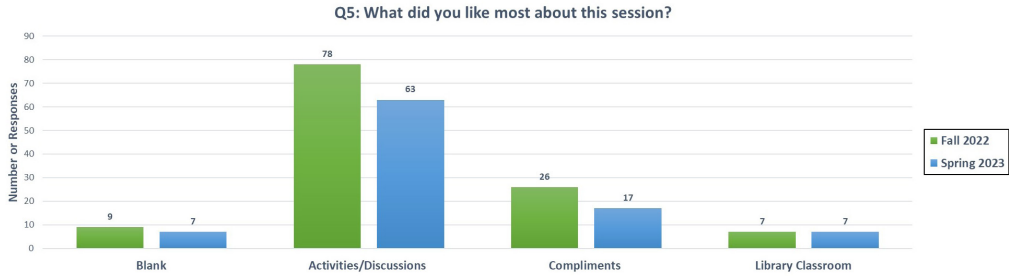


Figure 7.3. Analysis of responses to immediate instruction survey question five.

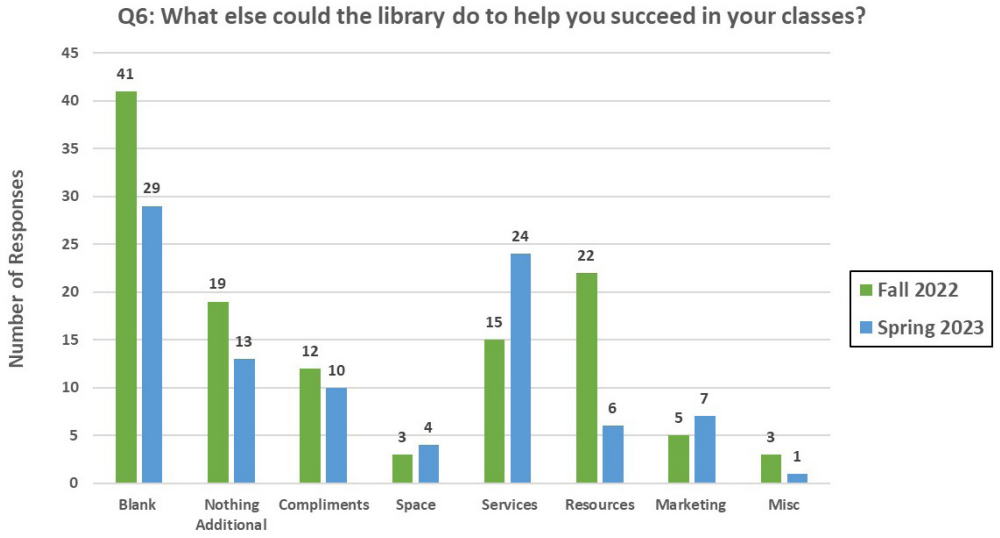


Figure 7.4. Analysis of responses to immediate instruction survey question six.

In 2022, the University of Houston-Clear Lake's Carnegie classification changed from the Master's Colleges and Universities category to the Doctoral/Professional Universities category. The timing of our adoption of ACRL Project Outcome was serendipitous (see Figure 7.5), as it allowed us to begin data comparison at our new Carnegie classification and national levels. When analyzing our data, the overall learning criteria scores were lower than the Carnegie classification and national averages (see Figure 7.6). This provided us with a new assessment goal for our PSYC 1100 instruction program: to make corrective changes.

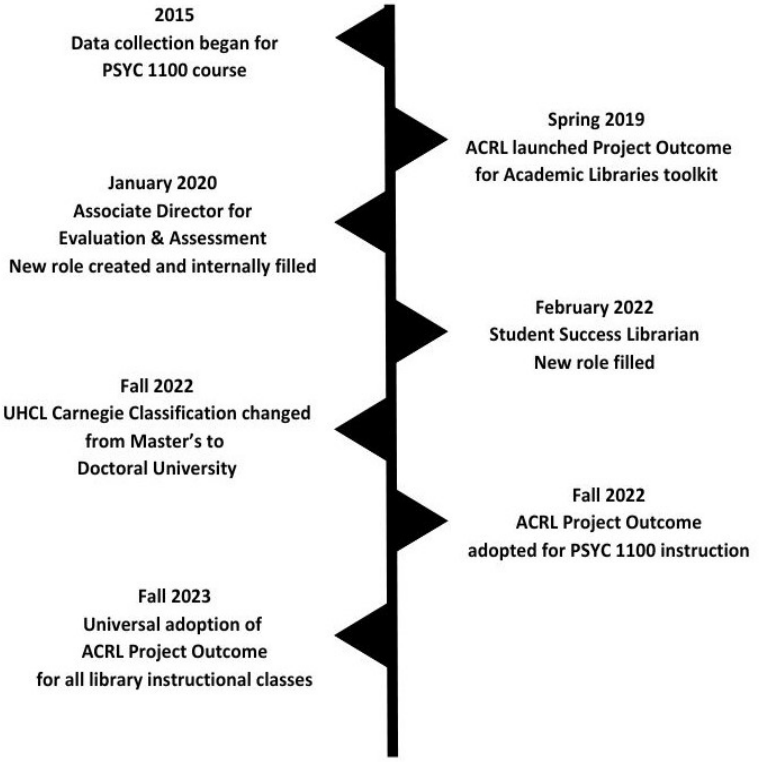


Figure 7.5. Implementation timeline of ACRL Project Outcome at UHCL.

	Selected Criteria Average	Instruction Topic Averages			
	Library	Library	Carnegie	National	Total
Knowledge	4.2	4.2	4.4	4.4	4.4
Confidence	4.0	4.0	4.2	4.2	4.2
Application / New Skills	4.2	4.3	4.4	4.4	4.4
Awareness of Resources	4.1	4.2	4.4	4.4	4.4

AVERAGES: Ranges from 1.0 (Strongly Disagree) to 5.0 (Strongly Agree)

Figure 7.6. Average learning criteria scores for library, national, and Carnegie categories.

TAKING ACTION

In order to collect instructor feedback and thoughts on the librarian-instructed classes, a Qualtrics follow-up survey link was created and sent to instructors via email. The ADEA and SSL deliberated using the ACRL Project Outcome teaching support survey; however, there was a preference to tailor our questions specifically to the newly integrated classroom activities. The email included a wrap-up message on the instruction session, attached the ACRL Project Outcome quick report with open-ended responses, and encouraged instructors to share their feedback in the provided Qualtrics survey link. We asked six questions: three open-ended, two Likert scale, and one multiple choice question about the content. Three of four faculty members who received the survey provided feedback. In order to obtain additional feedback, the SSL also met with the director of the first-year seminar program.

The data provided by faculty on the Qualtrics feedback form was predominantly positive, so drastic modification of the sessions was not needed. However, one question, "What is one thing that you would change about the library instruction session (e.g. more activities, vocabulary too challenging, etc.)?," provided helpful feedback which led to some small adjustments. One modification was to allocate additional time at the end of instruction to review the conceptual toolkit assignment. A second modification made was to define and review basic information literacy terminology.

Along with the ability to provide student feedback to faculty, ACRL Project Outcome allows librarians to create a customized report with statistical data and colorful visualizations to share with relevant stakeholders. Alfred R. Neumann Library found that sharing the report with faculty was an excellent way to validate instructional time with librarians and provided evidence of measurable growth. It illustrated our mission of contributing to student success by providing them lifelong learning tools and improving their information literacy. Faculty responded with not only appreciation for the report, but also saw its value as a preparation tool for their upcoming classes. This was a great opportunity to increase communication and deepen the relationship between librarian and instructor.

In addition, the ADEA shared the report findings with the Library Assessment Advisory Committee (LAAC), who had been introduced to the toolkit earlier in the year. Committee members and library administration were impressed with the report and the ability to share with various stakeholders. The report demonstrated the potential of using Project Outcome to measure and visualize the impact of other programs and services beyond instruction.

Now that the pilot program had provided such impactful data for the PSYC 1100 Learning Frameworks class, the SSL and ADEA shared with the research and instruction team the need to develop learning objectives and adopt the use of Project Outcome across the department. Librarians saw that using Project Outcome's instructional survey could assist them both on an individual level and team level. On an individual level, it

would allow research and instruction librarians to conduct a self-assessment of their own teaching abilities. On the team level, the survey would help align library instruction with broader institutional goals related to student success, retention, and learning outcomes.

NEXT STEPS

Much of the work accomplished over 2022 and 2023 set a foundation in instructional assessment. We learned that years of data collection with some quantitative feedback did not help our research and instruction team measure their teaching effectiveness. More important, we had no way of knowing if our methods were impactful and therefore, whether students were being properly supported. As a result, we learned the importance of setting benchmarks to determine how to best proceed. The data reports we created from ACRL Project Outcome showed that there are multiple opportunities for growth.

In preparation for the 2023-2024 academic year, the director of first-year seminar program and the SSL debriefed the 2022-2023 academic year sessions and discussed instructional needs. The SSL informed the PSYC 1100 instructors that librarians will prioritize two learning criteria, "Knowledge" and "Confidence," in future sessions. Also new for the 2023-2024 academic year, the student learning objectives were adjusted to incorporate the learning criteria language as well as the course's learning objectives. As a way to further measure these, the SSL proposed using Project Outcome's follow-up instruction survey at the end of the spring 2024 semester to check for any significant growth between the immediate and follow-up surveys for PSYC 1100.

Additionally, we will update our original lesson plan to add resources, edit talking points, and ensure the group activities help students to build their confidence and awareness of library resources. All research and instruction librarians have been trained to teach PSYC 1100 as well as how to use ACRL Project Outcome. Moving forward, the Research and Instruction department will work collaboratively with the ADEA to comprehensively assess all library instruction. One intended outcome is to generate individual self-assessment for each librarian in addition to a teamwide data report.

Lastly, in the upcoming academic years, Alfred R. Neumann Library will develop a new long-term strategic plan as well as create Research and Instruction department goals to align with the University of Houston-Clear Lake's newly created strategic plan.

FINAL TAKEAWAYS

The UHCL Alfred R. Neumann Library has made tremendous strides in assessment as well as transparency in many areas. Here are a few takeaways which were instrumental in improving our library and aligning our goals with the university's values.

- Implementing the ACRL Project Outcome toolkit changed the library's assessment approach to our largest instructional program. This led to standardization and consistency across the research and instruction department. As a result, the library established new learning criteria to better support first-year and new transfer students, many of whom identify as first generation and Hispanic/Latino.
- Comparing learning criteria scores against the national and Carnegie classification categories gave the library benchmark data to strive for improving our overall scores as a newly classified Doctoral/Professional Universities institution.
- Collaborating between the Research and Instruction department, the Evaluation and Assessment department, and the first-year seminar program enhanced the quality of information literacy instruction and assessment. Librarians demonstrably contributed to the development of students' critical thinking abilities with the intention of improving their overall academic success.
- Enhancing the value of library instruction in PSYC 1100 through the creation of ACRL Project Outcome data reports. Librarians can now share evidence of how library instruction directly impacts student learning and success. In addition, the open-ended responses allowed librarians to modify their subsequent instruction for improvement.

NOTES

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2. "State and Federal Data," State and Federal Data, Microsoft Edge, last modified March 8. 2024, https://www.uhcl.edu/about/administrative-offices/institutional-research/state-federal-data.
3. "First Generation Demographics," IR Data Portal, data visualization by University of Houston-Clear Lake, accessed June 15, 2023, https://apps.uhcl.edu/OIE/DataPortal/Reports/FirstGen?rptType=PR&rptCat=frstgen&rptName=FirstGen&rptDesc=First%20Generation.
4. "Enrollment Demographics – (Program Review)," IR Data Portal, data visualization by University of Houston-Clear Lake, accessed June 15, 2023, https://apps.uhcl.edu/OIE/DataPortal/Reports/OIE_Reports?rptType=PR&rptCat=enr&rptName=Enrollment_PR&rptDesc=Enrollment%20Demographics.
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CHAPTER 8

Establishing Baseline Data for Library Instruction Improvement

Laura D'Aveta

INTRODUCTION

Hiram College is a small, private, not-for-profit college located in northeastern Ohio. Our Carnegie designation is "Baccalaureate College", and we belong to the Great Lakes region of the Association of College & Research Libraries (ACRL). Hiram College emphasizes experiential learning opportunities, boasting a 95 percent post-graduation job placement rate. With a student-to-faculty ratio of 10:1, the college offers 33 undergraduate majors, 37 minors, pre-professional concentrations, and dual-degree opportunities.

In January 2022, the library recognized the importance of data-driven decision-making and implemented the Project Outcome instruction survey to establish a baseline of performance data related to library instruction. Before this initiative, the college had not collected any data on library instruction, making the survey crucial for assessing its effectiveness. The library needed baseline performance data so we could determine if there was an improvement in student performance following an administrative decision that made library instruction a mandatory part of the college's first year seminar courses.

METHODS

The library used an unmodified version of the Project Outcome instruction survey. For most of our library instruction sessions, the survey was printed out on a half-sheet of

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paper and distributed to students at the end of the session; in a few instances, a link to the survey was sent electronically to faculty after class for distribution to the students. We found that participation increased when the survey was distributed on paper, so that became our preferred method of administering the survey.

In advance of the 2022 school year, the library director and the instruction librarian implemented a traditional one-shot session as part of the Urgent Challenges Seminar (UCS) sequence that students take during their time at Hiram College. The UCS sequence consists of three courses: UCS 10101: Enduring Questions and UCS 20201: Addressing Urgent Questions, both of which are taken during the student's first year; and UCS 30301: Urgent Challenges Seminar, typically taken during a student's junior year. The theme of each course is determined by the faculty member teaching it, with the sequence being designed to develop the ability to write well, think critically, and engage with and evaluate resources in their research projects throughout their time at Hiram College and beyond. The library hoped that providing instruction to students two times during their first year—as part of UCS 10101 and UCS 20201—would provide a foundation of research skills that would support the goals of the Urgent Challenges Seminar sequence. The library instruction session during UCS 10101, which averaged 10-11 sections each fall semester with a cap of 25 seats per section, focused on the resources available through the library and basic research skills. The UCS 20201 library instruction sessions, offered each spring semester, included a brief review of the skills taught during UCS 10101, and then delved into more advanced research techniques, Boolean logic, and citation mining.

In addition to the library instruction sessions conducted during UCS courses, the library conducted sessions on a by-request basis for non-UCS courses. These instruction sessions were customized to address the unique requirements of assignments in each course. The library specifically reached out to the faculty of courses that had a major writing or capstone assignment to suggest a review of library research skills for their students.

The library implemented the Project Outcome survey in January of 2022; a total of three semesters' worth of data—spring 2022, fall 2022, and spring 2023—has been gathered at the time of this writing. At the end of the spring 2023 semester, we created a survey similar to the Project Outcome instruction survey for faculty who had library instruction as part of their course. We were interested in whether faculty were observing any improvement in student performance on research-related assignments when they had attended one or more library instruction sessions. Using the same Likert scale and open-ended question format as the student survey, we asked faculty the following questions:

- I noticed an improvement in the quality of sources students used for their assignments.
- I received fewer questions regarding finding sources for assignments.
- I noticed a difference in the quality of work between students who attended the session and students who missed the session.
- What did you and/or your students find most helpful about the session?

- What could we do to improve the session?
- Is there anything else you would like us to know?

The survey was printed on a half-sheet of paper and distributed to faculty at an all-faculty meeting at the end of spring 2023.

RESULTS

During spring 2022—the first semester we began administering the Project Outcome instruction survey—we distributed the surveys during seven separate instruction sessions and received a total of 63 responses. As previously stated, this first round of surveys was intended to get a baseline measurement of the impact our services had on student performance and confidence. Comparative analysis with peer institutions and the national average positioned Hiram College slightly above average, reinforcing its commitment to enhancing student learning experiences.

The responses were incredibly positive. When compared with our Carnegie peers we were slightly higher than average, and we were closely in line with the national average (see Figure 8.1). One area identified for improvement was student confidence. Although the results indicated positive responses, the library acknowledged that some students might still lack confidence in their research abilities. To address this, the library focused on

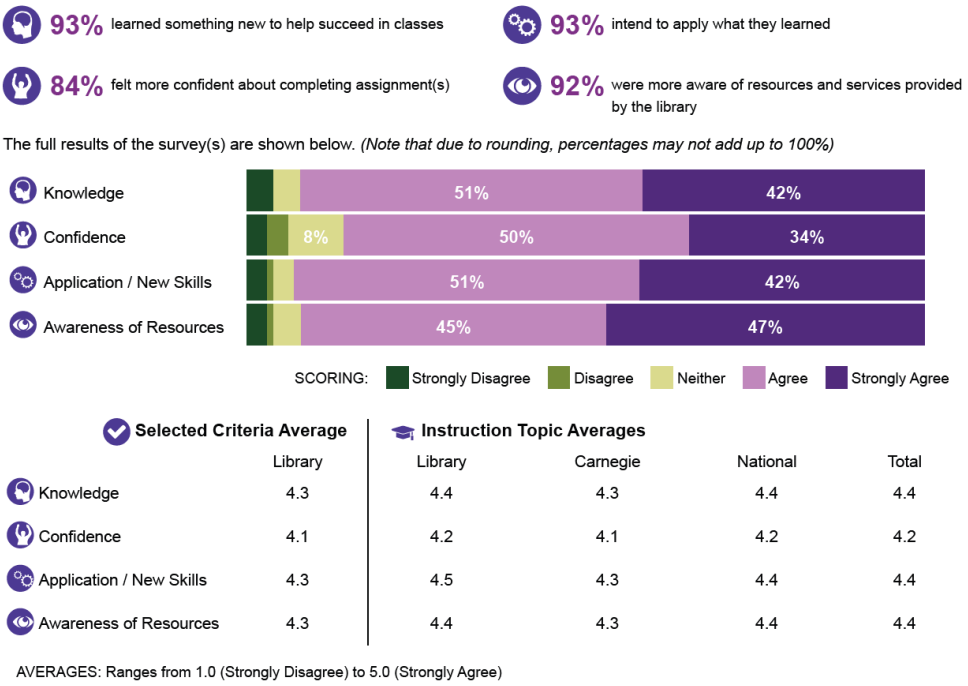


Figure 8.1. Spring 2022 instruction survey results.

designing instruction sessions that fostered a supportive and approachable environment, encouraging students to seek help when needed.

Prior to the 2022-2023 school year, library instruction was mandatory only for the UCS 20201 course. While many UCS 10101 instructors invited librarians for instruction, it was not consistent across the campus. Librarians believed that library instruction for UCS 20201 students would be most effective if we knew that students had received core information in their UCS 10101 class and could build on it. In the past, a significant amount of repetition was required to ensure that all students in the UCS 20201 course had the same knowledge of library resources as those who had an instruction session in UCS 10101. Thanks in part to positive responses from faculty and students about our library instruction sessions, conversations with the director of writing across the curriculum and the associate dean of academic affairs about this dilemma were well received. The library director asked to trial inclusion of a library instruction session requirement for both UCS 10101 and UCS 20201 for the 2022-2023 school year. Thanks to Project Outcome, we then had data to use that demonstrated the value of this new sequence at the end of the year.

Fall semester of 2022 saw a significant increase in the number of responses to the instruction survey. The library hosted 18 instruction sessions in fall 2022; 11 of them were UCS courses. This represented an increase in the number of non-UCS courses as well; there were only three library instruction sessions offered in spring 2022. A total of 226 student survey responses were collected in fall 2022; the results are shown in Figure 8.2.

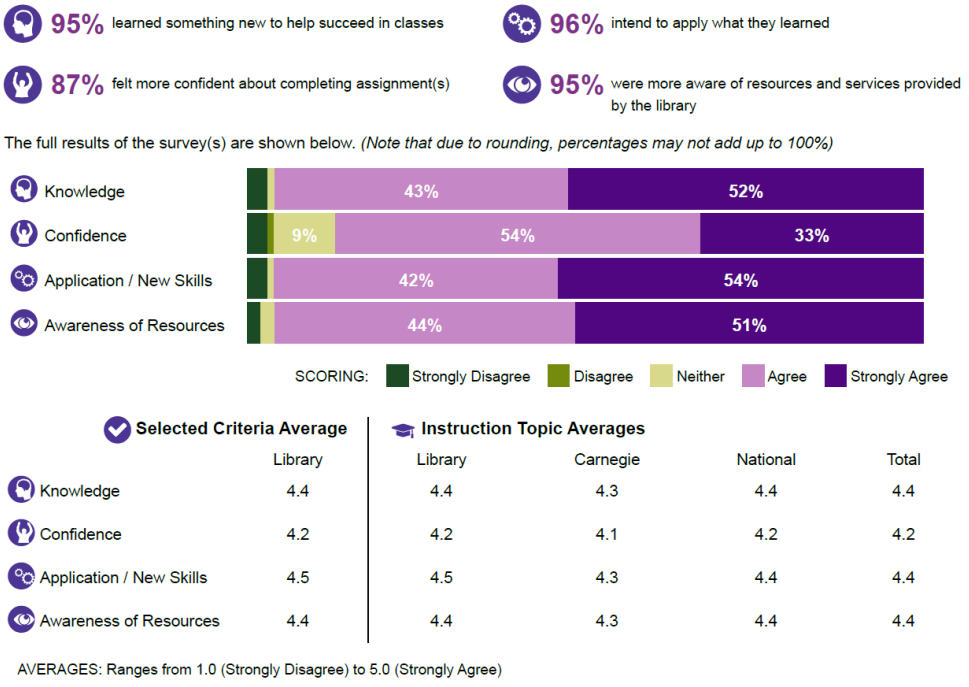


Figure 8.2. Fall 2022 instruction survey results.

The library experienced improvement in all its overall percentages on the Project Outcome instruction survey from spring 2022 to fall 2022, with the greatest improvement being in the awareness of resources category. These improvements might have been a function of the increased number of surveys collected; the surveys are anonymous, so there is no way to know if there were changes in a particular student's responses. The library did not intend to use the data at such a granular level; we were looking for evidence of overall improvement and treated the data as high-level and impressionistic for the purpose of advocating for library involvement in academic programming.

The natural rhythm of Hiram College's curriculum results in fewer library instruction sessions being offered during spring semesters because there are fewer sections in the UCS sequence being offered. Nine of 14 library instruction sessions offered in spring 2023 were courses in the UCS sequence. A total of 153 survey responses were collected; the results are shown in Figure 8.3 below.

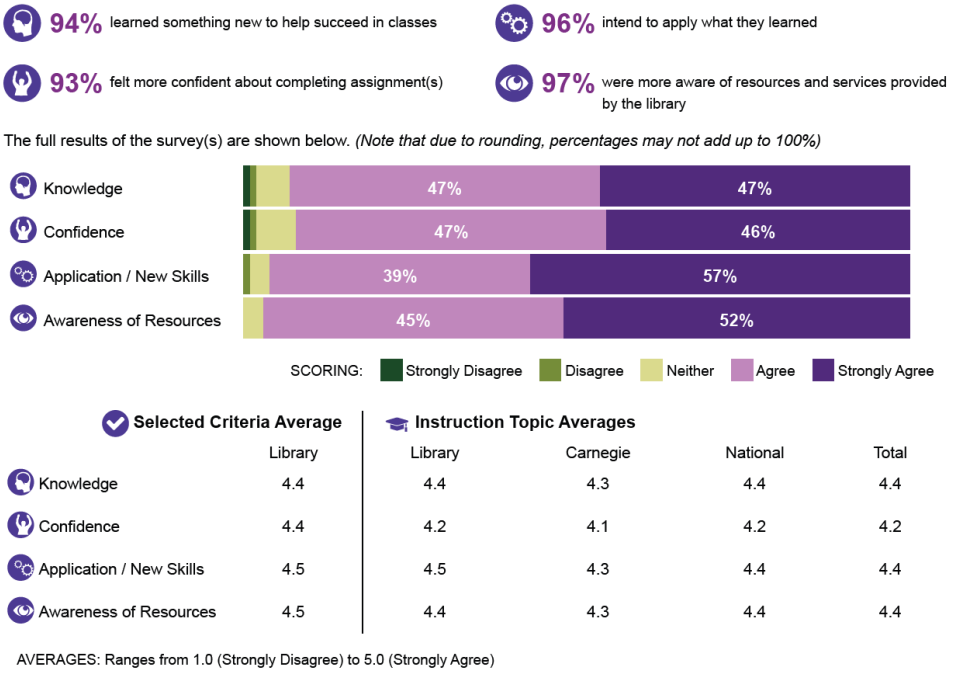


Figure 8.3. Spring 2023 instruction survey results.

The spring 2023 data showed a 3 percent decrease in knowledge but an improvement in the other three areas, including a 5 percent increase in student confidence. The library was consistently performing on par with or above both our Carnegie and national peers.

In addition to the four Likert Scale questions referenced in Figures 8.1-8.3 above, the Project Outcome instruction survey includes two open-ended questions: "What do you

like most about this session?" and "What else could the library do to help you succeed in your classes?" Responses to those questions varied wildly, with students often using the fields in a humorous attempt to ask for video game systems or fixes to the building's troubled HVAC system. There were several helpful themes that emerged from these responses, however. Students overwhelmingly appreciated practicing searching using their own assignments/topics, and they enjoyed realizing that there was a more efficient way to conduct their research. Even students who had previously had library instruction appreciated the review of library skills. A surprise was the popularity of our distribution of crayons and a library coloring page. Our main goal for these library instructions is to make sure students know that we are a friendly source of help, and we are successful in that area. Regarding ways the library could improve, most comments related to increasing library hours, providing additional study spaces, and additional modes of instruction (YouTube videos, additional workshops, tutoring).

The results of the faculty survey were also positive. Of the 19 responses received, 14 observed an improvement in the sources students used on their assignments; 10 observed a decrease in the number of questions asked by students; and 8 believed the quality of student work was higher when students attended the instruction session. One faculty member noted that the questions they received from students were more sophisticated than in the past. Faculty found customizing the instruction sessions to their specific courses/assignments and hands-on research practice to be the most helpful aspect of the instruction sessions, and the main recommendation for improvement was to spend more time on hands-on research practice.

TAKING ACTION

The Project Outcome instruction survey data provided the library with the foundation for data-driven decision-making. Armed with concrete evidence of the positive impact of library instruction, the college's administration found it easier to justify budget requests aimed at maintaining library services, allowing us to continue to provide high-quality instruction and support to students.

The positive feedback from students has led to making library instruction a mandatory part of both UCS 10101 and UCS 20201, ensuring that every student has at least two instruction sessions in their first year at Hiram College. We have also experienced an increase in library instruction session requests for non-UCS courses. The benefit of knowing each student has already had some exposure to library instruction via the UCS course sequence is that we can spend most of our time in other sessions on assignment-specific hands-on research practice, with just a brief refresher of what was covered in the UCS sessions. In addition, we have begun receiving requests from students for individual reference appointments, and we have started manually collecting data to track that metric going forward.

The faculty's positive feedback on the effectiveness of library instruction also strengthened the library's relationship with faculty members. Faculty perceived the library as a partner in student success rather than merely a support service. We have developed a library instruction menu that offers faculty a variety of ways to partner with the library to incorporate library instruction into their courses, including an embedded librarian model and a multi-session library instruction approach that staggers shorter sessions throughout the course's schedule to coincide with assignment benchmarks. Feedback from the faculty survey confirmed our suspicion that more than one approach to library instruction was needed to serve both students and faculty. The library's commitment to tailoring instruction to faculty-specific assignments and course goals further solidified this partnership, fostering a collaborative and supportive environment.

NEXT STEPS

The success of tailoring each library instruction session to the course and/or assignment, in combination with timing the session to occur close to the start of an assignment, has provided us with reassurance that while not ideal, one-shot library instruction sessions can be effective. When resources are limited, and the basics are all you can offer, make the most of that time and make sure that students know where to go for help—and that you are willing and able to provide it. While we would not change anything we have done to date, we would encourage faculty to involve the library more in their non-UCS courses so that library skills can be reinforced and built upon throughout each student's career.

FINAL TAKEAWAYS

The benefits of incorporating Project Outcome into our operational procedures far outweigh the limited investment of time needed to collect and analyze the data. Those benefits include:

- improved student performance on research projects.
- increased student confidence in research abilities.
- increased library usage.
- increased visibility and involvement of library staff.
- continued support for the library from administration and faculty.

One of the most important things you can do for your library is begin using Project Outcome. Hiram College's dedication to improving library instruction, combined with its commitment to faculty collaboration and data-driven decision-making, has resulted in enhanced student learning experiences. The success of the Project Outcome instruction survey has highlighted the importance of continuous assessment and adaptation to meet the evolving needs of students and faculty. Data is power; knowing whether your

current efforts are effective can help you to strengthen any requests to administration to increase—or even just maintain—your budget. Showing student feedback to faculty can help you convince them to share their limited class time with you and to see you as a partner, rather than a service provider. More than anything else, Project Outcome has helped us identify ways to help students succeed.

CHAPTER 9

Using the Instruction Follow-Up Survey to Advance Library Integration in the First-Year Experience Program

Courtney Drysdale and Paul Betty

INTRODUCTION

Regis University is a private, Jesuit Catholic, four-year doctoral professional university located in Denver, Colorado, with about 6,300 students. The Dayton Memorial Library at Regis University serves three colleges that offer degree programs, academic certificates, and areas of study in-person, online, and in a hybrid model.

The library's Research & Instruction (R&I) department actively solicits opportunities to deliver library and information literacy instruction across all three colleges. The largest concentrated instruction effort by the R&I department is with the first-year experience (FYE) program in Regis College, the traditional and primarily campus-based liberal arts college within the university. All incoming students to the college are required to participate in the FYE program, including enrollment and completion of RCC 200 – First Year Writing.

RCC 200 is a seminar-style course with each section covering a different topic or theme as selected by the instructor. Each fall semester, there are typically between 25 and 30 sections

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of RCC 200 offered. While all sections of RCC 200 share common learning objectives, the syllabi, required readings, and research assignments are highly varied between the different sections. Ideally, each section is paired with a librarian to provide library orientation and instruction; however, historically, not all instructors have chosen to engage with the library.

When approaching individual RCC 200 instructors about including library instruction in their section of the course, R&I librarians often discuss with faculty that the skills and experience gained from library instruction in the first year are foundational to student success and retention. The department believes students who do not receive library instruction in the first year are at greater risk of unsuccessful academic outcomes, including dropping out of school entirely. Anecdotal feedback from RCC 200 instructors via email or informal discussion has always indicated that students performed better on their assignments and in the course overall after receiving library instruction. Similar positive outcomes resulting from integrated library instruction in college courses have been documented in the literature, both generally[1] and more specifically within first-year experience programs.[2] The library believed—and research supports—that interaction with the library at the beginning of the college experience supports student success, and that failure to include library instruction across all sections of RCC 200 could result in educational inequity that inadvertently penalizes some students based solely on their section of RCC 200.

To build a stronger, evidence-based argument in favor of mandatory library instruction for all sections of the RCC 200 course, the R&I department required additional evidence and turned to surveying students and faculty directly. We have been conducting assessment and research on library instruction for six years by using student and faculty surveys. For the first three years, we created our own surveys and tweaked them each year. We enjoyed creating our own survey instruments but found that the inconsistency due to changing questions each year made it difficult to compare student cohorts and was a large amount of work. We needed a proven tool that we could use year after year to assess our instruction efforts.

After looking at the survey options in Project Outcome, we chose the instruction follow-up survey because it aligned with some of our past practice and would help us assess what we wanted to know: Did students' skills transfer to other classes after the immediate need in RCC 200?

METHODS

Since 2020, the R&I department has used the Project Outcome library instruction follow-up survey each year to gather feedback from first-year experience (FYE) students enrolled in RCC 200. We added a sixth question to the standard five question survey to identify the instructor and section in which the respondent was enrolled. This question was open-ended for the first two years of using the survey and students filled in their instructor's name. We switched it to a multiple-choice question with all the RCC 200

instructors listed for the fall 2022 survey. Using a selectable option instead of an open-ended response allows the R&I librarians to sort and see survey results from the sections they taught. This also lets us filter the responses by whether that section received library instruction and if they had more than one library instruction session.

After the completion of the RCC 200 course in the fall, emails with a survey link are sent to every FYE student in February or March of the spring semester from the general R&I email and signed by all of the research and instruction librarians. To encourage survey completion, we offer a chance to win a $25 Visa gift card to two randomly selected respondents. Respondents who want to be included in the drawing for the gift card are asked to provide their contact information using a separate Google form or Qualtrics survey that's linked automatically as the "thank you page" redirect in Project Outcome.

At least two research and instruction librarians review student responses to the two standard open-ended survey questions—"What did you find most helpful from this session?" and "What could the library do to help you learn more?"—to look for themes. We use an inductive coding process, meaning that we read through the responses before deciding on codes/themes, then go back through and assign codes to each response. For the first two years, two librarians coded the data together and discussed any disagreements. For the fall 2022 survey, the entire Research and Instruction department read the responses and developed the codes, then one librarian applied the codes to the responses and reported the results back to the department. These qualitative coding skills have developed within the department over time and are now being taught to others in the library through the Library Assessment Committee.

In addition to the Project Outcome survey, we send a separate Qualtrics survey in the spring semester to the RCC 200 instructors for feedback on both the instruction sessions and how their students did on any research assignments. Each year between eight and 11 instructors respond to the survey, which is about a third of the total sections. The anonymous, open-ended questions for this survey are:

- What was the most useful skill/information you think your students learned in this library instruction session?
- Are there any changes you could suggest that might have improved the library instruction session?
- In your opinion, what did students do well regarding research?
- In your opinion, what did students struggle with while doing research?
- Anything else you'd like to say about RCC 200 library instruction?

RESULTS

We now have three years of data from first-year students, and we plan to continue using Project Outcome for our assessment of this population. Based on enrollment information,

we have had between 450 and 500 first-year students the last three years and received 90 responses in 2020, 81 in 2021, and 76 in 2022. This amounts to between 15 and 20 percent of the student population. It has been interesting to look at the differences in responses over the three years through multiple contexts. These students have had very different first-year experiences due to the COVID-19 pandemic, shifting delivery of classroom instruction, integration of the library's subject and course research guides in the learning management system during spring 2020, and the launch of our first discovery search tool in December 2020.

As a department, we have tended to use the quantitative data for overall advocacy and promotion of the library instruction program, as well as a way to look at how we are doing as a group over time with different student cohorts. The open-ended qualitative data has been more informative to our individual instruction efforts and determining bigger picture department efforts around new offerings.

A helpful feature of Project Outcome is that you can combine data from multiple years to look at them together. For the quantitative data we look at each group of students each year, but in Figure 9.1 we have collected the three years together.

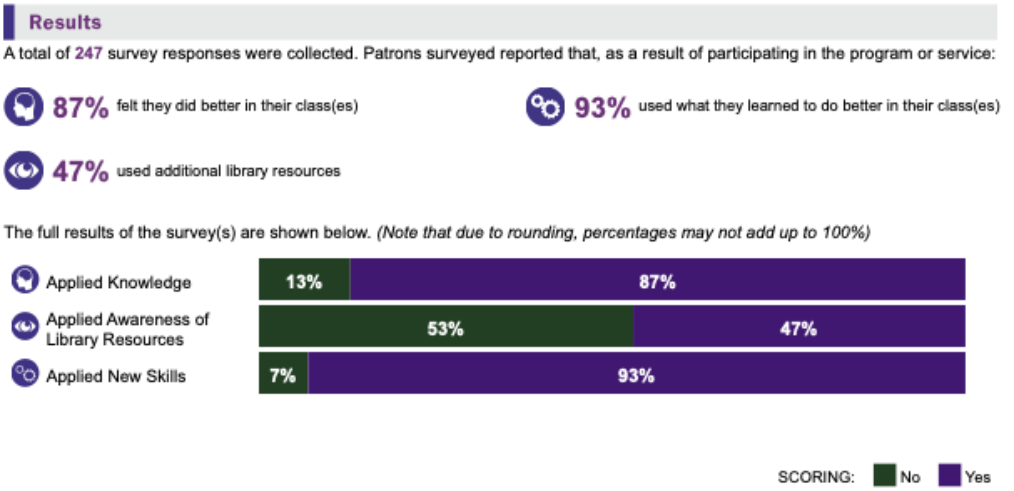


Figure 9.1. Project Outcome instruction follow-up survey results, 2020 through 2022.

Overall, the three-year results are positive, with 215 students (87 percent) reporting that they felt they did better in their classes and 230 students (93 percent) reporting that they used what they learned to do better in their classes. Since this survey is conducted just a couple of months after they finish their RCC 200 course, we hope that students will have more opportunities to use library resources in the future. However, less than half of the

students surveyed (47 percent) reported engaging with additional library resources. Looking at these percentages over time can help us know how our instruction is impacting students.

Tables 9.1, 9.2, and 9.3 describe the final codes/themes of the open-ended questions and resulting qualitative data, including the number of times each theme was mentioned each year. The codes are different for each year since they come directly from that year's responses and may or may not be the same for different groups of students.

Fall 2020						
What did you find most helpful from this session? N = 85				What could the library do to help you learn more? N = 81		
Code/Theme	# of Responses	% of Responses		Code/Theme	# of Responses	% of Responses
Navigating Website	21	24.71%		Nothing	30	37.04%
Finding sources	20	23.53%		Other	15	18.52%
Databases	18	21.18%		Tutorials	11	13.58%
Other	14	16.47%		More about website	10	12.35%
Learn about services	9	10.59%		Handouts	4	4.94%
Search strategy	9	10.59%		Physical materials	4	4.94%
Step by Step walkthrough	6	7.06%		Workshops	4	4.94%
Citation	4	4.71%		Physical space	3	3.70%
Didn't use it/didn't have one	3	3.53%		Search Strategy	1	1.23%
Reliable sources	3	3.53%				
Practice	2	2.35%				

Table 9.1. Final themes from the open-ended instruction follow-up survey questions, fall 2020.

Fall 2021						
What did you find most helpful from this session? N = 70				What could the library do to help you learn more? N = 68		
Code/Theme	# of Responses	% of Responses		Code/Theme	# of Responses	% of Responses
Searching	41	58.57%		Nothing or N/A	24	35.29%
General resources	11	15.71%		Finding books	11	16.18%
Website	11	15.71%		Other	8	11.76%
Other	7	10.00%		Searching related	7	10.29%
Citations	6	8.57%		More outreach/ resources	5	7.35%
				Website related	4	5.88%
				Other Regis services	4	5.88%
				Make it more engaging	3	4.41%
				Technology related	2	2.94%

Table 9.2. Final themes from the open-ended instruction follow-up survey questions, fall 2021.

Fall 2022						
What did you find most helpful from this session? N = 75				What could the library do to help you learn more? N = 76		
Code/Theme	# of Responses	% of Responses		Code/Theme	# of Responses	% of Responses
Website/Databases	27	36.00%		Nothing or N/A	29	38.16%
Finding things	15	20.00%		Workshops/Tutorials	16	21.05%
Nothing or N/A	9	12.00%		Warm Fuzzies	8	10.53%
Other	8	10.67%		Other	5	6.58%
Step by Step	6	8.00%		Printing	5	6.58%
Librarians/Research Help	6	8.00%		Books	4	5.26%
Quality Sources	5	6.67%		Outreach/ Awareness	4	5.26%
Services	4	5.33%		Website	4	5.26%
Keywords	3	4.00%				
Handouts	3	4.00%				

Table 9.3. Final themes from the open-ended instruction follow-up survey questions, fall 2022.

While different themes emerge each year, there are a few—with slight variations—that have been found in all three years of data. Students consistently answer that the library website, databases, and searching are what they find most helpful. These are all things that R&I librarians typically include in library instruction. Students also find the step-by-step walkthroughs, citation information, and learning about other services within the library particularly helpful.

For the open-ended question asking students what we could do to help them learn more, the most common answer is nothing! We are delighted that many students feel they already got what they needed from instruction. Other common suggestions for what would be helpful include offering more tutorials or workshops, increasing outreach and/or resources, and questions about finding physical materials/books.

KEY TAKEAWAYS

- Quantitative data helps support the outreach for and marketing of library instruction at the department level by providing evidence of its impact.
- Qualitative data helps with process improvement, lesson plan modifications, and identifying needed adjustments at the course/librarian level.
- A consistently high percentage of students indicate library instruction was useful; they did better in classes and used what they learned.
- About half of all students year-to-year report they did not use additional library resources.

- While syllabi often emphasize the use of academic journal articles, students indicated they also want to learn how to access books.
- Students are interested in additional library instruction opportunities outside of the class period (e.g. workshops and tutorials).
- Students indicated they wanted a library FAQ in the fall 2021 survey.
- We've seen and highlighted for stakeholders differences in identified themes between years when improvements to the online library user experience were made, including the launch of EBSCO Discovery Service and implementation of the Springshare LibApps LTI tool.

TAKING ACTION

The consistency observed in the Project Outcome survey results over three years appears to confirm that library instruction is important to student success or is at least perceived as such by the students themselves. These results are supported by research that indicates a librarian's participation in the student's first-year experience has a direct impact on improved information literacy performance[3] and achievement in student learning outcomes.[4] Similarly, multiple improvements in student outcomes associated with library use by students, including increased grade point averages for students enrolled in library courses, have been documented.[5] Results from the RCC 200 instructor's surveys generally support these arguments. To further validate these outcomes at Regis, R&I librarians will need access to additional university systems and data points related to student grades and enrollment. We intend to use Project Outcome survey results and the supporting research as a lever to open those opportunities and gain access to additional student data to produce a more robust assessment of the impact of collaboration between the library and the FYE program.

We are fortunate that the director and lead faculty in the FYE program are already supportive of the library and the efforts of the R&I librarians. The challenge lies in securing collaboration with the library across all sections of RCC 200 as a standardized practice and expectation. While the director of FYE frequently recommends library instruction to all instructors, in practice it has been left to the individual instructor's discretion whether—and how much—to include library instruction. There is also annual turnover among the RCC 200 instructors, and it is common for new faculty hires to be asked to teach a section. New faculty often look to their peers to help inform them about instruction norms and expectations, and recommendations from other RCC 200 instructors is one way that library instruction has been organically integrated into the FYE program. With the Project Outcome data, we can approach reluctant or new faculty members with evidence that their students are asking for library instruction. In this way, we can leverage the student data to compliment the peer pressure from fellow faculty members to collaborate with the

library. In addition, the open-ended student responses related to content covered (or not covered) in the library session is suggestive of a need for more time dedicated to library instruction. This is evidence we intend to use to advocate for multiple library instruction sessions over the course of the semester.

As we look to more wholly integrate library instruction into the FYE program, the data collected from the Project Outcome survey has provided R&I librarians with a demonstration of the benefits of standardized curricular practices related to library instruction and information literacy across the FYE program. This position is supported by research focused on outcomes associated with standardized curricular practices around information literacy in first year seminar courses.[6] What has recently evolved in our discussion with the directors and lead faculty in FYE, in part due to the evidence provided by the Project Outcome survey data, are conversations about librarians engaging in new duties or roles. These duties could include designations as formal co-instructors in RCC 200 and assisting in course redevelopment that explicitly includes library instruction in the syllabi, as well as the inclusion of information literacy outcomes for the course. These conversations have also included the possibility of developing a stand-alone single credit information literacy course (either offered as an elective or potentially as part of the core curriculum in Regis College) that all FYE students would either be encouraged or required to take. Currently, there is no credit-bearing information literacy course offered by the university, but the Project Outcome data indicates students are interested in additional library instruction opportunities. A credit-bearing information literacy or research skills course taught by a librarian could address this student interest. It would also be a paradigm shift, in that librarians are not currently expected to be the instructor of record for credit-bearing courses. We currently operate as invited guests of the instructor when delivering instruction. Being the formal instructors of a credit-bearing course could raise the stature of faculty librarians across the university and strengthen the notion of librarians as peers to the teaching faculty in the colleges.

While the Project Outcome results are helping advance programmatic change at the university, they have also helped inform the practice of R&I librarians. In particular, qualitative coding of student responses to the open-ended questions in the survey indicates that students could benefit from more instruction about searching for and accessing books (both print and electronic) from the library collection. Students have also indicated a desire for more instruction around citation generation and formatting, and a desire for more self-serve online instructional materials.

Given some of the current constraints related to time and limited opportunities for contact with students in the classroom environment (i.e., a higher percentage of "one-shot" instruction sessions), R&I librarians have not been able to give sufficient time to all the research topics and tools that students have expressed interest in while also addressing the primary information needs related to the RCC 200 course assignment. To help alleviate some of the limitations around time and contact with students, and in direct response to

a need identified by the students themselves, the R&I department developed a searchable library FAQ over the summer of 2022 that includes answers to common questions about the research process, search and retrieval, citation formatting, and other popular library instruction topics.

NEXT STEPS

The R&I department will continue to use the Project Outcome instruction survey in its assessment of library instruction in support of the FYE program. We expect the collaboration between Regis's FYE program and the R&I department in the library to grow and strengthen. If we are successful in integrating library and information literacy instruction as a standardized or required component of the FYE program, we anticipate tweaking the survey to address some of the institution-specific context and shared learning outcomes, perhaps using additional questions to ask a group of students about something in particular each year or relate the survey to departmental goals more directly.

Based on student feedback, we are piloting research workshops for the 2023-24 academic year, and we intend to use the immediate instruction survey for these workshops. Along with these workshops, we plan more general outreach to increase awareness and use of our services and resources. We hope to expand the use of instruction surveys to other courses where we teach information literacy. There is also an opportunity to use survey data when approaching instructors to request more time with students (e.g., scheduling multiple sessions, requiring one-on-one consultations, etc.) given all the library topics, search strategies, and research skills training of interest to students.

We plan to adjust our instruction practices and lesson plans using the feedback from the Project Outcome survey. Similarly, student responses to the survey have informed our library services and spaces outside of the instructional context, and we will continue to analyze the responses through multiple lenses and share with our colleagues in the library. The Project Outcome surveys continue to be a reliable means of gathering student feedback directly and will help us adjust our practices in accordance with the self-reported needs of the students.

FINAL TAKEAWAYS

Use of the Project Outcome survey:

- Demonstrates the importance and value of library instruction to students via direct feedback.
- Advances discussion about programmatic changes, standardizing library/information literacy practices, and holistically approaching student support and success. Bolstered by the data and evidence of the efficacy and importance of library instruction from multiple years of Project Outcome instruction surveys,

we have requested two mandatory library sessions for each section of RCC 200 for the fall 2023 semester.

- Informs the instructional practice of R&I librarians. The results of the Project Outcome surveys indicate that a scaffolded, multi-session approach is warranted to address the totality of research skills, information literacy, and orientation to the library desired by students, instructors, and librarians.
- Provides a data set that is complementary to other student success measures for a more robust analysis of factors associated with student achievement.
- Is so easy—jump in and try it!
- Documents the student side of things. Also surveying the instructors through Qualtrics is helpful, as well; sometimes faculty want to hear from other faculty, and sometimes they want the student feedback.

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CHAPTER 10

Using Project Outcome for an Undergraduate Library Orientation

Kaitlyn Clohosey

INTRODUCTION

Felician University is a small, private Catholic university in northern New Jersey with two campuses, located ten miles from New York City. The university falls in the Master's Colleges and Universities: Medium Programs Carnegie classification. There are four schools at Felician University: arts and sciences, business and information sciences, education, and nursing. The most popular programs are biology, business administration and management, criminal justice, mathematics, and registered nursing. The total enrollment in fall 2021 was 2,411 students, with the undergraduate student body accounting for 83 percent of the student population and graduate students accounting for the remaining 17 percent. Felician University is a Hispanic-Serving Institution, with 36 percent of the student population identifying as Hispanic/Latino and by 23 percent identifying as Black or African-American. Half of the student population that falls in the traditional undergraduate student body identifies as first-generation and 93 percent of all undergraduates receive some sort of financial aid; more than half are eligible for Pell grants.[1]

LIBRARY BACKGROUND

In fall 2022, the university oversaw two campus libraries—one on the Lodi campus, which is its own building, and one on the Rutherford campus that is part of an education

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commons. At this time, Felician University libraries employed four full-time librarians, with one serving as director. The remaining three librarians served as liaison librarians to the schools. The research and instructional services librarian served as the liaison librarian to the School of Nursing, the scholarly communications and outreach librarian served as the liaison to the School of Arts and Sciences, and the public services librarian and head of the archives served as the liaison librarian to the School of Business and Information Sciences and the School of Education.

PROJECT BACKGROUND

At Felician University, all first-year, first-semester students are required to take ENG 101: College Writing and Research I. This introductory-level writing and research course requires students to write three papers, with the second and third paper requiring the use of outside resources. In previous fall semesters, two or three ENG 101 professors out of around 15 asked librarians to do a research session in their classes. In spring 2022, the research and instructional services librarian approached the associate dean and the dean of the School of Arts and Sciences about adding a mandatory library orientation to the ENG 101 curriculum. This would ensure that all first-year, first-semester students receive a library orientation session, not just those whose professor arranged for a librarian to visit the class. The dean and associate dean approved this idea.

At the same time, a full-time English professor requested that the session have an information literacy component, and that the timing of the orientation and information literacy session coincide with the second essay. To cover the topics from both the librarian and faculty member point of view, the research and instructional services librarian created an online library orientation tutorial for ENG 101 professors to distribute before the library session for a flipped learning approach. The online library orientation tutorial, created in LibWizard, provides students with an overview of the library spaces, the librarians and their liaison roles, the library website, and how to use the catalog to locate articles and books. The research and instructional services librarian then coordinated an in-person session for each of the 15 sections of ENG 101 that ran in the fall 2022 semester, with each section capped at 25 students. The librarian scheduled these sessions to take place starting in mid-October and end by the beginning of November 2022. All four librarians taught from a universal lesson plan and administered a game to test students' recollection of the online tutorial, as well as the materials covered in class. The game is based off of The Research Race.[2] For the fall 2022 semester these sessions took place in-person, with the potential of the library orientation session moving to an online modality in the following academic years. The goal of the online library orientation tutorial, as well as the in-person information literacy session and game, was to increase first-year students' knowledge of library services and resources. To see if the learning materials were effective, there had to be an easy way to collect student feedback.

The research and instructional services librarian thought that using Project Outcome in the library orientation session for all first-year, first-semester students would both yield great data on how well students remembered information about the session and assess if their library awareness increased. The data generated by Project Outcome would also be helpful for the end-of-year report and to advocate for the permanency of the mandatory library session in ENG 101.

METHODS

For the ENG 101 librarian sessions, the research and instructional services librarian used Project Outcome's standard instruction survey for immediate use. The standard open-ended questions stayed, and the librarian did not add any customized questions (see Figure 10.1), wanting to keep it simple until there is greater mastery of the tool.

To collect as much data as possible, librarians distributed a paper survey in a packet with other teaching materials. Paper surveys allowed students to provide immediate feedback while the information and session was still fresh in their minds, without having to find a link. One section of ENG 101 was an online course; these students received an electronic link to the survey.



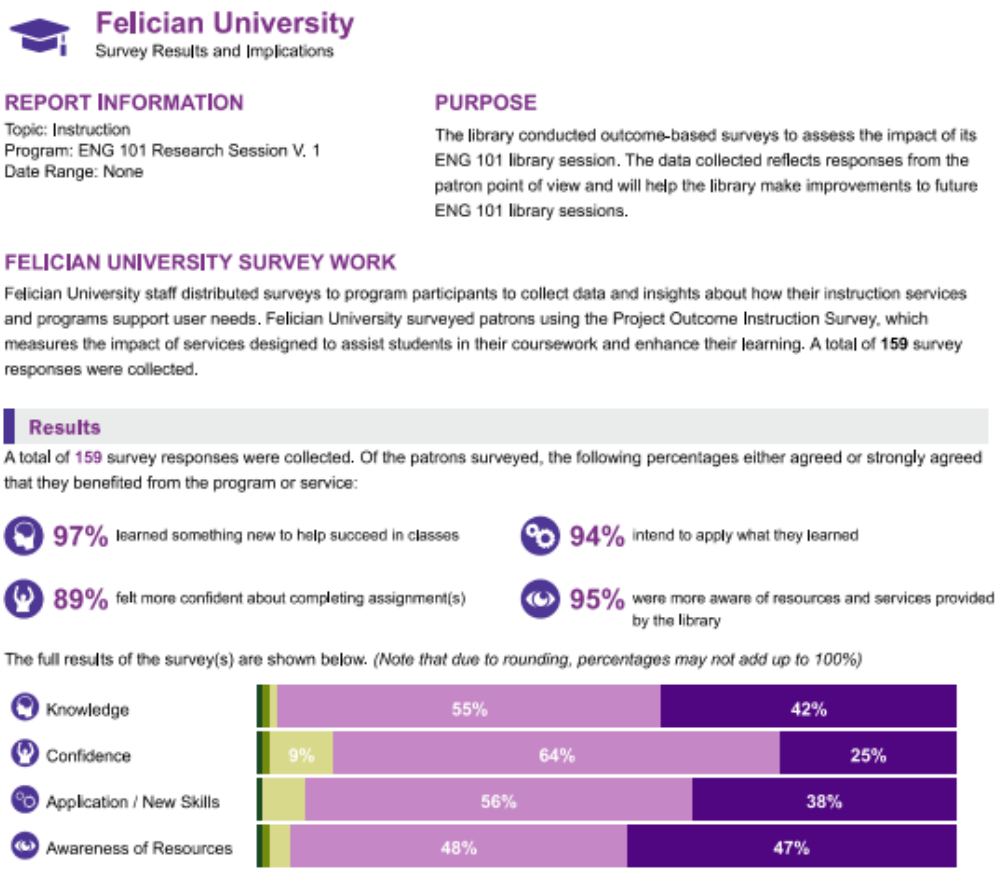
Please take a few minutes for this brief survey and let us know if, as a result of participating in the ENG 101 Research Session program...

1. **I learned something new that will help me succeed in my classes.**
   Strongly Disagree Disagree Neither Agree Strongly Agree N/A
2. **I feel more confident about completing my assignment(s).**
   Strongly Disagree Disagree Neither Agree Strongly Agree N/A
3. **I intend to apply what I just learned.**
   Strongly Disagree Disagree Neither Agree Strongly Agree N/A
4. **I am more aware of the library's resources and services.**
   Strongly Disagree Disagree Neither Agree Strongly Agree N/A
5. **What did you like most about this session?**

6. **What else could the library do to help you succeed in your classes?**

Figure 10.1. Project Outcome immediate instruction survey standard questions.

Of the 193 students who attended the ENG 101 library session, 159 completed the survey. The results were very positive, with each question having a high "agree/strongly agree" total. Based on the report generated by Project Outcome, 154 students (97 percent of respondents) said they learned something new to help them succeed in classes; 142 students (94 percent) intended to apply what they learned; 148 students (89 percent) felt more confident about completing assignments; and 150 students (95 percent) were more aware of resources and services provided by the library (see Figures 10.2 and 10.3).



Felician University
Survey Results and Implications

REPORT INFORMATION
Topic: Instruction
Program: ENG 101 Research Session V. 1
Date Range: None

PURPOSE
The library conducted outcome-based surveys to assess the impact of its ENG 101 library session. The data collected reflects responses from the patron point of view and will help the library make improvements to future ENG 101 library sessions.

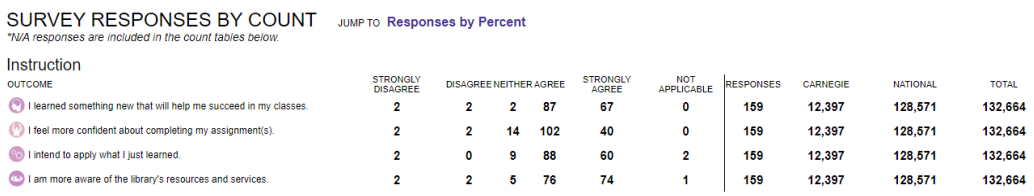
FELICIAN UNIVERSITY SURVEY WORK
Felician University staff distributed surveys to program participants to collect data and insights about how their instruction services and programs support user needs. Felician University surveyed patrons using the Project Outcome Instruction Survey, which measures the impact of services designed to assist students in their coursework and enhance their learning. A total of **159** survey responses were collected.

Results
A total of 159 survey responses were collected. Of the patrons surveyed, the following percentages either agreed or strongly agreed that they benefited from the program or service:

97% learned something new to help succeed in classes
94% intend to apply what they learned
89% felt more confident about completing assignment(s)
95% were more aware of resources and services provided by the library

The full results of the survey(s) are shown below. *(Note that due to rounding, percentages may not add up to 100%)*

Figure 10.2. Fall 2022 survey results.



SURVEY RESPONSES BY COUNT JUMP TO Responses by Percent
**N/A responses are included in the count tables below.*

Instruction

OUTCOME	STRONGLY DISAGREE	DISAGREE	NEITHER	AGREE	STRONGLY AGREE	NOT APPLICABLE	RESPONSES	CARNEGIE	NATIONAL	TOTAL
I learned something new that will help me succeed in my classes.	2	2	2	87	67	0	159	12,397	128,571	132,664
I feel more confident about completing my assignment(s).	2	2	14	102	40	0	159	12,397	128,571	132,664
I intend to apply what I just learned.	2	0	9	88	60	2	159	12,397	128,571	132,664
I am more aware of the library's resources and services.	2	2	5	76	74	1	159	12,397	128,571	132,664

Figure 10.3. Fall 2022 survey responses by count.

Using the word clouds generated by Project Outcome from the open-ended responses, we can see that students enjoyed the class activity, which was the Level Game. The Level Game, based off The Research Race,[3] asked students to locate library information and resources using the library's website and information presented in the LibWizard tutorial. Students competed in teams of two or three to see who could finish the game the fastest. The students liked that they could practice the skills the librarians covered in the session and to practice it with their classmates, as reported by 27 respondents (see Figure 10.4). The feedback from the open-ended questions showed that the students learned more about the library from the session. Based on the feedback gained from the second open-ended question, "What else could the library do to help you succeed in your classes?", the students wanted more: more librarian-led sessions whether workshops or class instruction, more workshops on research and citation, more resources, even more librarians and appointment availability (Figure 10.5).

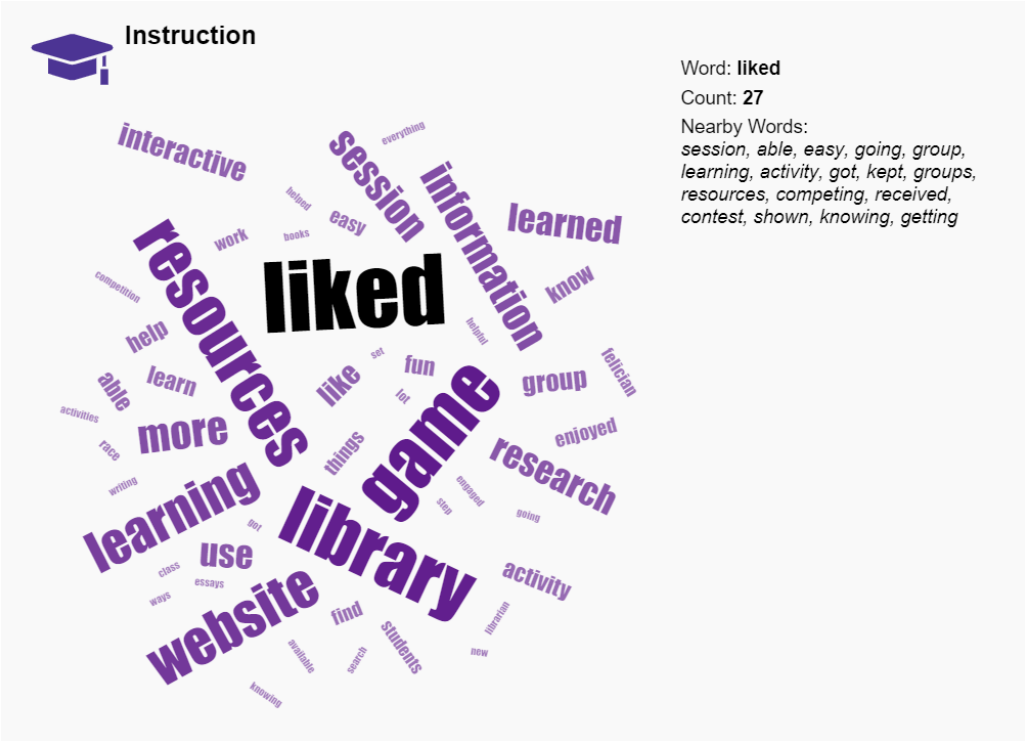


Figure 10.4. Word cloud created from responses to the open-ended question "What did you like most about this session?"

TAKING ACTION

The data collected from the survey was extremely useful, and Project Outcome compiled the data and created useful tables and graphics. The director of library services incorporated the graphics and data into the end-of-year report.

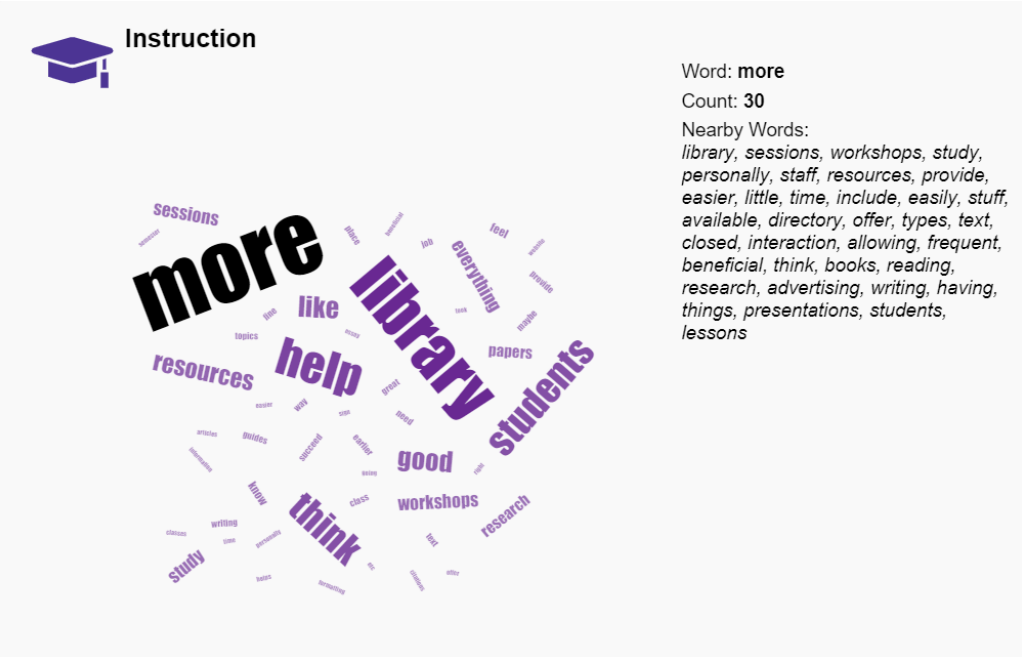


Figure 10.5. Word cloud created from responses to the open-ended question "What else could the library do to help you succeed in your classes?"

The data will guide session content revision and delivery. Since the qualitative data showed that many enjoyed the Level Game, it will stay, and the research and instructional services librarian will change any information that needs updating in the LibWizard tutorial.

When the library orientation and information literacy session was first proposed in spring 2022, there was the idea that it could be asynchronous, but based on the feedback from the open-ended questions the students liked the collaborative and synchronous work of the session. Research has shown that student-librarian interactions are meaningful and that there is a need to have some library sessions in-person in an environment that otherwise demands more asynchronous library materials.[4] Recognizing that students enjoyed this interaction and benefited from it, the research and instructional services librarian championed the success of the in-person session by sharing the reports and word clouds with the associate dean of arts and sciences at the end of the 2022-2023 academic year. The student feedback and data provided enough proof to keep the session in-person.

While the department has not used the feedback for larger library advocacy, the feedback from students highlighted a few things. One comment stated, "Just get more staff so you[']r[e] not stressing yourself out." Librarians did a lot of work for these sessions in a very short time span on top of other liaison and job-related duties, and students noticed. Since running the session, two full-time librarians left and, although one librarian's

position has been filled, the other position has not. Having another full-time librarian or two would ease the stress the librarians felt from doing so many library orientation sessions on top of the other instructional sessions they handle.

When asked "What else could the library do to help you succeed in your classes?", the word "more" came up 30 times (Figure 10.5). In the word cloud, the following words were nearby: sessions (six mentions), workshops (six mentions), papers (five mentions), and guides (three mentions). The word "resources" was mentioned eight times; however, it was not near the word more (Figure 10.5). The comments that mentioned more workshops or sessions indicated that these students may not realize that the library offers additional workshops throughout the academic year. While the majority of students indicated that their library awareness increased as a result of the session, the comments asking for more workshops or sessions highlights the need to put more of an emphasis on these library offerings in the online library orientation tutorial.

The comments that called for additional resources for their paper topics are not new to the librarians at Felician. Librarians know that students always want the most information possible; however, comments received about lack of resources have previously come from upper-level students pursuing higher-level research. This was the first time librarians saw comments regarding a lack of resources from first-year students.

NEXT STEPS

After reviewing the Project Outcome report, the research and instructional services librarian noted a few things. Librarians initially thought that student requests for more librarian instruction were a good thing. Upon closer examination, the LibWizard tutorial submissions totaled 112, a much smaller number than the 193 students who were present for the library sessions. Had a larger number of students reviewed the online library orientation tutorial, comments requesting more instruction may have been fewer. Going forward, to increase participation in the tutorial we may embed a link to it in the learning management course shell at the beginning of the semester; we might also offer an incentive to complete the tutorial. A higher tutorial completion rate could lead to a higher confidence score in Project Outcome, demonstrating a greater awareness of library resources and services.

The students also mentioned in the feedback that they enjoyed the interactivity of the session and the collaborative nature of the activity. If the session were to move to an asynchronous module, it would lose this interactive component and the students may not feel as much of a connection to the library. The feedback shows the need to keep the session a live event. Based on this feedback and with the associate dean's approval after reviewing the report generated by Project Outcome, the library orientation session will remain a part of the ENG 101 curriculum as an in-person session. The continued use of Project Outcome will allow the ENG 101 library session to remain synchronous if the student feedback continues to support this modality.

In the feedback, students mentioned they wanted more resources. While most ENG 101 sections used the topic of what it means to be educated, some ENG 101 professors allowed students to pick their own topics for the second paper. Without knowing which topics will be researched, it is difficult to determine what areas in the collection need additional resources.

Project Outcome was very useful in collecting feedback for the ENG 101 session. The librarians can implement Project Outcome surveys at the end of all instructional sessions to collect additional data on library instruction and use the information provided to track gaps in the collection, improve teaching styles, and update lesson materials.

One of the biggest lessons learned using Project Outcome is that a considerable amount of patience is required when uploading a bulk batch of responses. As most of the responses came through a paper survey, librarians added each survey response from their respective session to an Excel bulk upload sheet, which was later imported into Project Outcome. Project Outcome tells the user to be patient, as the upload could take some time. In this situation, the upload took so long that the research and instructional services librarian ended up clicking the upload button one too many times, which resulted in the responses being uploaded multiple times and over a thousand responses added to the survey. These responses had to be deleted and re-uploaded to reflect the correct number of survey responses.

FINAL TAKEAWAYS

While the thought of having to track an additional tool for end-of-year reporting can be overwhelming, the tools in Project Outcome make it easy to compile data. Project Outcome instantly created the graphics, saving the librarians the time and frustration of having to analyze and design the data. The graphics also add a nice visual touch to the end-of-year report and emphasize the highlights of the session, as well as student feedback. Overall, it can confirm what librarians think of a session, or help to pivot the direction of the class to be more useful to students.

SUMMARY POINTS

- Project Outcome data confirmed that an in-person library session creates a positive student-librarian interaction.
- The neat graphics and tables are helpful for advocacy and formal reports.
- Project Outcome can be used in all instruction sessions to track library knowledge retention.
- Patience is needed when uploading large amounts of surveys.

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CHAPTER 11

Assessing Information Literacy Programs for First-Year Writing Courses at a Public University

Kate Langan, LuMarie Guth, and Dylan McGlothlin

INTRODUCTION

This chapter discusses the findings of a study that used Project Outcome (PO) to assess the effectiveness of library programming for first-year writing courses in different colleges at a mid-sized public university. The purpose of this study was twofold. First, PO data was used to compare student feedback on library support across multiple college writing programs to measure the success of library programming for all incoming first-year students. Second, the PO data was used to set a benchmark for future assessment with the expansion of the library programming for a first-year gateway course. We are not concerned with benchmarking across institutions. For this study, data comes from library programming for first-year writing courses from the College of Engineering and Aviation (CEAS) and the College of Arts and Sciences (CAS), as well as data from a higher-level writing course for the Haworth College of Business (HCOB). The analysis looks at the quantitative and qualitative data captured in the PO immediate instruction survey. We used the immediate survey and not the follow up survey because we have more control over distributing

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them during class time and we get a higher response rate. For the purposes of this study, qualitative responses from question five on the survey— "What did you like most about this program/service?"—are used to interpret PO's quantitative data.

Institutional Background

Western Michigan University (WMU) is a publicly funded, doctoral-granting R2 university. Fall 2022 semester enrollment was 14,397 undergraduate students and 3,438 graduate students. 65 percent of undergraduates identify as white and 35 percent identify as another race and/or international students.[1] In the fall of 2022, over half of the undergraduate student body came from three Michigan regions representing a blend of urban, suburban, and rural communities: 4,173 (29 percent) came from southwest Michigan, 2,430 (17 percent) came from metro Detroit, and 2,029 (14 percent) came from west Michigan.[2] In fall 2022, 23.3 percent of full-time undergraduate students were Pell Grant recipients.[3]

WMU is committed to student success and degree completion. We offer many student success support programs, such as the "Reclaim the W" scholarship for students who have returned to the university after being academically at risk. WMU also participates in many federally and state-funded student success programs for students who are from historically underserved and underrepresented populations, which includes but is not limited to programs for first-generation post-secondary students (TRIO scholars), migrant students (CAMP scholars), and students who aged out of the foster care system (SEITA Scholars). WMU is based in Kalamazoo, Michigan, home of The Kalamazoo Promise scholarship program, which has provided free or reduced college to Kalamazoo public school graduates since 2006.[4]

Targeted Needs

The WMU Libraries Department of Instruction and Outreach (I&O), comprised of eight subject liaison librarians, prioritizes information literacy instruction to support the curriculum by targeting courses typically taken during the first year of study. One way to assure that we are reaching students from underrepresented populations or who are academically at risk is via a program for all first-year/college level writing courses. The Western Essential Studies (WES) core curriculum program requires students satisfy the college level writing requirement by taking one of three courses: BCM 1420: Introduction to Business Communication (BCM:1420) (required for all majors in the College of Business and health administration majors in the College of Education), ENGL 1050: Thought and Writing (ENGL 1050) (required for all majors in the College of Arts and Sciences), and IEE 1020: Technical Writing (IEE 1020) (required for all majors in the College of Engineering and College of Aviation). Library instruction has been embedded in ENGL1050 and IEE 1020 for over fifteen years and began to be embedded in BCM 1420 in fall 2023, after BCM 1420 was made a required course for business majors.

Embedded library instruction, particularly in first-year writing courses, ensures exposure to library resources and research skills early in students' academic careers.[5] The WES program has multiple learning outcomes including information literacy (IL). The program recognized that the "ability to synthesize and use information while contributing to scholarly, professional, civic or personal conversations" helps students achieve their academic and professional goals.[6] The three first-year writing courses are identified in WES as having an IL learning outcome, and course instructors are required to assess students using the rubric in Appendix A. For the purposes of this study, we looked at ENGL 1050 and IEE 1020 PO data. Since data from BCM 1420 is not available at the time of this writing, data from BCM 3700: Professional Business Writing (BCM 3700), another business writing course, is included in this analysis.[7] The data was used in the decision-making process that led to embedding IL in BCM 1420 beginning in fall 2023.

Course Assignments and Lesson Plan Descriptions

In fall 2022 and spring 2023, 629 students were enrolled in BCM 3700, 1,501 in ENGL 1050, and 409 in IEE 1020.[8] Eighty-seven BCM 3700 students and 279 IEE 1020 students had in-class library instruction, and 867 ENGL 1050 students received library instruction through an out-of-class library lab (see Table 11.1). All library instruction is mapped to broader course outcomes, though the IEE 1020 library lesson plan is aligned to a specific assignment.

	Enrollment On Census Day	Attended a Library Session
BCM 3700	629	87 (14%)
ENGL 1050	1501	867 (58%)
IEE 1020	409	279 (68%)

Table 11.1. Course enrollment.

Library instruction for BCM 3700 is taught by the business librarian during scheduled class time. The librarian was not given a copy of the final assignment but met with the course instructor to develop learning outcomes for the library session which included keyword development, advanced search techniques (Boolean operators, result filters, and search limiters), finding full text, and how to cite sources. Databases such as ProQuest ABI/INFORM Complete, IBISWorld, and Statista are covered. Students were also given a short tour of the physical library. By the time of the library instruction session, students have created a resume and cover letter. The assignment covered in the instruction session builds on this previous work in the form of an employment research report (see Table 11.2).

	BCM 3700	ENGL 1050	IEE 1020
Instructors	Business librarian	I&O librarians	Engineering librarian
Modality	In-class instruction	Library lab	In-class instruction
Enrollment and Participation Data	• 629 students enrolled • 87 (14 %) received IL instruction • 16 survey respondents	• 1,501 students enrolled • 867 (58 %) received IL instruction, • 778 survey respondents	• 409 students enrolled • 279 (68%) received IL instruction • 92 survey respondents
IL Concepts Taught	• Keyword development • Advanced search • Result filters and search limiters • Finding full text • Citing sources	• Authority and bias • Keyword development • Advanced search • Result filters and search limiters • Finding full text • Citing sources	• Keyword development • Advanced search • Result filters and search limiters • Finding full text
Library Resources Taught	• ABI/INFORM Complete • IBISWorld • Statista • Mini tour of physical library	• Library Search	• Engineering Village • Science Direct
Prior to the Library Session Students Have:	• Created a resume and cover letter	• Completed an autoethnography reflecting on their identity within a community and an equity issue in that community	• Developed keyword brainstorming
Assignment Description (length, sources, topics)	• Unknown	• 6-page position paper building on their autoethnography • Topic on power, privilege, or policy associated with an equity issue • No official source type or limit	• Students are required to write a 12- to 14-page thesis paper • Topic related to aeronautics or engineering paper • Use 10 sources, five from the last two years

Table 11.2. Research assignments by course.

Library instruction for ENGL 1050 is taught by all I&O librarians due to the size of the course and uses a library lab model.[9] In the lab, the librarians use a standardized lesson plan which covers authority and bias in research; keyword development; advanced search techniques including Boolean operators, result filters, and search limiters; and how to cite a peer-reviewed journal article. The lab emphasizes "Library Search," WMU's Primo federated search tool, over subject-specific databases. No subject-specific databases are covered because of the wide range of student topics and Library Search's ability to search almost every database. By the time students attend a lab, they have already completed an autoethnography paper about their individual identity within a community identity

and an equity issue that affects the community. The autoethnography continues into the assignment covered in the lab: a six-page position paper about the power, privilege, or policy associated with the equity issue from their autoethnography. There are no specific requirements for source types or number of sources to include; however, through collaboration with the ENGL 1050 program director, it was decided that peer-reviewed journal articles would be the source type covered during the lab (see Table 11.2).

Library instruction for IEE 1020 is taught by the engineering librarian during scheduled class time. Students have already been given an assignment to brainstorm keywords prior to coming to instruction. During the library session the librarian covers advanced search techniques including Boolean operators, result filters, and search limiters, as well as finding full text. The databases Engineering Village and Science Direct are the library resources covered. This leads to the thesis paper assignment covered in the library session. Students "select a topic related to their majors, gather background information, and develop what is called a purpose statement. The purpose statement describes a technical problem and the proposed solution for that problem."[10] The final submission is a twelve to fourteen-page paper requiring ten sources, five of which are from the last two years (see Table 11.2).

METHODS

This study was reviewed by the Western Michigan University Institutional Review Board and approved under the project number IRB-2023-106 "FYE and ENGL 1050 Programming." Students from BCM 3700, ENGL 1050, and IEE 1020 who participated in library instruction for these courses were invited to complete the PO immediate instruction survey. No modifications to the survey were made and no other assessment tool was used in addition to the PO Survey. As part of the informed consent, students could either complete or not complete the survey. All responses to the surveys were anonymous. A paper copy of the survey was administered to students at the end of the library instruction in the ENGL 1050 library lab, and an electronic survey was given to students at the end of the BCM 3700 and IEE 1020 library sessions. The scope of this study is limited to the 2022-2023 academic year. The analysis is limited to qualitative question five, "What did you like most about this session?", with a brief overview of the quantitative charts and coding of student responses. Question five student responses provide insight into the quantitative responses. The survey was given to all students who received library instruction: 87 BCM 3700 students, 867 ENGL 1050 students, and 279 IEE 1020 students (see Table 11.1).

RESULTS

The results section is presented in two parts. The first looks at the quantitative data results and the second at the qualitative data with an explanation of coding. Completed surveys

for each course are as follows: for BCM 3700, 16 of 87 (18 percent); for ENGL 1050, 778 of 867 (90 percent), and for IEE 1020, 92 of 279 (33 percent) (see Table 11.3). Response rates for BCM 3700 and IEE 1020 are low because completed surveys using alternate lesson plans were excluded and surveys were distributed electronically. ENGL 1050 distributed the survey in paper format and IEE 1020 and BCM 3700 both distributed the survey electronically. In the process of administering the PO surveys, the authors learned paper surveys received a much higher response rate than electronic surveys (see Table 11.3). Electronic surveys are easier to administer and manage, requiring no extra work from the librarian, but the benefits of such a high response rate outweighed the minimal time needed to manually enter responses into Project Outcome. PO provided the raw data for quantitative and qualitative data that we were then able to download and analyze with multiple tools, including Project Outcome quantitative tools, Excel, and Voyant.

	Attended	Completed Surveys	Completion Rate	Delivery of Assessment
BCM 3700	87	16	18%	Electronic survey
ENGL 1050	867	778	90%	Paper survey
IEE 1020	279	92	33%	Electronic survey

Table 11.3. Instruction survey response rates by course.

Quantitative Analysis

Quantitative analysis includes the responses to the four Likert scale questions in the survey. For the purposes of this study, we looked at the combined totals for "agree" and "strongly agree" (A/SA) for each of the four questions. Across the three courses, the A/SA ratings are mostly in the ninetieth percentile except for two questions. Across all four questions, there is a relatively small spread in percentage points. The largest spread is 15 percent for question two and the smallest is 7 percent for question 1 (see Figures 11.1, 11.2, and 11.3). Regardless of the low sample size for BCM 3700, the high number of A/SA ratings indicates that students were satisfied with the lesson plan.

Question one— "I learned something new that will help me succeed in my classes"—shows little variation between the three classes for A/SA responses. ENGL 1050 and IEE 1020 had 93 percent and 94 percent A/SA respectively. BCM 3700 had a 100 percent A/SA rating. The high ratings across all three courses indicate that, overall, library instruction is contributing to student success and meeting student needs.

The analysis of question two, "I feel more confident about completing my assignment(s)", reveals that this is the statement with the largest percentage spread. BCM 3700 had a 100 percent A/SA rating, while ENGL 1050 had 85 percent and IEE 1020 had 89 percent. One explanation for the lower ratings may be the time between the date of library instruction and the deadline for the actual assignment; ENGL 1050 has a larger

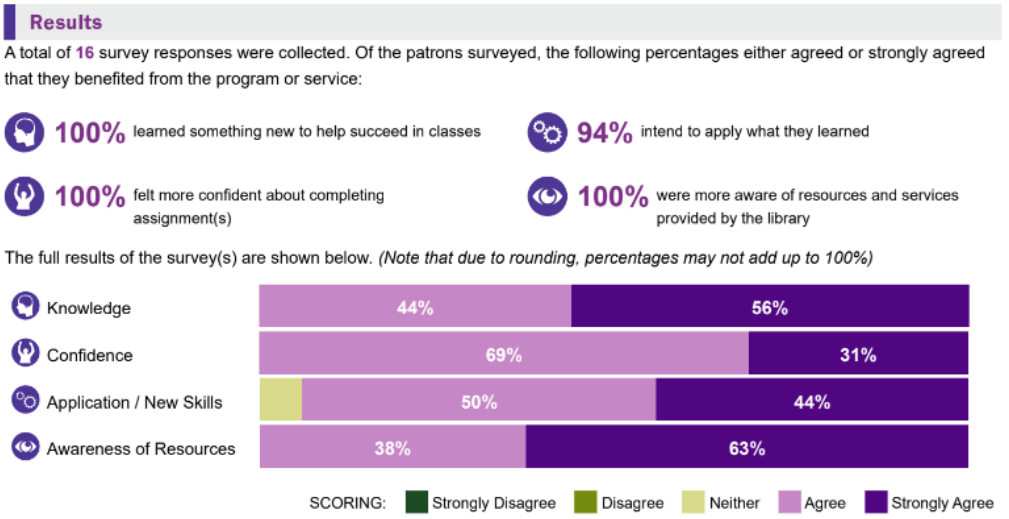


Figure 11.1. BCM 3700 quantitative survey results (n=16).

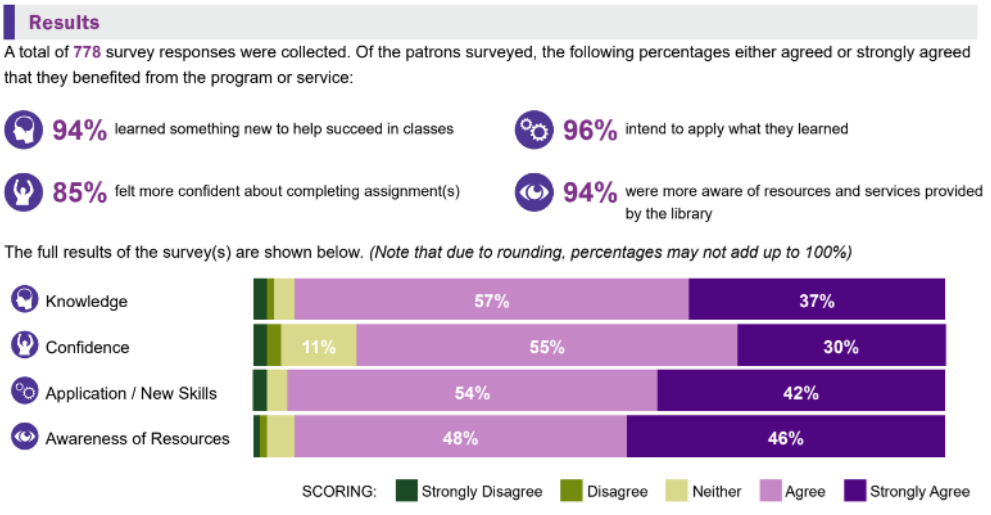


Figure 11.2. ENGL 1050 quantitative survey results (n=778).

time gap between library instruction and the assignment deadline than IEE 1020. The lack of subject-specific resources in the ENGL 1050 lesson plan due to the personal and open-ended nature of the research paper could also contribute to fewer A/SA ratings.

Analysis of question three, "I intend to apply what I just learned", shows a four-point spread in responses across the three courses. BCM 3700 had a 94 percent A/SA rating, ENGL 1050 92 percent, and IEE 1020 96 percent. The lower rating for ENGL 1050 is likely due to the nature of the library lab. The lab focuses on a worksheet which is meant to prepare students for a research assignment that comes later in the semester. The lab is

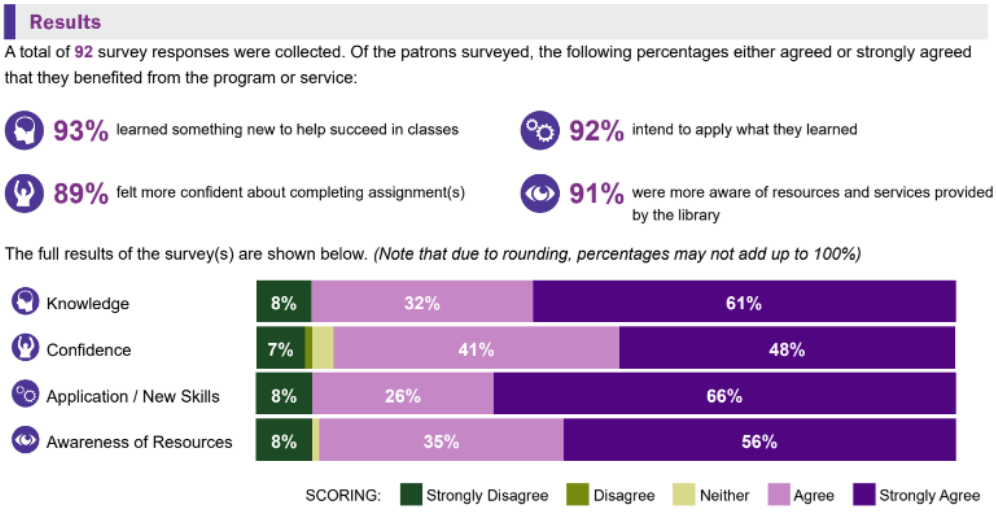


Figure 11.3. IEE 1020 quantitative survey results (n=92).

also not a graded assignment. The connection between library instruction and the final paper may not be entirely clear to students at the time of library instruction.

"I am more aware of the library's resources and services", question four, also had a high A/SA rating. BCM 3700 had 100 percent, ENGL 1050 94 percent, and IEE 1020 91 percent. It is believed that the BCM 3700 course had a higher rating due to the mini tour of the physical library as part of its lesson plan. It may be that students perceive resources and services as physical or tangible things in the library; they might not consider "Library Search" or databases to be a service or resource.

Qualitative Data Analysis

The response rate to question five, "What did you like most about this session?", returned 16 responses from BCM 3700, 778 responses from ENGL 1050, and 87 responses from IEE 1020 since some students left question five blank (see Table 11.3). We coded the smallest dataset (BCM 3700), identified common themes, and agreed on five categories to code the comments. One librarian coded the remaining datasets. We then reviewed the datasets and spot-checked the codes before proceeding with the analysis. Though many comments addressed multiple codes, selecting one code per comment made ENGL 1050's large dataset more manageable and simplified the comparison between datasets to draw broad conclusions about the overall program (see Table 11.4). Once all the student comments were coded, the similarities and differences among all three courses were clearer and the results helped to explain some of the differences between courses exhibited in the quantitative data. For the purposes of this study, question one, which measures new information gained, is mapped to the code "Instructional Design." Question two measuring confidence

is mapped to the codes "Assignment/Topic" and "Student Affect." Question three, which measures applicability, is mapped to "Assignment/Topic," and question four measuring awareness of resources is mapped to "Library Resources." The code "Librarians" is not mapped to any PO question, but we include it because student comments mention the positive experience with a specific librarian or the presence of a librarian in general.

Code	Mapped to PO Questions 1-4	Definition and Examples
Instructional Design	Q1 Knowledge	Comments addressing learning outcomes or the method of instruction
Library Resources	Q4 Awareness	Comments referencing library resources including databases, spaces, library search (catalog search), library chat reference, books, articles, etc.
Librarians	N/A	Comments about the librarians encountered during the session
Assignment/Topic	Q2 Confidence Q3 Applicability	Comments about the course assignment or working with their topic
Student Affect	Q2 Confidence	Emotional responses

Table 11.4. Qualitative data code definitions.

"Instructional Design" was the code most used, representing almost half of all responses for each course: 53 percent, 49 percent, and 49 percent for BCM, ENGL, and IEE respectively. Across all three courses, students acknowledged that they learned something new and relevant (see Figure 11.4). This can explain the high ratings in question one measuring new knowledge (see Table 11.5).

	"Instructional Design"	Q1 Knowledge
BCM 3700	53%	100%
ENG 1050	49%	94%
IEE 1020	49%	93%

Table 11.5. "Instructional Design" code mapped to question one.

Following is an unedited sample of student comments relating to "Instructional Design" by course.

BCM 3700

> "Interacting and using the resources rather than just looking at them on a screen."

ENGL 1050

> "how we learned to use every topic and not bias"

> "i like how we got to follow along and actually go along with the steps"

IEE 1020

> "I liked how he showed us examples of what he was talking about and walked us through it."
>
> "How it was tailored to the majority of the class's major —> aviation/engineering"

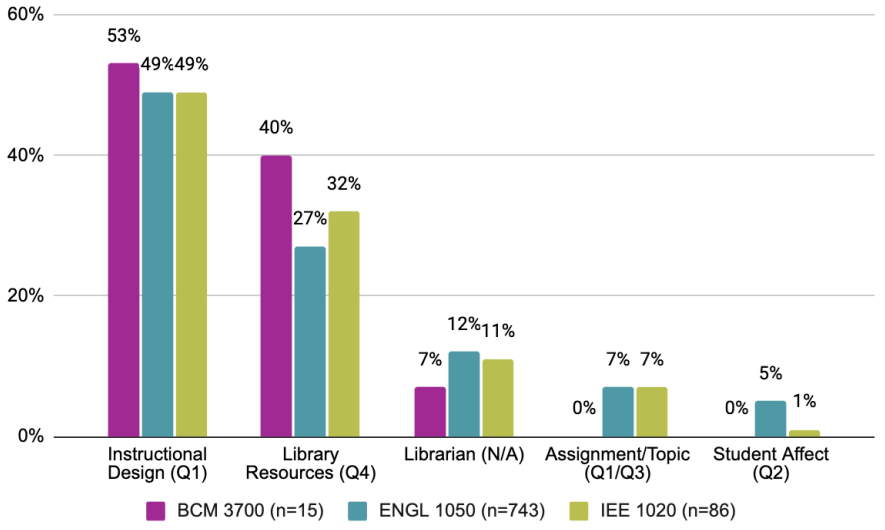


Figure 11.4. Prevalent codes by course.

The code "Library Resources" appears in qualitative student responses more frequently in BCM 3700 and IEE 1020, at 40 percent and 32 percent respectively, than it does in ENGL 1050 with 27 percent (see Figure 11.4). This was likely due to the variation in lesson plans. Both BCM 3700 and IEE 1020 demonstrate subject-specific databases and ENGL 1050 does not. This shows that students in more discipline-specific writing courses appreciate the customization to their unique information needs and may explain the difference in confidence reported in the quantitative data. "Library Resources" is mapped to question four, which measures awareness (see Table 11.6).

	"Library Resources"	Q4 Awareness
BCM 3700	40%	100%
ENG 1050	27%	94%
IEE 1020	31%	91%

Table 11.6. "Library Resources" code mapped to question four.

Here is an unedited sample of student comments relating to "Library Resources" by course.

BCM 3700

> "I really enjoyed the new resources that were provided to us: LinkedIn, IBISworld, Statista, ABI/INFORM Complete, & Marketline. The tour of the library was also helpful in showing me new, physical resources."

ENGL 1050

> "helping me use the library search instead of google scholar"

> "i liked learning about the advanced search feature available"

IEE 1020

> "I was able to improve my knowledge of the library databases."

> "The thing that I liked most about this session was that I learned where to find scholarly journals."

Seven percent of ENGL 1050 and 8 percent of IEE 1020 responses to question five commented on the relevance of the instruction to their assignments and were coded "Assignment/Topics," while none of the comments for BCM 3700 addressed this (see Figure 11.4). This may be due to the low number of responses to BCM 3700 or because the librarian was not provided a copy of the final assignment to align with the lesson. "Assignment/Topics" is mapped to questions two and three, which measure confidence and applicability (see Table 11.7).

	Assignments/Topics	Q2 Confidence	Q3 Applicability
BCM 3700	0%	100%	94%
ENG 1050	7%	85%	96%
IEE 1020	8%	89%	92%

Table 11.7. "Assignments/Topics" code mapped to questions two and three.

Some unedited student comments relating to "Assignments/Topics" follow.
ENGL 1050

> "i liked how it is directly related to the in-class work and is meant to help with that"

IEE 1020

> "The session lets me work on my own topic and search for articles I will actually use"

ENGL 1050 stood out from BCM 3700 and IEE 1020 with the prevalence of comments showing an emotional response, tagged "Student Affect (see Figure 11.4). This may be

due to the personal nature of the topics in ENGL 1050. ENGL 1050 also had the highest percentage of comments about librarians (12 percent for ENGL 1050 versus 11 percent and 7 percent for IEE 1020 and BCM 3700). We realized students were researching and writing about their concept of their identity and an issue impacting it for the ENGL 1050 assignment. Students may have identified more with the librarian teaching this session than those helming the other library instruction programs; the BCM 3700 research assignment is about future career exploration while the IEE 1020 assignment is about a student's major. "Student Affect" and "Library Staff/Instructor" are mapped to question two, which measures confidence (see Table 11.8).

	"Student Affect"	"Librarian"	Q2 Confidence
BCM 3700	0 percent	7 percent	100 percent
ENG 1050	5 percent	12 percent	85 percent
IEE 1020	1 percent	11 percent	89 percent

Table 11.8. "Student Affect" and "Librarian" code mapped to question two.

Here is an unedited sample of student comments relating to "Student Affect" or "Librarian" by course.

BCM 3700

"Very personable and we worked in groups, very good instructor."

ENGL 1050

"it felt casual and not intimidating"

"the environment and how welcoming everyone was"

"It was my first time at the library and loved it"

IEE 1020

"Meeting our Engineering Liberian was pretty neat."

"How personal the help was"

While there are no apparent patterns or relevant connections between the frequency of key terms as they map to questions one, two, three, and four, the comparison provides insight into how students talk about knowledge, confidence, applicability, and awareness in their own terms. The analysis of the qualitative data also reveals themes that are not captured by the qualitative questions but that proved important to students, such as the high number of student comments about librarians who positively impacted their library experience. The variations in comments between each course could also reveal the student culture within each subject. As these programs change and more students take BCM 1420

due to the WES requirements, it will be interesting to see if these trends continue or even out across the courses.

Student comments were further analyzed using Voyant Tools (https://voyant-tools.org/) to identify any additional themes. When analyzing word frequency, the authors observed the common use of words using the word stems learn*, informat*, help*, search*, and eas*/simp* (see Figure 11.5). These are the terms that best match the concepts of learning, confidence, awareness, and applicability measured in PO quantitative questions one through four.

When analyzed by course, mentions of learn/learned/learning were prevalent across all courses, ranging from 20 to 24 percent of the comments. This is an encouraging takeaway.

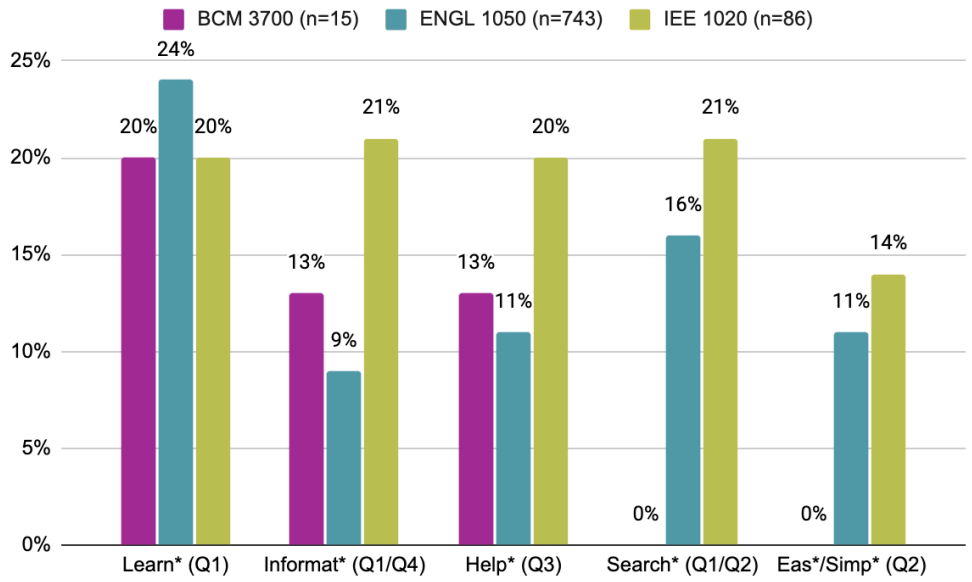


Figure 11.5. Prevalent words in the comments by course.

The terms "information/informative", "help/helping/helped/helpful", and "search/searches/searched/searching" were mentioned frequently in IEE 1020 and less so in the other courses. This may reflect the predominance of comments that mentioned the word "research" in IEE 1020; 14 percent versus only 4 percent in ENGL 1050. IEE 1020 has the most demanding research assignment of the three courses. The IEE 1020 library lesson plan refers to a specific research assignment, while ENGL 1050 refers more immediately to the worksheet completed in the lab. Since the lab worksheet is preparation for a larger assignment for ENGL 1050 students, these factors could explain why IEE 1020 students are more oriented to the assignment.

Mentions of the words "easy/easier" or "simple/simplicity" were present in both ENGL 1050 and IEE 1020, in many cases demonstrating the effectiveness of the library assignment and its role in facilitating the completion of coursework. These terms represent 14 percent and 11 percent respectively for all responses for IEE 1020 and ENGL 1050, though these terms never appear in the student responses for BCM 3700.

TAKING ACTION

The quantitative and qualitative data from BCM 3700, ENGL 1050, and IEE 1020 has helped us assess modes of instruction, learning outcomes, and student affect. It has also helped in the development of a program for BCM 1420. The recent addition of BCM 1420 to the requirements of the business major has directly contributed to an increase from two to 14 sections a semester. The business librarian opted to use the BCM 3700 lesson plan taught in fall 2022 for the fall 2023 BCM 1420 sections. The business instructors were transitioning the BCM 3700 assignment to BCM 1420. The business librarian was further encouraged to adopt the lesson plan because of its similarities with the concepts covered in the other first-year writing courses.

Student comments are shared with instructors to inform them of student feedback and show that students understand and value library programming. It proves especially helpful for ENGL 1050; with the library lab model, student registration and attendance are heavily dependent on the course instructor's promotion of the IL sessions. Though feedback on the library lab model is not explicitly addressed in the questions of the PO survey, students mention enjoying the lab or workshop-like instruction setting.

IEE 1020 library instruction has been refined over several years and regularly re-evaluates its instructional design. The engineering librarian is interested in the results of the PO surveys following the integration of BCM 1420 data for insights into the differences between the courses and evidence for integrating new instructional practices. The information gathered from this data will be used to assess any future changes in our IL instruction in the first-year writing courses, such as timing of courses so that they can be scaffolded from First Year Experience and First Year Writing across multiple colleges. Now that BCM 1420 and IEE 1020 are required within their colleges, we will continue to use PO to assess the overall program and determine if changes need to be made to our lesson goals and/or student outcomes.

NEXT STEPS

The business and engineering librarians plan to incorporate the use of paper surveys to collect a more robust dataset for comparison to ENGL 1050 in the coming year. Student feedback indicates a need to keep subject-specific resources in the BCM 1420 and IEE 1020 lesson plans—BCM 3700 and IEE 1020 students valued database instruction and

discipline-specific library resources. As more business students are enrolled in their college's first-year writing courses instead of the more interdisciplinary ENGL 1050 course, it will be interesting to see how student responses change over time.

FINAL TAKEAWAYS

The takeaways from this project fall into three categories: student perspectives, program planning, and assessment administration.

- Student Perspectives
  - Student feedback indicates that they prefer contextualizing the assignment within IL instruction. We recommend mapping library instruction directly to an assignment.
- Program Planning
  - PO can be used to evaluate new models of IL instruction delivery. We recommend using PO to evaluate and compare library instruction across multiple disciplines.
  - PO can be used to evaluate new models of IL instruction delivery such as the library lab model.
  - Clearly communicating the value of IL instruction by sharing student feedback with instructors is easily done through the quick reports.
- Assessment Administration
  - Paper surveys receive more responses. We recommend administering paper PO surveys.

APPENDIX A. RUBRIC FOR INFORMATION LITERACY ESSENTIAL STUDIES STUDENT LEARNING OUTCOME[11]



Learning Outcome: Demonstrate and apply information literacy.

Criteria	Exemplary	Proficient	Developing	Beginning
Students are consumers of information. Students engage with, locate, and evaluate information appropriate to a question, topic, or research need. Students consume information from a variety of sources (e.g. text, data, oral, in both traditional and emerging information formats).	Students will: • Build on topic to **generate increasingly complex** questions and lines of inquiry. • **Critique** information to recognize what voices and perspectives are *not* represented in available resources.	Students will: • **Construct** a topic that is appropriate in scope and suits a specific information need. • **Evaluate** information to find and select the *most* appropriate resources to fill the need.	Students will: • **Explore and refine** topic and/or need. • **Select focused** sources of information, specifically related to information need.	Students will: • **Define a general** information need. • **Select general** sources of information.
Students are creators of information. Students synthesize and use information to negotiate meaning and contribute to scholarly, professional, civic, and/or personal conversations. Students create information in a variety of sources (e.g. text, data, oral, in both traditional and emerging information formats).	Students will: • **Critique** connections and relationships between different sources of information. • **Create and share** *new* knowledge based on synthesized information in appropriate information format.	Students will: • **Construct** increasingly complex relationships between different sources of information, and connect sources of information to a larger body of knowledge. • **Draw and communicate conclusions** based on synthesized information.	Students will: • **Explain** relationships and connections between different sources of information, whether and how sources agree with or contradict each other. • **Communicate relationships** and connections between sources of information.	Students will: • **Identify** sources of information, but do not make connections between them. • **Communicate information** from sources without explaining relationships between sources.

Definition of information literacy: The set of integrated abilities encompassing effective and reflective discovery of information, the understanding of how information is produced and valued, the use of information in creating new knowledge, and participating ethically in communities of learning.

(From WMU Libraries "Information Literacy Instruction," http://www.wmich.edu/library/services/instruction retrieved October 30, 2017)

NOTES

1. "Enrollment Data," Western Michigan University, accessed January 23, 2024, https://wmich.edu/institutionalresearch/interactivedashboards/enrollment/data.
2. "Enrollment Map," Western Michigan University, accessed January 23, 2024, https://wmich.edu/institutionalresearch/interactivedashboards/enrollment/maps.
3. "Full-Time Undergraduate Student Body Diversity," Western Michigan University, accessed January 23, 2024, https://wmich.edu/institutionalresearch/interactivedashboards/enrollment/diversity.
4. "Kalamazoo Promise," Kalamazoo Promise Foundation, accessed January 23, 2024, https://www.kalamazoopromise.com/.
5. Anna M. Van Scoyoc, "Reducing Library Anxiety in First-Year Students: The Impact of Computer-Assisted Instruction and Bibliographic Instruction," *Reference & User Services Quarterly* 42, no. 4 (2003): 329-341, https://www.jstor.org/stable/20864059 found that composition students who received instruction from a library staff member scored lower on Bostick's Library Anxiety Scale than students who took an online tutorial; Erin E. Rinto and Elisa I. Cogbill-Seiders, "Library Instruction and Themed Composition Courses: An Investigation of Factors that Impact Student Learning," *The Journal of Academic Librarianship* 41, no. 1 (2015): 14-20, https://doi.org/10.1016/j.acalib.2014.11.010 found that students in a first-year composition course who received library instruction scored higher on an annotated bibliography assignment; Xiaorong Shao and Geraldine Purpur, "Effects of Information Literacy Skills on Student Writing and Course Performance," *The Journal of Academic Librarianship* 42, no. 6 (2016): 670-678, https://doi.org/10.1016/j.acalib.2016.08.006 found students in first-year writing courses who scored higher on the ACRL information literacy skills test had higher writing assignment scores and final course scores; and Holly Leutkenhaus et al., "Measuring Library Impacts through First Year Course Assessment," *Communications in Information Literacy* 11, no. 2 (2017): 339-353, https://doi.org/10.15760/comminfolit.2017.11.2.6 found that first-years who attended at least one library instruction session scored higher in argument building, source type integration, and ethical source citation elements of their research papers than their counterparts.
6. "Information Literacy," Western Michigan University, accessed January 23, 2024, https://wmich.edu/essentialstudies/info-literacy.
7. In addition to the first-year writing requirement, all students are required to take a higher-level baccalaureate writing course tailored to their disciplines; BCM 3700 is the baccalaureate writing requirement for all business majors. In discussions with the BCM 1420 instructors, the decision was made to adopt this lesson plan into BCM 1420 due to the positive feedback from students in the PO data and the similarity of the assignment.
8. "WMU Course Offerings," Western Michigan University, accessed January 23, 2024, https://www.wmich.edu/classlookup/wskctlg.wskctlg_menu.
9. Students register and attend a one-shot IL session/lab taught outside of regularly scheduled class time. Labs are scheduled at various times Monday-Friday over three weeks. For a background of ENGL 1050 instruction using a traditional one-shot model, see Kathleen A. Langan and Dianna E. Sachs, "Mapping Information Literacy to a First-Year Writing Curriculum," in *Creative Instructional Design: Practical Application for Libraries*, ed. Brandon West, Kimberly Hoffman, and Michelle Costello (Association of College and Research Libraries, 2017), 55-71.
10. Edward J. Eckel, "Topic Development in the Freshman Engineering Paper: Finding a Focus," *Science & Technology Libraries* 38, no. 2 (2019): https://doi.org/10.1080/0194262X.2019.1566042.
11. "Learning Outcome: Demonstrate and apply information literacy," Western Michigan University, October 30, 2017, https://wmich.edu/sites/default/files/attachments/u370/2018/information-literacy_0.pdf.

APPENDICES

APPENDIX A

Project Outcome 101

GETTING STARTED

What is the goal of Project Outcome?

The goal of Project Outcome is to help libraries understand and share the impact of essential library programs and services. As librarians, we see every day that library services put people on the path to literacy, technological know-how, or a better job—what we are often missing is the data to support it. Project Outcome provides simple tools and an easy-to-use process for measuring outcomes and collecting insights about all the ways we are meeting the needs of our communities. This understanding can improve the way libraries do business—from allocating existing resources more efficiently to advocating for new resources more effectively.

How does Project Outcome work?

Project Outcome provides simple survey instruments that library staff can use to help them measure the outcomes of their library services. Any library that chooses to participate in Project Outcome will have access to patron surveys developed by a team of library leaders, plus additional resources to help administer the surveys and use the results for advocacy and strategic decision-making. In return, participants will report their data using a simple online survey management tool and be able to see their results in an interactive data dashboard.

How is outcomes measurement different from other types of performance measurement?

Libraries commonly measure services in terms of "how much we do," focusing on the volume of outputs like the number of activities or services offered and number of patrons reached. Project Outcome seeks to capture the benefits (or outcomes) of library services in terms of "what good we do," like changes in library users' knowledge, confidence, behavior

change, or awareness. For example, while outputs may tell us how many library patrons attended a resume-writing workshop, outcomes measurement can tell us whether those users feel more confident, informed, and equipped to seek a new job. For more information, please visit What is Outcome Measurement.*

Why is measuring outcomes important for my library?

Measuring and understanding outcomes is important for a number of reasons. At the most basic level, data about the impact of your library services provides insights you can use to 1) demonstrate the value of those services, 2) make plans to improve them, and 3) decide how to allocate limited resources. The same data can inform long-term planning about the services your library will offer in the future. Outcomes data can also play a critical role in strengthening library advocacy, providing library leaders with a compelling case for increased library funding and supportive policies. For more information, please visit What is Outcome Measurement.†

Who can participate in Project Outcome for Academic Libraries?

Project Outcome is available for free for all academic and research libraries, as well as library school students! If you do not fit in these categories, you may still register for free to learn more about outcomes measurement and access our online resources. The survey and data analysis tools will not be available for non-library users. If you are from a public library, you can sign up for Project Outcome for Public Libraries at https://www.projectoutcome.org/.

What kind of commitment is involved with participating in Project Outcome?

We recognize that library staff have a long list of responsibilities, so Project Outcome is designed to align with the work you are already doing and fit easily into your schedule. You can create and administer surveys based on your library's needs, and collecting your own data means you can see and start to use the findings immediately. There is no minimum requirement for participating. Libraries can choose their participation based on time and staff capacity. The surveys and tools were designed to be easy enough for libraries of all sizes to participate and benefit from the outcomes collected.

* https://acrl.projectoutcome.org/surveys-resources/what-is-outcome-measurement

† https://acrl.projectoutcome.org/surveys-resources/what-is-outcome-measurement

Do I have to attend a training before I can begin?

No formal training is required to participate in Project Outcome. Visit our free online resources and archived webinars to get the most of your Project Outcome experience! If you are interested in hands-on training for a conference or event as part of a consortium, ACRL/ALA chapter, or regional or state association, you can find more information about our workshops here.‡ Contact acrl@projectoutcome.org to book a training workshop.

How many staff members can register under one library? How do I see who is registered under my library?

Project Outcome allows multiple users to register under a single library account. An individual user will sign up with their unique email address to create their account. All users registered under a library will have full access to that library's data and surveys. We recommend that libraries make sure the appropriate staff are creating accounts for and accessing Project Outcome to avoid data inaccuracy. You can view who else is registered under your library by clicking Account at the top of the page, and clicking the Users tab. You can report any users who should not have access to your library's account by clicking "Report User." Visit the Data Collection Team resource to organize your staff's roles and responsibilities within Project Outcome.

What library service areas does Project Outcome measure and what are the surveys like?

The Project Outcome for Academic Libraries surveys cover seven essential library service areas: Digital & Special Collections, Events / Programs, Instruction, Library Technology, Research, Space, and Teaching Support. There are two types of surveys to choose from: immediate and follow-up. The immediate surveys use Likert-scale responses to measure immediate benefits gained from a program or service. The follow-up surveys use Yes/No/Please explain responses to measure whether or not patrons continued to benefit or changed their behavior as a result of a program or service. All Project Outcome surveys are designed to help libraries capture their impact on the community.

Do I need to administer a survey for every service area in my library?

No. You can pick and choose which service areas are priorities for your library, or start with one and add more later. If your library has a strategic plan or you have identified

‡ https://acrl.projectoutcome.org/pages/4

goals you would like to achieve, the service area(s) you choose to survey should align with those plans. For help, please visit Choosing the Right Survey.*

Why were surveys chosen as the primary tool to measure outcomes?

Project Outcome's approach is to provide the simplest format to engage libraries in measuring outcomes and to ensure the library is capturing feedback consistently. Early pilot testing showed the shorter and more concise the survey, the more patrons were willing to respond. However, survey implementation is up to the library. As long as they stay consistent with the standardized questions, libraries may choose paper or online surveys, phone or in-person interviews, or group discussions to gather responses.

What will happen to the outcomes data that my library collects?

Once survey results are submitted into the survey management tool, you will be able to access your raw data and build custom reports. The data will be aggregated on the data dashboard for further analysis and reporting. Individual library data submitted will not be made public, rather, Project Outcome will aggregate and analyze the data to demonstrate the total impact of programs and services nationally.

I want to measure surveys across several libraries as a single library group. How do I do that?

Project Outcome offers group accounts for libraries, which allow a central account to act as administrator for their member libraries, and will allow administrators to create and enter responses for surveys, as well as create templates for member libraries to use on their own. To create a group account, CONTACT US at acrl@projectoutcome.org.

I am from a library outside the United States. How will my data be aggregated?

Project Outcome is available to all academic libraries internationally. International library data will be aggregated at the country level.

* https://acrl.projectoutcome.org/surveys-resources/choosing-the-right-survey-s-for-your-library

SURVEYS & SURVEY MANAGEMENT

Where do I access the surveys?

All surveys are created and published in the survey management tool. The survey tool allows libraries to create their surveys based on their program needs. All Project Outcome surveys are standardized and un-editable, but libraries have the ability to customize their survey with the program name, library logo, custom messaging, and up to three additional questions. Preview the surveys here.†

Why aren't my locations showing up?

Check your Account page to add or edit library locations. The Account page can be accessed by clicking the Account link in the upper-right corner of the homepage. If there are no locations available, click the "Add a Location" Button. Any locations in your Account will display as a response location option in your survey.

Are the surveys available in other languages?

The surveys can be administered in English, Spanish, or French. Google Translate is integrated throughout the site to allow you to view content in other languages. ACRL and PLA are working on assessing the field's demand for other language translations of the surveys. If you have translated a survey and are willing to share your translation, please email us at acrl@projectoutcome.org and we can add it to the resources for other users.

Can we change the survey questions? If not, why?

The survey questions have been designed and tested specifically to be reliable measures of perceived impact. The goal is to ensure that Project Outcome can collect consistent baseline data from all participating libraries. The standardized surveys cannot be changed, but libraries have the ability to add up to three additional questions of their own.

How do I add questions to the survey?

Adding questions to Project Outcome's standardized surveys is allowed but discouraged. Libraries can add up to three questions and can choose from a pre-determined list of commonly asked questions or write their own open-ended questions. The surveys have been designed to maximize patron response rates through simple wording, survey design and formatting. Adding questions to your survey may negatively impact patron

† https://acrl.projectoutcome.org/surveys-resources/outcome-measurement-survey-questions

participation rates and increase survey abandonment rates. For more information, please visit Writing Open-Ended Survey Questions.*

What number of respondents is considered a successful data collection? Should my library offer an incentive?

Project Outcome is designed to collect insights and data about the ways libraries are meeting the needs of our communities without rigorous data collection requirements. Regardless of number of surveys completed, your goal should always be to achieve high response rates (based on total attendance). The more data you collect, the better; however, you can report any of your findings as long as you are transparent about the number of responses received and attendance totals. Incentives were not offered in the pilot-test stages of the PLA Project Outcome surveys and libraries still received a high response rate. The response rate typically drops when conducting follow-up surveys, which libraries should be transparent about. If incentives are given to patrons, make sure your library includes that when reporting the data results. For more information, please visit the Survey Best Practices resource.

My institution has several libraries. Are the survey responses tied to library locations?

Yes, all survey responses will be assigned to a location. Libraries will have the option to pre-set all survey responses to a specific location or allow patrons to select their location. Libraries can also set custom locations, such as for an off-site event. This is done under Step 2: Program Information in the survey creation process. Responses by location can be found in the raw data .CSV download and in the report builder and outlet selection menu of the data dashboard.

I am part of a Group Account. What's the difference between a survey and a template? How do I create a survey for my Group account?

Project Outcome for Academic Libraries has two types of group users: template-only access, and full access. Both types allow group users to create and administer surveys on behalf of their member libraries. They also have the option to create templates to publish to their member libraries to administer on their own. Groups can edit and control surveys they create, and surveys created from their templates. Template-only access users will only

* https://acrl.projectoutcome.org/surveys-resources/writing-open-ended-questions

be able to view data results from surveys created by their member libraries. Full access users can view *all* data from their member institutions.

Survey templates allow groups to create a locked survey template that member libraries can use. Templates include "initiatives" that act as the group's program name. All surveys created from the same template will aggregate up to the initiative name.

Group users have the ability to create surveys on behalf of their member libraries. This is helpful if the group libraries that don't have the capacity to create surveys and enter data on their own, or if the group's members are separate institutions but their libraries are administered centrally (for example, a community college district). Groups also have the ability to create surveys for their own accounts. To do so, they will select their group name in the location selection menu for the survey or template. This will tie all data from that survey to the group account and will not be assigned to any member libraries.

Can I upload survey responses?

There are four ways of entering survey response data in Project Outcome:

1. Use a URL to administer the surveys, and data automatically will appear in the system as soon as users respond.
2. Enter responses from paper surveys individually.
3. Submit multiple survey responses at once using batch entry.
4. Bulk upload survey responses from Excel. If doing this, you must download and use the template specific to the survey you created and responses must exactly match the formatting programmed into that template.

How often should I survey an ongoing program?

It is up to the library to determine survey periods. Depending on how you want to report your data later, you may want to try a few techniques until you find what works best for you and your staff. A few examples of how you may want to make that decision: 1) If you want to use the reports for a monthly update on the program (generally), you may want to measure several programs each month; 2) If you are trying to measure for a grant application, you may want to measure that program frequently during the time period leading up to the grant application due date.

How do I combat survey fatigue?

Libraries should emphasize the importance of measuring outcomes to patrons. If they have answered surveys in the past about technology use or collections, explain how Project Outcome surveys measure the actual benefits (outcomes) of programs and services, rather than numbers, and how feedback will help improve your impact on the community. In addition, you should never survey the same person twice for a single survey period, so

we recommend making a surveying schedule. For more information, please visit How to Talk to Patrons.*

What should I do if I suspect a patron has entered their survey answers incorrectly?

First and foremost, the best way to ensure that surveys are completed accurately is to give clear instructions to patrons about how to fill out the surveys. Regardless, with any user-reported responses you may not always have 100% complete or accurate answers. If you and your team suspect a patron misunderstood the Likert scale or wrote their scores incorrectly, you can either enter the data as it has been answered or discard the survey in its entirety and not include it in the total response rate. Do not try to edit or interpret the answers.

How do I delete a survey or responses?

Once a response is submitted to a survey, individual responses cannot be edited or deleted. The whole survey needs to be deleted. If a library has entered the data incorrectly, they can start over by copying the survey, re-entering responses, and deleting the original survey and its responses. To delete a survey, change the survey status to "Deleted." Underneath each survey name, there is a survey status listed. Below, there is a dropdown menu that allows a library to change the survey status (Draft, Archived, Deleted). Use the filters on the survey management tool to find deleted surveys. Survey statuses can be changed at any time.

How do I edit my survey?

To edit a survey, go to the survey management page and click the title of the survey. You'll be brought to the survey detail page, where you can find different tabs to edit your survey's information. You can edit a survey until a response is collected for that survey. Once responses are collected, the editable fields of the survey will be locked.

How should I schedule my follow-up survey? Do I need to collect the immediate survey before doing a follow-up?

The immediate survey will serve as a nice baseline of data if you are following up with patrons. However, immediate surveys are not required to use follow-up surveys. We recommend following up with patrons 4-8 weeks after a program or service is completed,

* https://acrl.projectoutcome.org/surveys-resources/how-to-talk-to-patrons-about-project-outcome-surveys

but this may vary depending on the program or service you are measuring. For recommendations and planning support, visit the Follow-Up Survey Protocol.†

REPORTS & DATA DASHBOARD

How do I build a report for my survey?

Click "Build a Report" on the Project Outcome homepage or on the survey management tool to start building a PDF summary report. The report builder allows libraries to build custom reports to fit their needs. Use the filters to aggregate and disaggregate your data by the following criteria: survey topic, survey type, date range, program name, or survey name. To get a quick summary report for a specific survey, click "Get Report" next to the survey name on the survey management tool. This will allow you to skip choosing your report criteria and will only pull data from that specific survey.

What do the averages on my summary report represent?

Filtered Average / Library (left of graph): These numbers represent the averages of survey topic + outcome type (knowledge, confidence, application, awareness) for any filtered criteria selected for that report. For example, if you created a report for the Instruction immediate survey and applied a date range and selected English 101 to run under that report, the filtered average library will display the average of all of those combined results from your library's datasets - Instruction + immediate + date range + English 101.

Topic/Outcome Averages / Library, Carnegie Class, National (right of graph): Library average represents the total average score from all survey responses for your library under the survey topic + outcome type (e.g. Instruction + Knowledge). The Carnegie Class and national averages represent total results for that survey topic + outcome type across all institutions in that basic class and across the country. The averages displayed to the right of the graph are not affected by any filters you put on the report (like date range or program name); they represent all data aggregated by survey topic & outcome type.

How do I save my summary report and data dashboards as a PDF?

To save a report as a PDF, click "Preview Report," then "Confirm & Print." Instead of printing, choose Adobe PDF as your printer to be able to save the report as a PDF. Once "Confirm & Print" is selected for a report, it is automatically saved in the Report Builder

† https://acrl.projectoutcome.org/surveys-resources/follow-up-survey-protocol

under “Print Past Reports.” To save a data dashboard page as a PDF, go to print the page and save as PDF instead.

Where are my open-ended responses?

You can find open-ended responses in the .CSV raw data download available in the survey management tool, or in the “Filter & Export Open-Ended Responses” button on the Detail data dashboard. This button will allow you to filter by survey topic, type, program name, survey name, or date. Export options at the top of the window allow you to export all or select a few comments to export as a .CSV file. The results from open-ended responses can be added to the PDF Summary Reports by clicking the “Filter & Export Open-Ended Responses” button on the Print Report page.

How do I filter my data dashboards by library location?

Use the library location dropdown menu at the top of the page to select all libraries or a specific library location to filter results. The dropdown menu should appear below the dashboard tabs (Dashboard Home, Build Report, Overview, Matrix, Detail, Map, Library Info).

Why is no data showing up in my Map dashboard?

Check your Account page to add or edit your library locations. The Account page can be accessed by clicking the Account link in the upper-right corner of the homepage. Locations must have a physical address to display properly on the Map dashboard.

APPENDIX B

Introduction to Project Outcome



Project Outcome is a FREE online toolkit designed to help libraries understand and share the impact of essential library programs and services by providing simple surveys and an easy-to-use process for measuring and analyzing outcomes.

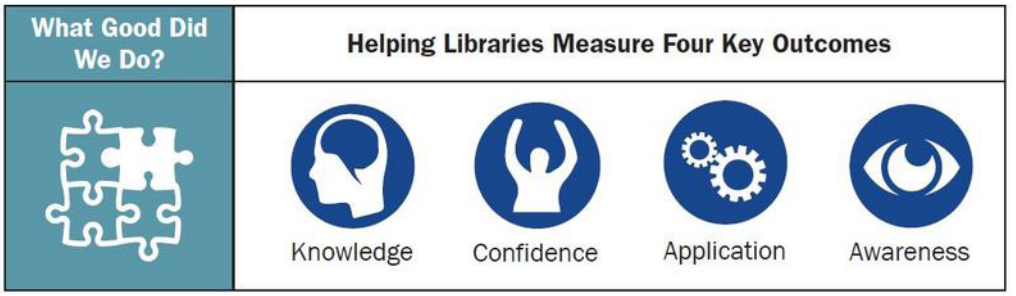
Academic libraries have free access to standardized outcome measures that allow them to measure consistently within their institution and library and with participating institutions across the country, being able to compare data over time and a national level.

While many academic libraries collect data about their programs and services, like needs assessments, patron satisfaction surveys and output measures, what is often lacking are the data to support the learning that happens in libraries and the benefits programs and services provide to patrons. That's where measuring outcomes can help.

An outcome is a *specific benefit* a patron receives from a library program or service. Outcomes can be quantitative or qualitative, and are expressed as changes that individuals perceive in themselves.

Project Outcome's surveys help libraries quantify what good they did by measuring if patrons learned something new, gained confidence, changed their behavior, or increased awareness of library resources. Libraries will also learn what their patrons liked most and what the library can do to improve.

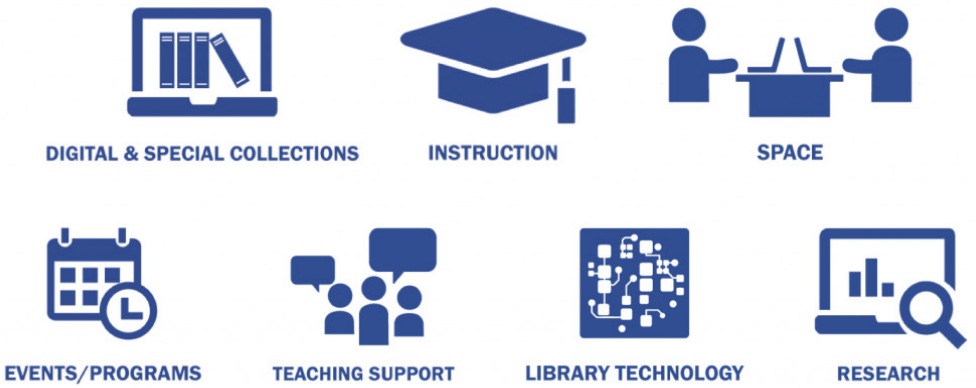
Combined, these outcomes help libraries determine whether and how their programs and services benefit patrons and make an impact on their lives.



SURVEYS

Project Outcome provides two types of surveys for libraries to use.

- *Immediate surveys* are designed to be administered immediately after a program or service is completed and help libraries better understand their immediate impact on patrons.
- *Follow-up surveys* are designed to be administered after a period of time has passed, and help libraries understand if patrons have changed their behavior or continued to benefit as a result of the program or service.



Project Outcome for Academic Libraries provides surveys in seven key library service areas: Instruction, Events/Programs, Research, Teaching Support, Digital & Special Collections, Space, and Library Technology.

These service areas allow libraries to measure a variety of programs and services—from classroom instruction, to reference consultations, to group study rooms, to laptop checkouts.

Academic libraries that register for Project Outcome have access to a variety of tools and resources, including the standardized patron surveys, the survey management tool, data dashboard visualizations, the custom report builder, training resources, a peer discussion board, and in-person and online learning opportunities.

The survey and data analysis tools are available for all academic and research libraries and LIS students globally. Participants who do not work at a library have access to the free resources and peer discussion board, where they can learn more about outcome measurement and adapt it in their organizations.

All the standardized surveys can be accessed from within the *survey management tool*. Libraries create their surveys* by choosing the survey topic and survey type, which will pre-load the survey with the standardized Project Outcome questions.

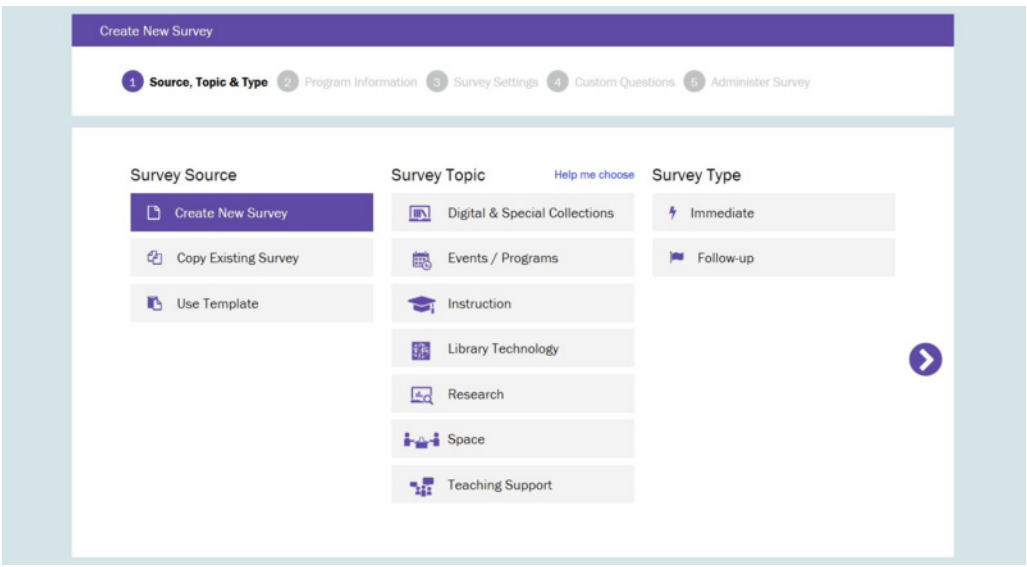
Libraries will be given options to customize their surveys by inserting their specific program information, library logo, custom messaging, and by including up to three custom questions. Choose from a list of pre-populated additional questions† or write your own.‡

The survey management tool makes it easy for libraries to manage their survey activity, edit surveys, and get patron surveys. From the tool, they can download paper surveys, and access links to online surveys they have created. Once surveys have been administered, libraries can easily enter survey data and access their results.

* https://acrl.projectoutcome.org/modyules/122

† https://acrl.projectoutcome.org/surveys-resources/additional-survey-questions

‡ https://acrl.projectoutcome.org/surveys-resources/writing-open-ended-questions



REVIEWING AND SHARING RESULTS



Survey results are automatically aggregated into several reporting tools, making it easy to review, interpret, analyze and share them.

Reports come in two forms: get a quick PDF summary report* directly from the survey management tool or create a custom PDF report† from the report builder tool in the data dashboard. For the latter, the data can be filtered and aggregated based on your needs.

Interactive data visualizations are available in the *data dashboards*. These provide different ways for you to analyze and interpret your data based on survey topic, outcome type, program name and date. Academic libraries see how their results compare to other users, aggregated by Carnegie Class and national averages.

You can also choose to download raw data in CSV format for deeper analysis.

Project Outcome provides a wide variety of training and information resources‡ full of helpful tips, guidelines, case studies, and videos.

Project Outcome users can participate in a peer discussion board§ to interact, share ideas and resources, ask questions, and share their experiences with other users.

* https://acrl.projectoutcome.org/modyules/141

† https://acrl.projectoutcome.org/modyules/142

‡ https://acrl.projectoutcome.org/surveys-resources

§ https://connect.ala.org/acrl/communities/community-home?CommunityKey=58122b3e-c4d1-4388-be25-018636f037aa

APPENDIX C

Survey Text (Immediate Surveys)

INSTRUCTION

Please take a few minutes for this brief survey and let us know if, as a result of participating in the [program name] session…

1. I learned something new that will help me succeed in my classes.
2. I feel more confident about completing my assignment(s).
3. I intend to apply what I just learned.
4. I am more aware of the library's resources and services.
5. What did you like most about this session?
6. What *else* could the library do to help you succeed in your classes?

EVENTS / PROGRAMS

Please take a few minutes for this brief survey and let us know if, as a result of participating in the [program name] event/program…

1. I learned something new from this event or program.
2. I feel more confident about using what I just learned.
3. I intend to discuss or share with others what I just learned.
4. I am more aware of the library's programs and services.
5. What did you like most about this event / program?
6. What *other* types of events or programs would you be interested in attending?

RESEARCH

Please take a few minutes for this brief survey and let us know if, as a result of getting research assistance from the library…

1. I learned something new that will help me with my research.
2. I feel more confident about my ability to conduct my research.
3. I intend to apply what I learned to my research.
4. I am more aware of the library's resources and services.
5. What did you like most about this research service?
6. What *else* could the library do to help you with your research?

TEACHING SUPPORT

Please take a few minutes for this brief survey and let us know if, as a result of getting teaching support from the library…

1. I learned something new that will be helpful in developing my course(s).
2. I feel more confident about incorporating library resources into my course(s).
3. I will apply the techniques and/or information I learned to future courses.
4. I am more aware of the library's resources and services.
5. What did you like most about this program/service?
6. What *else* could the library do to help you with your teaching?

DIGITAL & SPECIAL COLLECTIONS

Please take a few minutes for this brief survey and let us know if, as a result of using this collection or resource…

1. I feel more knowledgeable about the topic supported by the resource I used.
2. I feel more confident about my ability to use this resource or collection.
3. I intend to apply what I just learned from the resource I used.
4. I am more aware of the library's resources and collections.
5. What did you like most about this resource or collection?
6. What *else* could the library do to help you better use its resources and collections?

SPACE

Please take a few minutes for this brief survey and let us know if, as a result of using this space…

1. This space contributed to my ability to learn something new.
2. Using this space makes me feel more confident about my ability to achieve my goals.
3. I am likely to use this space again in the future.
4. After using this space, I am more aware of the library resources and services available to me.
5. What did you like most about this space?

6. What could the library do to improve this space?

LIBRARY TECHNOLOGY

Please take a few minutes for this brief survey and let us know if, as a result of using this technology…

1. I gained experience that contributed to my knowledge of this technology.
2. I feel more confident about integrating technology into my assignments or projects.
3. I intend to use technology the library provides again in future.
4. I am more aware of the library's technology resources and services.
5. What did you like most about this technology?
6. What could the library do to help you better use technology?

APPENDIX D

Survey Text (Follow-Up Surveys)

INSTRUCTION

As a result of participating in this library instruction session…

1. …I used what I learned to complete my assignment(s).
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
2. …I did better in my class(es).
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I used other library resources or services.
   - ○ Yes
     - □ What other resource or service did you use?
   - ○ No
   - ○ N/A

   Please Explain:
4. What did you find most helpful from this session?
5. What could the library do to help you learn more?

EVENTS / PROGRAMS

As a result of participating in this event or program...

1. …I used what I learned to try something new or different.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
2. …I discussed or shared what I learned with others.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I attended another library event / program.
   - ○ Yes
     - □ What other event(s) or program(s) did you attend?
   - ○ No
   - ○ N/A

   Please Explain:
4. What did you like most about this event / program?
5. What other types of events / programs would you be interested in attending?

RESEARCH

As a result of getting research assistance from the library…

1. …I used what I learned to complete a research task or goal.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
2. …I applied what I learned to other research tasks or goals.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I used another library resource or service.
   - ○ Yes

- □ What other resource or service did you use?

- ○ No
- ○ N/A

Please Explain:

4. What did you like most about this research service?
5. What could the library do to continue to help you with your research?

TEACHING SUPPORT

As a result of getting teaching support from the library…

1. … I incorporated library resources into my course(s).
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
2. …I used what I learned to enhance my course(s).
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I used another library resource or service.
   - ○ Yes
     - □ What other resource or service did you use?
   - ○ No
   - ○ N/A

   Please Explain:
4. What was most helpful from this teaching support service?
5. What could the library do to continue to help support your teaching?

DIGITAL & SPECIAL COLLECTIONS

As a result of using this resource or collection…

1. …I used what I learned to complete a task or goal.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:

2. …I incorporated it into my research or coursework.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I used another library resource or collection.
   - ○ Yes
     - □ What other resource or collection did you use?
   - ○ No
   - ○ N/A

   Please Explain:
4. What did you like most about this resource or collection?
5. What could the library do to help you learn more?

SPACE

As a result of using this space…

1. …I was better able to complete a task or goal.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
2. …I returned to use this space again.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I used other library resources or services.
   - ○ Yes
     - □ What other resource or service did you use?
   - ○ No
   - ○ N/A

   Please Explain:
4. What did you like most about this space?
5. What could the library do to help you continue to complete your tasks or goals?

LIBRARY TECHNOLOGY

As a result of using this technology…

1. …I was able to do my work more effectively.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
2. …I incorporated it into an assignment or project.
   - ○ Yes
   - ○ No
   - ○ N/A

   Please Explain:
3. …I used other technology resources or services the library provides.
   - ○ Yes
     - □ What other technology resource or service did you use?
   - ○ No
   - ○ N/A

   Please Explain:
4. What was most helpful about this technology?
5. What could the library do to help you learn more?

APPENDIX E

List of Schools Who Have Used Project Outcome

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Abilene Christian University	Texas	United States	Doctoral Universities	Doctoral/Professional Universities
Adelphi University	New York	United States	Doctoral Universities	Doctoral/Professional Universities
AdventHealth University	Florida	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Agnes Scott College	Georgia	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Alabama State University	Alabama	United States	Doctoral Universities	Doctoral/Professional Universities
Alamance Community College	North Carolina	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
American College of Thessaloniki		Greece	International	
American InterContinental University-Online	Illinois	United States	Master's College & Universities	
American University	District of Columbia	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
American University in Bulgaria		Bulgaria	International	
American University in Cairo		Egypt	International	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
American University of Sharjah		United Arab Emirates	International	
Anderson University-Anderson	Indiana	United States	Master's College & Universities	
Anderson University-Indiana	Indiana	United States of America	Associate's Colleges / 2-Year Institutions	
Appalachian State University	North Carolina	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Arcadia University	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Aristotle University of Thessaloniki		Greece	International	
Arizona State University-Tempe	Arizona	United States of America	Associate's Colleges / 2-Year Institutions	
Arizona Western College	Arizona	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Arkansas Tech University	Arkansas	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Augsburg College	Minnesota	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Augusta University	Georgia	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Aurora University	Illinois	United States of America	Doctoral Universities	Doctoral/Professional Universities
Austin College	Texas	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Baldwin Wallace University	Ohio	United States of America	Associate's Colleges / 2-Year Institutions	
Ball State University	Indiana	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Barton College	North Carolina	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Bastyr University	Washington	United States	Doctoral Universities	Doctoral/Professional Universities
Bates College	Maine	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Baylor University	Texas	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Bellevue University	Nebraska	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Belmont University	Tennessee	United States of America	Doctoral Universities	Doctoral/Professional Universities
Bentley University	Massachusetts	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Berea College	Kentucky	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Bergen Community College	New Jersey	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Berkeley College-New Jersey	New York	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Berkeley College-New York	New York	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Berry College	Georgia	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Bethel College-North Newton	Kansas	United States	Baccalaureate Colleges	
Bethel University-McKenzie	Tennessee	United States	Master's College & Universities	
Bethel University-Saint Paul	Minnesota	United States of America	Doctoral Universities	
Biola University	California	United States	Doctoral Universities	Doctoral/Professional Universities
Black Hills State University	South Dakota	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Bloomsburg University of Pennsylvania	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Blue Mountain Community College	Oregon	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Boston University	Massachusetts	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Botswana University of Agriculture and Natural Resources				

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Bowling Green State University-Main Campus	Ohio	United States of America	Associate's Colleges / 2-Year Institutions	
Bradley University	Illinois	United States of America	Doctoral Universities	Doctoral/Professional Universities
Brescia University	Kentucky	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Briar Cliff University	Iowa	United States of America	Doctoral Universities	Doctoral/Professional Universities
Brigham Young University-Idaho	Idaho	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Brightpoint Community College	Virginia	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Brock University	Ontario	Canada	International	
Bryan College-Dayton	Tennessee	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Bryant & Stratton College-Online	New York	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Cabrini University	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Caldwell Community College and Technical Institute	North Carolina	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
California Institute of the Arts	California	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
California Polytechnic State University-San Luis Obispo	California	United States of America	Associate's Colleges / 2-Year Institutions	
California State University-Dominguez Hills	California	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
California State University-Long Beach	California	United States of America	Associate's Colleges / 2-Year Institutions	
California State University-Monterey Bay	California	United States of America	Associate's Colleges / 2-Year Institutions	
Calvin University	Michigan	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Campbell University	North Carolina	United States of America	Doctoral Universities	Doctoral/Professional Universities
Canisius College	New York	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Cardiff University	Cardiff	United States		
Carnegie Mellon University	Pennsylvania	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Carnegie Mellon University in Qatar		Qatar	International	
Carolina University	North Carolina	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Casper College	Wyoming	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Catawba Valley Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Catholic University of America	District of Columbia	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Cedarville University	Ohio	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Cégep Limoilou	Québec	Canada	International	
Centenary College of Louisiana	Louisiana	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Central Connecticut State University	Connecticut	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Central Piedmont Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Central Wyoming College	Wyoming	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Centre College	Kentucky	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Chaffey College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Chandler-Gilbert Community College	Arizona	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Charleston Southern University	South Carolina	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Chattanooga State Community College	Tennessee	United States of America	Associate's Colleges / 2-Year Institutions	
Chowan University	North Carolina	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
City Colleges of Chicago-Harry S Truman College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
City Colleges of Chicago-Kennedy-King College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Clarion University of Pennsylvania	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Clayton State University	Georgia	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Clemson University	South Carolina	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Cleveland Institute of Art	Ohio	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Arts, Music & Design Schools
Colgate University	New York	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
College of Central Florida	Florida	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
College of Charleston	South Carolina	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
College of Saint Benedict	Minnesota	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
College of Southern Nevada	Nevada	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
College of the Albemarle	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
College of the Sequoias	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
College of William and Mary	Virginia	United States of America	Associate's Colleges / 2-Year Institutions	
Colorado Mountain College	Colorado	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Colorado State University-Fort Collins	Colorado	United States of America	Associate's Colleges / 2-Year Institutions	
Colorado Technical University-Online	Colorado	United States of America	Master's Colleges & Universities	
Columbia Theological Seminary	Georgia	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Columbus State Community College	Georgia	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Community College of Aurora	Colorado	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Community College of Rhode Island	Rhode Island	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Conception Seminary College	Missouri	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Concordia University-Saint Paul	Minnesota	United States of America	Associate's Colleges / 2-Year Institutions	
Conestoga College	Ontario	Canada	International	
Connecticut College	Connecticut	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Connecticut State Colleges & Universities Library Consortium	Connecticut	United States of America	Not Classified	
Contra Costa College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Covenant Theological Seminary	Missouri	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Crafton Hills College	California	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Craven Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
CUNY Hostos Community College	New York	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Curry College	Massachusetts	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Cuyamaca College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Dallas Baptist University	Texas	United States of America	Doctoral Universities	Doctoral/Professional Universities
Dallas Theological Seminary	Texas	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Davidson College	North Carolina	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Davidson County Community College	North Carolina	United States	Associate's Colleges / 2-year Institutions	
Daymar College - Murfreesboro	Tennessee	United States of America	Baccalaureate Colleges	
Delta State University	Mississippi	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Denmark Technical College	South Carolina	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-High Nontraditional
Denver Seminary	Colorado	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
DeSales University	Pennsylvania	United States of America	Doctoral Universities	Doctoral/Professional Universities
DeVry University-Illinois	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	
Diablo Valley College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Dickinson College	Pennsylvania	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Drake University	Iowa	United States of America	Doctoral Universities	Doctoral/Professional Universities
Duke University	North Carolina	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
D'Youville College	New York	United States of America	Doctoral Universities	Doctoral/Professional Universities
Earlham College	Indiana	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
East Carolina University	North Carolina	United States	Doctoral Universities	Doctoral Universities: High Research Activity
East Central University	Oklahoma	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
East Georgia State College	Georgia	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
East Los Angeles College	California	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Eastern Arizona College	Arizona	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Eastern Connecticut State University	Connecticut	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Eastern Illinois University	Illinois	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Eastern Kentucky University	Kentucky	United States	Doctoral Universities	Doctoral/Professional Universities
Eastern New Mexico University-Main Campus	New Mexico	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Edinboro University of Pennsylvania	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
El Centro College	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
Elizabethtown College	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Elmhurst University	Illinois	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Elon University	North Carolina	United States	Doctoral Universities	Doctoral/Professional Universities
Emory University	Georgia	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Emporia State University	Kansas	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Erie Community College	New York	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Everett Community College	Washington	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Fairfield University	Connecticut	United States	Doctoral Universities	Doctoral/Professional Universities
Fairleigh Dickinson University-College at Florham	New Jersey	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Farmingdale State College	New York	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Fashion Institute of Technology	New York	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Felician University	New Jersey	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Finger Lakes Community College	New York	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Finlandia University	Michigan	United States of America	Baccalaureate Colleges	
Florida Gulf Coast University	Florida	United States	Doctoral Universities	Doctoral/Professional Universities
Florida Southern College	Florida	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Fordham University	New York	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Framingham State University	Massachusetts	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Franklin College	Indiana	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Fresno City College	California	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Gannon University	Pennsylvania	United States of America	Doctoral Universities	Doctoral/Professional Universities
Garrett-Evangelical Theological Seminary	Illinois	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Gaston College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Gateway Community College-New Haven	Connecticut	United States of America	Associate's Colleges / 2-Year Institutions	
GateWay Community College-Phoenix	Arizona	United States of America	Associate's Colleges / 2-Year Institutions	
Gateway Seminary	California	United States of America	Not classified	
Gateway Technical College	Wisconsin	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-Mixed Traditional/ Nontraditional
Genesee Community College	New York	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
George Mason University	Virginia	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
George Washington University	District of Columbia	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Georgetown College	Kentucky	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Georgetown University in Qatar		Qatar	International	
Georgia College and State University	Georgia	United States of America	Associate's Colleges / 2-Year Institutions	
Georgia Highlands College	Georgia	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Georgia Southern University	Georgia	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Georgia State University	Georgia	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Gettysburg College	Pennsylvania	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Glen Oaks Community College	Michigan	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical- High Nontraditional
Golden Gate University-San Francisco	California	United States of America	Associate's Colleges / 2-Year Institutions	
Golden West College	California	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer- Mixed Traditional/ Nontraditional
Goshen College	Indiana	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Grace College and Theological Seminary	Indiana	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Grand Canyon University	Arizona	United States of America	Doctoral Universities	Doctoral/Professional Universities
Grand Rapids Community College	Michigan	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical- High Traditional
Gulf Coast State College	Florida	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Gwinnett Technical College	Georgia	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical- Mixed Traditional/ Nontraditional
Habib University		Pakistan	International	
Hagerstown Community College	Maryland	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical- Mixed Traditional/ Nontraditional
Hanover College	Indiana	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Harding University	Arkansas	United States of America	Doctoral Universities	Doctoral/Professional Universities

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Harford Community College	Maryland	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Harrisburg Area Community College-Harrisburg	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
Harrisburg University of Science and Technology	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Harvard University	Massachusetts	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Hawaii Pacific University	Hawaii	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Hazard Community and Technical College	Kentucky	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-High Nontraditional
Heartland Community College	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Hilbert College	New York	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Hill College	Texas	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Hillsborough Community College	Florida	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Hillsdale College	Michigan	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Hiram College	Ohio	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Hollins University	Virginia	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Hong Kong Baptist University	Hong Kong	China	International	
Hope College	Michigan	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Howard University	District of Columbia	United States	Doctoral Universities	Doctoral Universities: High Research Activity

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Huntington University	Indiana	United States of America	Master's Colleges & Universities	
IE University	MÃ¡laga			
Illinois Central College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Illinois State University	Illinois	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Indian River State College	Florida	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Indiana University-Kokomo	Indiana	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Institute of American Indian and Alaska Native Culture	New Mexico	United States of America	Associate's Colleges / 2-Year Institutions	
Iowa State University	Iowa	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Ithaca College	New York	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Ivy Tech Community College	Indiana	United States of America	Associate's Colleges / 2-Year Institutions	
J Sargeant Reynolds Community College	Virginia	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-High Nontraditional
Jackson State Community College	Tennessee	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Jacksonville University	Florida	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
James Sprunt Community College	North Carolina	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Jefferson College	Missouri	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
John Cabot University		Italy	International	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Johnson & Wales University-Providence	Rhode Island	United States of America	Associate's Colleges / 2-Year Institutions	
Judson University	Illinois	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Juniata College	Pennsylvania	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Kalamazoo College	Michigan	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Kankakee Community College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Kansas State University	Kansas	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Kapiolani Community College	Hawaii	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Kaskaskia College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Kent State University at Stark	Ohio	United States of America	Associate's Colleges / 2-Year Institutions	
Keuka College	New York	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Khawarizmi International College		United Arab Emirates	International	
Kirkwood Community College	Iowa	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-Mixed Traditional/ Nontraditional
Kishwaukee College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Kutztown University of Pennsylvania	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
La Salle University	Pennsylvania	United States	Doctoral Universities	Doctoral/Professional Universities

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
LaGrange College	Georgia	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Lake Forest College	Illinois	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Lake Superior State University	Michigan	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Langara College	British Columbia	Canada	International	
Lansing Community College	Michigan	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Larkin University	Florida	United States	Not classified	
Lawson State Community College-Birmingham Campus	Alabama	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Lebanon Valley College	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Lehigh University	Pennsylvania	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Leiden University	Iles Loyaute			
Lenoir Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Lewis University	Illinois	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Liberty University	Tennessee	United States	Doctoral Universities	Doctoral/Professional Universities
Lille Institute of Political Studies		France	International	
Lincoln Christian University	Illinois	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Linköping University		Sweden	International	
Lone Star College System	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
Longwood University	Virginia	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Los Angeles City College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Louisiana State University and Agricultural & Mechanical College	Louisiana	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Louisiana State University-Shreveport	Louisiana	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Loyola University Chicago	Illinois	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Loyola University New Orleans	Louisiana	United States	Doctoral Universities	Doctoral/Professional Universities
Lycoming College	Pennsylvania	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Machakos University		Kenya	International	
Manchester Community College-Manchester	Connecticut	United States	Associate's Colleges / 2-year Institutions	
Manchester Community College-Manchester, NH	New Hampshire	United States of America	Associate's Colleges / 2-Year Institutions	
Manchester University	Indiana	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Mandalay University	Mandalay			
Mary Baldwin University	Virginia	United States of America	Doctoral Universities	Doctoral/Professional Universities
Maryland Institute College of Art	Maryland	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Arts, Music & Design Schools
Marymount University	Virginia	United States	Doctoral Universities	Doctoral/Professional Universities
Massachusetts College of Liberal Arts	Massachusetts	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Massey University		New Zealand	International	
Mayland Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
McDaniel College	Maryland	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
McLennan Community College	Texas	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
McNeese State University	Louisiana	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
MCPHS University	Massachusetts	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Medical University of Graz	Steiermark			
Mercer County Community College	New Jersey	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Mercer University	Georgia	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Mercyhurst University	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Messiah College	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
Methodist College	Illinois	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
MGH Institute of Health Professions	Massachusetts	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Miami University-Middletown	Ohio	United States of America	Baccalaureate Colleges	
Michigan Technological University	Michigan	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Mid-Atlantic Christian University	North Carolina	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Middle Tennessee State University	Tennessee	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Middlesex Community College-Bedford	Massachusetts	United States of America	Associate's Colleges / 2-Year Institutions	
Middlesex Community College-Middletown	Connecticut	United States of America	Associate's Colleges / 2-Year Institutions	
Midland College	Texas	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Midland University	Nebraska	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Mid-South Christian College	Tennessee	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Midwestern Baptist Theological Seminary	Missouri	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Midwestern University-Downers Grove	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	
Millersville University of Pennsylvania	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Millikin University	Illinois	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Mississippi State University	Mississippi	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Mitchell College	Connecticut	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Mitchell Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Monmouth College	Illinois	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Montana State University	Montana	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Montgomery College	Maryland	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Moody Bible Institute	Illinois	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Moore College of Art and Design	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
Morton College	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-Mixed Traditional/ Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Mount Aloysius College	Pennsylvania	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Muranga University of Technology		Kenya	International	
Muskingum University	Ohio	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Nash Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
National Louis University	Illinois	United States	Doctoral Universities	Doctoral/Professional Universities
National University	California	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
National University of Villa MarÃa	CÃ³rdoba		Not classified	
Naugatuck Valley Community College	Connecticut	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Nebraska Wesleyan University	Nebraska	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Nevada State College	Nevada	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
New College of Florida	Florida	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
New England College of Optometry	Massachusetts	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
New Mexico Institute of Mining and Technology	New Mexico	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
New Mexico State University-Dona Ana	New Mexico	United States of America	Associate's Colleges / 2-Year Institutions	
New Mexico State University-Main Campus	New Mexico	United States of America	Associate's Colleges / 2-Year Institutions	
New York Institute of Technology	New York	United States of America	Doctoral Universities	
Niagara County Community College	New York	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Nicholls State University	Louisiana	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Norco College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Norfolk State University	Virginia	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
North Carolina Central University	North Carolina	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
North Carolina State University at Raleigh	North Carolina	United States	Doctoral Universities	
North Carolina Wesleyan University	North Carolina	United States of America	Baccalaureate Colleges	
North Central Michigan College	Michigan	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
North Greenville University	South Carolina	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
North Seattle College	Washington	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Northeastern State University	Oklahoma	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Northern Illinois University	Illinois	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Northern Kentucky University	Kentucky	United States of America	Doctoral Universities	Doctoral/Professional Universities
Northshore Technical Community College	Louisiana	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-Mixed Traditional/ Nontraditional
NorthWest Arkansas Community College	Arkansas	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Northwest Missouri State University	Missouri	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Northwestern College-Southwestern Campus	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Northwestern Health Sciences University	Minnesota	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Northwestern State University of Louisiana	Louisiana	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Northwestern University	Illinois	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Norwalk Community College	Connecticut	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Notre Dame de Namur University	California	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Notre Dame University-Louaize		Lebanon	International	
Nova Southeastern University	Florida	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Oakland University	Michigan	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Oberlin College	Ohio	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Ohio State University-Mansfield Campus	Ohio	United States of America	Associate's Colleges / 2-Year Institutions	
Ohio University-Main Campus	Ohio	United States of America	Associate's Colleges / 2-Year Institutions	
Oklahoma State University Institute of Technology	Oklahoma	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Oklahoma State University-Main Campus	Oklahoma	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Old Dominion University	Virginia	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Olympic College	Washington	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Orange County Community College	New York	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Oregon Health & Science University	Oregon	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Research Institution
Oregon Institute of Technology	Oregon	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Otero Junior College	Colorado	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-Mixed Traditional/ Nontraditional
Pace University-New York	New York	United States of America	Doctoral Universities	
Pacific Northwest University of Health Sciences	Washington	United States	Special Focus 4-Year Institutions: Medical	Special Focus Four-Year: Medical Schools & Centers
Palm Beach Atlantic University	Florida	United States	Doctoral Universities	Doctoral/Professional Universities
Palm Beach State College	Florida	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Palomar College	California	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Paradise Valley Community College	Arizona	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Paris Institute of Political Studies (Sciences Po)	ÃŽle-de-France			
Parker University	Texas	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Peirce College	Pennsylvania	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Pellissippi State Community College	Tennessee	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Pennsylvania College of Technology	Pennsylvania	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Pennsylvania Highlands Community College	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Pennsylvania State University-College of Medicine	Pennsylvania	United States of America	Special Focus 4-Year Institutions: All Other	
Pennsylvania State University-Main Campus	Pennsylvania	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Pennsylvania State University-Penn State Abington	Pennsylvania	United States of America	Baccalaureate Colleges	
Pennsylvania State University-Penn State Beaver	Pennsylvania	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Pennsylvania State University-Penn State Erie-Behrend College	Pennsylvania	United States of America	Master's Colleges & Universities	
Pennsylvania State University-Penn State Harrisburg	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
Pennsylvania State University-Penn State York	Pennsylvania	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Peru State College	Nebraska	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Piedmont College	Georgia	United States	Master's College & Universities	
Pikes Peak Community College	Colorado	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Pima Community College	Arizona	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Plymouth State University	New Hampshire	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Point Park University	Pennsylvania	United States	Doctoral Universities	Doctoral/Professional Universities
Polk State College	Florida	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Polytechnic University of Hauts-de-France		France	International	
Pontificia Universidad Javeriana	BogotÃ¡ D.C.	United States	Not classified	
Presbyterian College	South Carolina	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Purdue University-Calumet Campus	Indiana	United States of America	Master's Colleges & Universities	
Quinnipiac University	Connecticut	United States of America	Doctoral Universities	Doctoral/Professional Universities
Quinsigamond Community College	Massachusetts	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Radford University	Virginia	United States of America	Doctoral Universities	Doctoral/Professional Universities
Randolph College	Virginia	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Rangsit University	Pathum Thani			
Red Rocks Community College	Colorado	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Regis University	Colorado	United States of America	Doctoral Universities	Doctoral/Professional Universities
Richard Bland College of the College of William and Mary	Virginia	United States	Associate's Colleges / 2-Year Institutions	Special Focus Two-Year: Other Fields
Richland College	Texas	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Rider University	New Jersey	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Rio Hondo College	California	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Rio Salado College	Arizona	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
River Valley Community College	New Hampshire	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-High Nontraditional
Riverside City College	California	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Roanoke-Chowan Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Robeson Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-High Traditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Rochester Institute of Technology	New York	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Rock Valley College	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Rockhurst University	Missouri	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Rockingham Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Rocky Mountain College of Art and Design	Colorado	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Rose-Hulman Institute of Technology	Indiana	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Engineering and Other Technology-Related Schools
Roseman University of Health Sciences	Nevada	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Rosemont College	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Rowan College at Gloucester County	New Jersey	United States of America	Associate's Colleges / 2-Year Institutions	
Rowan-Cabarrus Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Roxbury Community College	Massachusetts	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Saint Ambrose University	Iowa	United States of America	Doctoral Universities	Doctoral/Professional Universities
Saint Cloud State University	Minnesota	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Saint John Fisher College	New York	United States	Doctoral Universities	Doctoral/Professional Universities
Saint Joseph's University	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Saint Louis Community College	Missouri	United States of America	Associate's Colleges / 2-Year Institutions	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Saint Mary's College of California	California	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Salisbury University	Maryland	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Salt Lake Community College	Utah	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Sam Houston State University	Texas	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Samford University	Alabama	United States	Doctoral Universities	Doctoral/Professional Universities
San Antonio College	Texas	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
San Diego State University	California	United States	Doctoral Universities	Doctoral Universities: High Research Activity
San Francisco State University	California	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
San Jose State University	California	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Santa Barbara City College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Santa Fe College	Florida	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Santa Fe Community College	New Mexico	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Santa Rosa Junior College	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional
Saskatchewan Polytechnic	Saskatchewan			
Schenectady County Community College	New York	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Seminole State College of Florida	Florida	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Seneca College	Ontario	Canada		

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Seton Hill University	Pennsylvania	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Seward County Community College and Area Technical School	Kansas	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Career & Technical-Mixed Traditional/ Nontraditional
Sheridan College	Wyoming	United States	Associate's Colleges / 2-year Institutions	
Simmons College of Kentucky	Kentucky	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Simon Fraser University	British Columbia	Canada	International	
Singapore Management University	Central Singapore		Not classified	
South Dakota State University	South Dakota	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Southcentral Kentucky Community and Technical College	Kentucky	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Southeast Missouri State University	Missouri	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Southeastern University	Florida	United States	Doctoral Universities	Doctoral/Professional Universities
Southern Alberta Institute of Technology	Alberta	Canada	International	
Southern Connecticut State University	Connecticut	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Southern Methodist University	Texas	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Southern Utah University	Utah	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Southern Wesleyan University	South Carolina	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Southwest Baptist University	Missouri	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Southwestern College-Chula Vista	California	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Spalding University	Kentucky	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Spartanburg Community College	South Carolina	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Spokane Community College	Washington	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Springfield College	Massachusetts	United States of America	Doctoral Universities	Doctoral/Professional Universities
St Andrews University	North Carolina	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
St Catherine University	Minnesota	United States of America	Doctoral Universities	Doctoral/Professional Universities
St Charles Community College	Missouri	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
St Mary's College of Maryland	Maryland	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
St Petersburg College	Florida	United States of America	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Mixed Baccalaureate/ Associate's
Stanly Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Stevens Institute of Technology	New Jersey	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Stone Child College	Montana	United States of America	Tribal Colleges & Universities	Tribal Colleges and Universities
Stony Brook University	New York	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Stratford University	Virginia	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Sullivan University	Kentucky	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
SUNY Broome Community College	New York	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-Mixed Traditional/ Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
SUNY College at Brockport	New York	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
SUNY College at Cortland	New York	United States of America	Associate's Colleges / 2-Year Institutions	
SUNY College at Oswego	New York	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
SUNY College of Technology at Delhi	New York	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
SUNY Maritime College	New York	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
Susquehanna University	Pennsylvania	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Sweet Briar College	Virginia	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Tarleton State University	Texas	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Tarrant County College District	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
Texas A & M University-College Station	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
Texas A & M University-Corpus Christi	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
Texas State University	Texas	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Texas Tech University	Texas	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Texas Wesleyan University	Texas	United States of America	Doctoral Universities	Doctoral/Professional Universities
Texas Woman's University	Texas	United States of America	Doctoral Universities	Doctoral/Professional Universities
The Chicago School of Professional Psychology	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	
The College of Saint Scholastica	Minnesota	United States	Doctoral Universities	Doctoral/Professional Universities
The National Willa Cather Center	Nebraska	United States	Not classified	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
The University of Findlay	Ohio	United States	Doctoral Universities	Doctoral/Professional Universities
The University of Hong Kong	Hong Kong			
The University of Montana	Montana	United States	Doctoral Universities	
The University of Queensland	Queensland			
The University of Tennessee-Martin	Tennessee	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
The University of Texas at Austin	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
The University of Texas at El Paso	Texas	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
The University of Texas MD Anderson Cancer Center	Texas	United States of America	Associate's Colleges / 2-Year Institutions	
The University of West Florida	Florida	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Tohono O'Odham Community College	Arizona	United States	Tribal Colleges & Universities	Tribal Colleges and Universities
Touro College	New York	United States of America	Doctoral Universities	
Towson University	Maryland	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Trevecca Nazarene University	Tennessee	United States	Doctoral Universities	Doctoral/Professional Universities
Trident Technical College	South Carolina	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
Triton College	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Trocaire College	New York	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Tulsa Community College	Oklahoma	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Tunxis Community College	Connecticut	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Traditional
Ulster County Community College	New York	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Union of Concerned Scientists	Massachusetts	United States	Research Library	
Union Presbyterian Seminary	Virginia	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Faith-Related Institutions
Union University	Tennessee	United States of America	Doctoral Universities	Doctoral/Professional Universities
United Arab Emirates University	Abu Dhabi		Not classified	
United States Army War College	Pennsylvania	United States	Special Focus 4-year Institutions	
Unitek College	California	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
Universal College of Learning - Whanganui	Manawatu-Wanganui		Not classified	
Université du Québec - Montréal	Quebec	Canada	International	
University of Akron Wayne College	Ohio	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
University of Alabama at Birmingham	Alabama	United States of America	Associate's Colleges / 2-Year Institutions	
University of Alaska Southeast	Alaska	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
University of Arkansas	Arkansas	United States of America	Associate's Colleges / 2-Year Institutions	
University of Arkansas at Little Rock	Arkansas	United States of America	Associate's Colleges / 2-Year Institutions	
University of Arkansas Community College-Morrilton	Arkansas	United States of America	Associate's Colleges / 2-Year Institutions	
University of Calgary	Alberta	Canada	International	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
University of California-Berkeley	California	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of California-Davis	California	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of California-Irvine	California	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of California-Merced	California	United States	Doctoral Universities	Doctoral Universities: High Research Activity
University of California-Riverside	California	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of California-San Diego	California	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of California-Santa Barbara	California	United States of America	Associate's Colleges / 2-Year Institutions	
University of Central Arkansas	Arkansas	United States of America	Doctoral Universities	Doctoral/Professional Universities
University of Central Florida	Florida	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Central Missouri	Missouri	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of Colorado Boulder	Colorado	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Colorado Colorado Springs	Colorado	United States	Doctoral Universities	Doctoral Universities: High Research Activity
University of Colorado Denver	Colorado	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Dallas	Texas	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of Dayton	Ohio	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
University of Detroit Mercy	Michigan	United States of America	Doctoral Universities	Doctoral/Professional Universities
University of Guelph-Humber	Ontario	Canada	International	
University of Hawaii at Manoa	Hawaii	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
University of Hawaii Maui College	Hawaii	United States of America	Associate's Colleges / 2-Year Institutions	
University of Houston-Clear Lake	Texas	United States of America	Doctoral Universities	
University of Idaho	Idaho	United States	Doctoral Universities	Doctoral Universities: High Research Activity
University of Illinois at Chicago	Illinois	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Illinois at Springfield	Illinois	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of Illinois at Urbana-Champaign	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	
University of Iowa	Iowa	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Lethbridge	Alberta	Canada	International	
University of Louisville	Kentucky	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Luxembourg - Belval	Esch-sur-Alzette			
University of Maryland-College Park	Maryland	United States of America	Associate's Colleges / 2-Year Institutions	
University of Massachusetts Global	California	United States of America	Doctoral Universities	Doctoral/Professional Universities
University of Massachusetts-Boston	Massachusetts	United States of America	Associate's Colleges / 2-Year Institutions	
University of Miami	Florida	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Michigan-Flint	Michigan	United States of America	Doctoral Universities	
University of Minnesota-Twin Cities	Minnesota	United States of America	Associate's Colleges / 2-Year Institutions	
University of Mississippi	Mississippi	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Missouri-Kansas City	Missouri	United States of America	Associate's Colleges / 2-Year Institutions	

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
University of Mount Olive	North Carolina	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
University of Nevada-Reno	Nevada	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of New Brunswick	New Brunswick			
University of New Hampshire-Main Campus	New Hampshire	United States of America	Associate's Colleges / 2-Year Institutions	
University of New Mexico-Main Campus	New Mexico	United States of America	Associate's Colleges / 2-Year Institutions	
University of New Orleans	Louisiana	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
University of North Carolina at Chapel Hill	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	
University of North Carolina at Charlotte	North Carolina	United States	Doctoral Universities	Doctoral Universities: High Research Activity
University of North Carolina at Pembroke	North Carolina	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of North Dakota	North Dakota	United States	Doctoral Universities	Doctoral Universities: High Research Activity
University of North Georgia	Georgia	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of North Texas	Texas	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Northern Iowa	Iowa	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of Northwestern-St Paul	Minnesota	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
University of Oklahoma-Health Sciences Center	Oklahoma	United States	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Research Institution
University of Oklahoma-Tulsa	Oklahoma		Doctoral Universities	
University of Oregon	Oregon	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
University of Ottawa	Ontario	Canada	International	
University of Pennsylvania	Pennsylvania	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Pittsburgh-Pittsburgh Campus	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
University of Prince Edward Island	Prince Edward Island	Canada	International	
University of Rochester	New York	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Saint Francis-Fort Wayne	Indiana		Master's College & Universities	
University of Saint Joseph	Connecticut	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of Saint Mary	Kansas	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
University of Saskatchewan	Saskatchewan	Canada	International	
University of Scranton	Pennsylvania	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
University of South Carolina-Aiken	South Carolina	United States	Master's Colleges & Universities	Master's Colleges & Universities: Small Programs
University of South Carolina-Columbia	South Carolina	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of South Carolina-Upstate	South Carolina	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
University of South Florida-Main Campus	Florida	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of St Augustine for Health Sciences	Florida	United States of America	Special Focus 4-Year Institutions: All Other	Special Focus Four-Year: Other Health Professions Schools
University of the District of Columbia	District of Columbia	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
University of the Incarnate Word	Texas	United States	Doctoral Universities	Doctoral/Professional Universities
University of the Philippines Los Banos	Laguna			

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
University of the West Indies	Tunapuna-Piarco		Not classified	
University of Toledo	Ohio	United States	Doctoral Universities	Doctoral Universities: High Research Activity
University of Toronto	Ontario	Canada	International	
University of Tulsa	Oklahoma	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
University of Utah	Utah	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Virginia-Main Campus	Virginia	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Wisconsin-Eau Claire	Wisconsin	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
University of Wisconsin-Madison	Wisconsin	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
University of Wisconsin-Oshkosh	Wisconsin	United States of America	Associate's Colleges / 2-Year Institutions	
University of Wisconsin-Parkside	Wisconsin	United States of America	Associate's Colleges / 2-Year Institutions	
University of Wisconsin-Stout	Wisconsin	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Ursuline College	Ohio	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Utah State University	Utah	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Utah Valley University	Utah	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Utica College	New York	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Valley City State University	North Dakota	United States	Baccalaureate Colleges	Baccalaureate Colleges: Diverse Fields
Valparaiso University	Indiana	United States of America	Doctoral Universities	Doctoral/Professional Universities
Vanderbilt University	Tennessee	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Virginia Commonwealth University	Virginia	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
Virginia Union University	Virginia	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Wabash College	Indiana	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Wake Technical Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Traditional
Warren Wilson College	North Carolina	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Washington & Jefferson College	Pennsylvania	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Washington State University	Washington	United States of America	Associate's Colleges / 2-Year Institutions	
Washington University in St Louis	Missouri	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Washtenaw Community College	Michigan	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Waubonsee Community College	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional
Wayne State University	Michigan	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Weber State University	Utah	United States	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
Wellesley College	Massachusetts	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Wesleyan University	Connecticut	United States	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
West Chester University of Pennsylvania	Pennsylvania	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
West Shore Community College	Michigan	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-High Nontraditional

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
West Virginia University	West Virginia	United States of America	Doctoral Universities	Doctoral Universities: Very High Research Activity
Western Carolina University	North Carolina	United States of America	Doctoral Universities	Doctoral/Professional Universities
Western Connecticut State University	Connecticut	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Western Kentucky University	Kentucky	United States of America	Doctoral Universities	Doctoral/Professional Universities
Western Michigan University	Michigan	United States of America	Doctoral Universities	Doctoral Universities: High Research Activity
Western University	Ontario	Canada	International	
Westminster College-New Wilmington	Pennsylvania	United States of America	Associate's Colleges / 2-Year Institutions	
Westminster College-Salt Lake City	Utah	United States of America	Associate's Colleges / 2-Year Institutions	
Wheaton College (MA)	Massachusetts	United States of America	Baccalaureate Colleges	
Wheaton College-Wheaton	Illinois	United States of America	Associate's Colleges / 2-Year Institutions	
Wichita State University	Kansas	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Wilfrid Laurier University	Ontario	Canada	International	
Wilkes Community College	North Carolina	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: Mixed Transfer/ Career & Technical-Mixed Traditional/ Nontraditional
William Jessup University	California	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
William Paterson University of New Jersey	New Jersey	United States of America	Master's Colleges & Universities	Master's Colleges & Universities: Larger Programs
William Rainey Harper College	Illinois	United States	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
William Woods University	Missouri	United States of America	Doctoral Universities	Doctoral/Professional Universities
Windward Community College	Hawaii	United States of America	Associate's Colleges / 2-Year Institutions	Associate's Colleges: High Transfer-High Nontraditional
Wingate University	North Carolina	United States of America	Doctoral Universities	Doctoral/Professional Universities

Organization	State/Province	Country	Carnegie Class	Carnegie Class Detailed
Winona State University	Minnesota	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Wittenberg University	Ohio	United States of America	Baccalaureate Colleges	Baccalaureate Colleges: Arts & Sciences Focus
Worcester Polytechnic Institute	Massachusetts	United States	Doctoral Universities	Doctoral Universities: High Research Activity
Xavier University of Louisiana	Louisiana	United States	Master's Colleges & Universities	Master's Colleges & Universities: Medium Programs
Yakima Valley Community College	Washington	United States	Baccalaureate Colleges	Baccalaureate/ Associate's Colleges: Associate's Dominant
Yale University	Connecticut	United States	Doctoral Universities	Doctoral Universities: Very High Research Activity
York College Pennsylvania	Pennsylvania	United States of America	Baccalaureate Colleges	
Zayed University	Dubai		Not classified	

About the Editor

Gena Parsons-Diamond is the program manager, data and research at the Association of College and Research Libraries and supports ACRL's data products: Project Outcome for Academic Libraries, Benchmark: Library Metrics and Trends, and the Threshold Achievement Test for Information Literacy. She holds a Masters in Arts Management from Columbia College Chicago.

About the Authors

Jennifer Arnold (she/her) is the director of library services at Central Piedmont Community College in Charlotte, NC. She has served on a variety of ACRL committees, including chairing the Project Outcome for Academic Libraries Editorial Board. She currently serves as the community college representative on North Carolina's statewide consortium's (NC LIVE) executive committee. Jennifer has published and presented on a number of topics, including library administration and management, outcomes assessment, and library buildings. In addition to her MLIS from the University of South Florida, Jennifer has a Master of Public Administration from UNC-Chapel Hill.

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Dr. Laura D'Aveta is the Digital Resources and Graduate Liaison Librarian for Baldwin Wallace University in Berea, Ohio. She has five years of experience in academic libraries, with expertise in the areas of instruction, information literacy, and collection management. Her research interests involve children's literature and the integration of AI tools in the humanities classroom. She received her Master's of Library and Information Science from Kent State University.

Greg Davis (retired) most recently served as the manager of operations and planning and as the head of library assessment for the Iowa State University Library. He served on the initial ACRL Project Outcome Editorial Board that was responsible for the design and development of the Project Outcome for Academic Libraries toolkit. He also served as a visiting program officer for the Association of Research Libraries (ARL) Research Library Impact Factor (RLIF) initiative, where he worked to connect Project Outcome work with the ARL RLIF. He holds a Bachelor of Science degree in Mathematics, a Master of Science Degree in Curriculum and Education Technology, and a PhD in Education Leadership and Policy Studies, all from Iowa State University.

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Sara Goek is project manager, data and research at the Association of College and Research Libraries and the Public Library Association, divisions of the American Library Association. She is responsible for leading data-related initiatives and developing related research reports and training materials for public and academic libraries. She first came to ALA as Mellon/ACLS Public Fellow at ACRL, contributing to efforts to improve research about academic library contributions to student learning and success. She holds a Ph.D. in History / Digital Arts & Humanities from University College Cork.

LuMarie Guth (she/her)is the business librarian and associate professor at Western Michigan University in Kalamazoo, Michigan. Her research focuses on library instruction best practices. Her two-part study with colleagues at Western Michigan University and Wayne State University on disciplinary faculty perspectives on the ACRL Framework and faculty attitudes toward collaboration with librarians, published in portal, was recognized on the Library Instruction Round Table Top 20 list in 2018 and 2020 respectively. LuMarie holds a Master of Science in Library and Information Studies from the University of Wisconsin-Madison and an MBA from Columbus State University

Retha Hall is a senior librarian and manages the Cato campus library at Central Piedmont Community College. She has over 20 years experience in academic libraries, including both two-year and four-year institutions. Her duties include managing daily operations, program development and outreach and information literacy/instruction. In addition, she is a member of the NCCCLA executive board and serves as the nominating committee chair. Retha received her Master of Library Science degree from San Jose State University.

Martha Hood is the Associate Director for Evaluation and Assessment at the University of Houston-Clear Lake. She began her career as a cataloger working at both public and academic libraries. With 20 years of experience, she has held management and leadership roles within technical services. In 2020, she transitioned into a newly created role focused on assessment and data visualization. In addition to her MLS from the University of North Texas, Martha has a Master of Science in Occupational Education from the University of Houston.

Dr. Karen F. Kaufmann is an assistant professor of instruction at the University of South Florida School of Information. Her research interests center around information literacy, socio-cognitive user relevance, and information literacy as a paradigm for transfer student success. Before joining the USF iSchool faculty, she taught information literacy at a state college for over ten years. During this time, she initiated the development of a formal information literacy program for the academic library and developed an information literacy course for the Bachelor of Health Science program. Kaufmann is an educator, researcher, author, and presenter of information literacy in higher education. She is currently working with a global collaboration of colleagues on the ILIAD (Information

Literacy is a Discipline) project. The project includes her work as co-editor of the forthcoming publication titled: The Information Literacy Handbook: Charting the Discipline, with an anticipated publication date in 2025.

Kathleen Langan (she/her) is the engagement librarian and professor at Western Michigan University. Her research focuses on social and emotional engagement in higher education outside and bias in the information literacy landscape. She has served as chair and vice-chair of the ACRL Project Outcome Editorial Board and as president of the Michigan chapter of ACRL (Mi-ALA). Kate holds a Master of Science in Library and Information Studies from the University of Wisconsin-Madison and a Master of Arts in French Literature from the University of Oklahoma. She earned her Ph.D. in Interdisciplinary Studies from Western Michigan University. Her dissertation is titled "Flourishing in the Infosphere: Confidence, Community, and Belonging in Academic Libraries."

Kara Malenfant (she/her) is an assistant professor in the School of Information Studies at Dominican University in River Forest, IL. She previously worked at the Association of College and Research Libraries (ACRL, a division of the American Library Association) for nearly 18 years coordinating government relations advocacy, scholarly communication activities, ACRL's Value of Academic Libraries initiative, and leading the ACRL Consulting Services Program. Prior to working at ACRL, Kara was an academic librarian for 6 years and worked in international development. She earned a Ph.D. in leadership and change from Antioch University, a M.S. in library and information science from the University of Illinois–Urbana-Champaign, and a B.A. in English at Allegheny College.

Karoline Manny (she/her) is the reference, instruction and assessment librarian at Centre College in Danville, Kentucky. She also serves as the co-chair of the Information Literacy Work Group for the Kentucky Virtual Library. She has published and presented on effective library assessment and instruction practices, combating misinformation, and tools to support digital humanities. She has 30 years of experience teaching in higher education and ten years of experience in academic libraries. She holds a Master of Library and Information Science from the University of South Florida and a PhD in Spanish from the University of Kentucky.

Dylan McGlothlin (he/they) is the humanities librarian at Western Michigan University. His library research includes information literacy instruction and visualizing research. Their work has been published in *Envisioning the Framework: A Graphic Guide to Information Literacy* under the chapter "Applying Visuals to Frames in Music Information Literacy Education." Dylan holds a Master of Library and Information Science from Kent State University and a Master of Music from Central Washington University.

Katy Miller is department head for Student Learning and Engagement and the Connect Libraries at the University of Central Florida. She has 15 years of experience in academic libraries, with a focus on information literacy, student engagement, and online learning. She received her Master's of Library and Information Science from the University of North Texas.

Joe Pirillo is an Information Literacy and online learning librarian at the University of Wisconsin-Oshkosh. He has 20 years of experience in academic libraries, primarily in information literacy instruction. Currently, he is the library representative to UWO's College of Business. He has served as a member of ACRL's Value in Academic Libraries committee, where he chaired a sub-committee on Learning Analytics and privacy that produced ACRL's online learning analytics toolkit. He received his Master's of Library and Information Science degree from the University of Wisconsin, Milwaukee and his Master's in Education from Lakeland University. He also holds a lifetime education license in broad field social studies for grades 9-12 in Wisconsin. Prior to being an academic librarian, he was a high school teacher in Milwaukee, Wisconsin.

Dr. Rachel Edford Trnka is an instruction & engagement librarian at the University of Central Florida. She has an interdisciplinary background with a PhD in English, an MA in Liberal Studies, and an MLIS. Dr. Trnka has experience working in archives and public libraries as well as academic libraries. Prior to joining UCF Libraries, she worked as a college composition and English faculty at colleges and universities in Oregon and Florida. Trnka's research interests include theories and methods of information literacy instruction, intersections between library and information science and rhetorical studies, and the history of libraries and librarianship. Her work has appeared in The Journal of Academic Librarianship, the New Review of Academic Librarianship, Library and Information History, and College Literature.